Complex Analysis

Lecture Notes following Stein–Shakarchi (Ch. 1–10, Appendices A–B)

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Chapter 1. Preliminaries to Complex Analysis

Overview

We review the algebraic and analytic properties of complex numbers and some topology of the complex plane. We then define *holomorphicity*, the complex analytic version of differentiability, which leads to the Cauchy–Riemann equations and to power series. Finally we define curves and integrals along them, and prove a first result in the direction of Cauchy's theorem: if f has a primitive (a holomorphic F with $F' = f$), then $\int_\gamma f(z)\,dz = 0$ for every closed curve γ.

1 Complex numbers and the complex plane

1.1 Basic properties

A complex number has the form $z = x+iy$ with x, y real and $i^2 = -1$; $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$ are its **real part** and **imaginary part**. Real numbers are those with zero imaginary part; a number with zero real part is **purely imaginary**. The set of complex numbers is $\mathbb{C}$, identified with $\mathbb{R}^2$ by $z = x + iy \leftrightarrow (x, y)$; the x- and y-axes are the **real axis** and **imaginary axis**.

Treating all numbers as real and using $i^2 = -1$: if $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$,

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \qquad z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2).$$

These operations are commutative, associative and distributive. Addition is vector addition in $\mathbb{R}^2$; multiplication is a rotation composed with a dilation (see the polar form below); multiplication by i is rotation by $\pi/2$.

The **absolute value** of $z = x+iy$ is $|z| = (x^2+y^2)^{1/2}$, the euclidean distance from 0 to (x, y). The triangle inequality $|z + w| \le |z| + |w|$ holds, and also $|\mathrm{Re}(z)| \le |z|$, $|\mathrm{Im}(z)| \le |z|$ and

$$\big||z| - |w|\big| \le |z - w|,$$

which follows from $|z| \le |z - w| + |w|$ and $|w| \le |z - w| + |z|$.

The **complex conjugate** of $z = x + iy$ is $\overline{z} = x - iy$ (reflection across the real axis). z is real iff $z = \overline{z}$, purely imaginary iff $z = -\overline{z}$, and

$$\mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}, \qquad |z|^2 = z\overline{z}, \qquad \frac{1}{z} = \frac{\overline{z}}{|z|^2}\ (z \neq 0).$$

Any non-zero z has a **polar form** $z = re^{i\theta}$ with $r > 0$, $\theta \in \mathbb{R}$, where $e^{i\theta} = \cos\theta + i\sin\theta$. Since $|e^{i\theta}| = 1$, $r = |z|$; θ is the **argument** $\arg z$, defined up to a multiple of 2π, the counterclockwise angle from the positive real axis to the half-line through z. If $z = re^{i\theta}$ and $w = se^{i\varphi}$ then $zw = rse^{i(\theta+\varphi)}$: multiplication is a homothety (a rotation composed with a dilation).

1.2 Convergence

A sequence $\{z_n\}$ **converges** to $w \in \mathbb{C}$ if $\lim_{n\to\infty}|z_n - w| = 0$; we write $w = \lim z_n$. Since $|\cdot|$ on $\mathbb{C}$ is the euclidean distance on $\mathbb{R}^2$, this is convergence of the points in the plane; equivalently, the real and imaginary parts of z_n converge to those of w. $\{z_n\}$ is a **Cauchy sequence** if $|z_n - z_m| \to 0$ as $n, m \to \infty$, i.e. for every $\epsilon > 0$ there is N with $|z_n - z_m| < \epsilon$ for $n, m > N$. $\mathbb{R}$ is complete (every Cauchy sequence of reals converges), and $\{z_n\}$ is Cauchy iff its real and imaginary parts are; hence:

Theorem 1.1

$\mathbb{C}$, the complex numbers, is complete.

1.3 Sets in the complex plane

For $z_0 \in \mathbb{C}$ and $r > 0$, the **open disc** is $D_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\}$, the **closed disc** is $\overline{D}_r(z_0) = \{z : |z - z_0| \leq r\}$, and their boundary is the circle $C_r(z_0) = \{z : |z - z_0| = r\}$. The unit disc is denoted $\mathbb{D} = \{z : |z| < 1\}$.

Given $\Omega \subset \mathbb{C}$, z_0 is an **interior point** of Ω if $D_r(z_0) \subset \Omega$ for some $r > 0$; the **interior** of Ω consists of its interior points. Ω is **open** if every point of Ω is an interior point, and **closed** if $\Omega^c = \mathbb{C} - \Omega$ is open. z is a **limit point** of Ω if there are $z_n \in \Omega$, $z_n \neq z$, with $z_n \to z$; a set is closed iff it contains all its limit points. The **closure** $\overline{\Omega}$ is Ω together with its limit points, and the **boundary** $\partial\Omega$ is the closure minus the interior. Ω is **bounded** if $|z| < M$ for $z \in \Omega$; then $\operatorname{diam}(\Omega) = \sup_{z,w\in\Omega}|z - w|$. Ω is **compact** if it is closed and bounded.

Theorem 1.2

The set $\Omega \subset \mathbb{C}$ is compact if and only if every sequence $\{z_n\} \subset \Omega$ has a subsequence that converges to a point in Ω.

Proof (supplied). If Ω is closed and bounded, a sequence in Ω has bounded real and imaginary parts; by Bolzano–Weierstrass (twice) a subsequence converges, and its limit lies in Ω since Ω is closed. Conversely, if Ω were unbounded, choose $z_n \in \Omega$ with $|z_n| > n$; no subsequence converges. If Ω were not closed, some limit point $z \notin \Omega$ is the limit of $z_n \in \Omega$, and every subsequence converges to $z \notin \Omega$. ■

An **open covering** of Ω is a family of open sets $\{U_\alpha\}$ with $\Omega \subset \bigcup_\alpha U_\alpha$.

Theorem 1.3

A set Ω is compact if and only if every open covering of Ω has a finite subcovering.

Proof (supplied). Suppose Ω is compact and $\{U_\alpha\}$ is an open covering with no finite subcovering. Ω lies in a square Q_0; subdivide it into four equal squares; the parts of Ω in at least one of them, Q_1, cannot be finitely covered. Repeating, we get nested squares Q_n with $\operatorname{diam}(Q_n) \to 0$ such that $\Omega \cap Q_n$ is not finitely covered. Pick $z_n \in \Omega \cap Q_n$; $\{z_n\}$ is Cauchy, converges to some $w \in \Omega$ (Ω is closed), and $w \in U_\alpha$ for some α. Since U_α is open and $\operatorname{diam}(Q_n) \to 0$, $Q_n \subset U_\alpha$ for large n, a contradiction. Conversely, if every open covering has a finite subcovering, the discs $D_n(0)$ cover Ω, so Ω is bounded; and if $z \notin \Omega$ were a limit point, the sets $\{w : |w - z| > 1/n\}$ would cover Ω with no finite subcovering. ■

Proposition 1.4

If $\Omega_1 \supset \Omega_2 \supset \cdots \supset \Omega_n \supset \cdots$ is a sequence of non-empty compact sets in $\mathbb{C}$ with the property that $\operatorname{diam}(\Omega_n) \to 0$ as $n \to \infty$, then there exists a unique point $w \in \mathbb{C}$ such that $w \in \Omega_n$ for all n.

Proof. Choose $z_n \in \Omega_n$. The condition $\operatorname{diam}(\Omega_n) \to 0$ says precisely that $\{z_n\}$ is Cauchy, so it converges to some w. Since each Ω_n is compact and contains z_m for $m \geq n$, $w \in \Omega_n$ for all n. If $w' \neq w$ had the same property, then $|w - w'| > 0$ would violate $\operatorname{diam}(\Omega_n) \to 0$. ■

An open set Ω is **connected** if it cannot be written as $\Omega = \Omega_1 \cup \Omega_2$ with Ω_1, Ω_2 disjoint non-empty open sets. A connected open set in $\mathbb{C}$ is a **region**. A closed set F is connected if it cannot be written as $F = F_1 \cup F_2$ with F_1, F_2 disjoint non-empty closed sets. Equivalently, an open set Ω is

connected iff any two points of Ω can be joined by a curve in Ω.

2 Functions on the complex plane

2.1 Continuous functions

f defined on Ω is **continuous** at $z_0 \in \Omega$ if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(z) - f(z_0)| < \epsilon$ whenever $z \in \Omega$, $|z - z_0| < \delta$; equivalently, $z_n \to z_0$ implies $f(z_n) \to f(z_0)$. f is continuous on Ω if continuous at every point. Sums and products of continuous functions are continuous. A function of $z = x + iy$ is continuous iff it is continuous as a function of the two real variables x, y. If f is continuous, so is $z \mapsto |f(z)|$. f **attains a maximum** at z_0 if $|f(z)| \leq |f(z_0)|$ for all $z \in \Omega$ (reversed for a minimum).

Theorem 2.1

A continuous function on a compact set Ω is bounded and attains a maximum and minimum on Ω.

(The proof is the same as for functions of a real variable.)

2.2 Holomorphic functions

Definition (holomorphic function)

Let Ω be open and f complex-valued on Ω. f is **holomorphic at** $z_0 \in \Omega$ if the quotient

$$\frac{f(z_0 + h) - f(z_0)}{h} \tag{1}$$

converges to a limit as $h \to 0$ ($h \in \mathbb{C}$, $h \neq 0$, $z_0 + h \in \Omega$). The limit is the **derivative** $f'(z_0)$. f is **holomorphic on** Ω if it is holomorphic at every point of Ω; if C is closed, f is holomorphic on C if it is holomorphic on some open set containing C; f holomorphic on all of $\mathbb{C}$ is **entire**.

In the limit h may approach 0 from any direction. The terms *regular* or *complex differentiable* are also used. Holomorphic functions turn out to be infinitely complex differentiable and *analytic* (they have power series expansions), in contrast with real-differentiable functions.

Example 1

$f(z) = z$ is holomorphic on any open set with $f'(z) = 1$; any polynomial $p(z) = a_0 + a_1 z + \cdots + a_n z^n$ is entire, with $p'(z) = a_1 + \cdots + n a_n z^{n-1}$ (by Proposition 2.2).

Example 2

$1/z$ is holomorphic on any open set not containing 0, with $f'(z) = -1/z^2$.

Example 3

$f(z) = \overline{z}$ is not holomorphic: $\dfrac{f(z_0 + h) - f(z_0)}{h} = \dfrac{\overline{h}}{h}$, which has no limit as $h \to 0$ (it is 1 for real h and -1 for purely imaginary h).

Clearly f is holomorphic at z_0 iff there is $a \in \mathbb{C}$ with

$$f(z_0+h) - f(z_0) - ah = h\psi(h), \tag{2}$$

where ψ is defined for small h and $\lim_{h\to 0}\psi(h) = 0$; then $a = f'(z_0)$. Hence f is continuous wherever it is holomorphic. Arguing as for one real variable, using (2):

Proposition 2.2

If f and g are holomorphic in Ω, then: (i) $f+g$ is holomorphic in Ω and $(f+g)' = f'+g'$. (ii) fg is holomorphic in Ω and $(fg)' = f'g + fg'$. (iii) If $g(z_0) \neq 0$, then f/g is holomorphic at z_0 and $(f/g)' = \dfrac{f'g - fg'}{g^2}$. Moreover, if $f : \Omega \to U$ and $g : U \to \mathbb{C}$ are holomorphic, the chain rule holds: $(g\circ f)'(z) = g'(f(z))f'(z)$ for all $z \in \Omega$.

Proof (supplied, chain rule). Write $f(z+h) - f(z) = f'(z)h + h\psi_1(h)$ and $g(w+k) - g(w) = g'(w)k + k\psi_2(k)$ with $\psi_j \to 0$, $w = f(z)$. With $k = f(z+h) - f(z) \to 0$, $g(f(z+h)) - g(f(z)) = g'(w)k + k\psi_2(k) = g'(w)f'(z)h + h[g'(w)\psi_1(h) + (f'(z)+\psi_1(h))\psi_2(k)]$, and the bracket tends to 0. (i)–(iii) are similar. ∎

Complex-valued functions as mappings. $\overline{z}$ corresponds to $F : (x,y) \mapsto (x,-y)$, which is linear, hence real-differentiable to all orders; so real differentiability does not imply holomorphy. To $f = u + iv$ associate $F(x,y) = (u(x,y), v(x,y))$. F is **differentiable** at $P_0 = (x_0,y_0)$ if there is a linear $J : \mathbb{R}^2 \to \mathbb{R}^2$ with

$$\frac{|F(P_0+H) - F(P_0) - J(H)|}{|H|} \to 0 \quad \text{as } |H| \to 0,\ H \in \mathbb{R}^2; \tag{3}$$

then J is the Jacobian matrix $J_F = \begin{pmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{pmatrix}$. If f is holomorphic at z_0, taking $h = h_1$ real in (1) gives $f'(z_0) = \frac{\partial f}{\partial x}(z_0)$, and taking $h = ih_2$ purely imaginary gives $f'(z_0) = \frac{1}{i}\frac{\partial f}{\partial y}(z_0)$. Hence $\frac{\partial f}{\partial x} = \frac{1}{i}\frac{\partial f}{\partial y}$, and separating real and imaginary parts (using $1/i = -i$) gives the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Define the operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y}\Big), \qquad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\Big).$$

Proposition 2.3

If f is holomorphic at z_0, then

$$\frac{\partial f}{\partial \overline{z}}(z_0) = 0 \qquad \text{and} \qquad f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2\frac{\partial u}{\partial z}(z_0).$$

Also, if we write $F(x,y) = f(z)$, then F is differentiable in the sense of real variables, and $\det J_F(x_0,y_0) = |f'(z_0)|^2$.

Proof. Taking real and imaginary parts, the Cauchy–Riemann equations are equivalent to $\partial f/\partial\overline{z} = 0$. By the observation above, $f'(z_0) = \frac{1}{2}\big(\frac{\partial f}{\partial x}(z_0) + \frac{1}{i}\frac{\partial f}{\partial y}(z_0)\big) = \frac{\partial f}{\partial z}(z_0)$, and the Cauchy–Riemann equations give $\partial f/\partial z = 2\partial u/\partial z$. For differentiability of F: if $H = (h_1,h_2)$ and $h = h_1 + ih_2$, the Cauchy–Riemann equations imply

$$J_F(x_0,y_0)(H) = \Big(\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}\Big)(h_1 + ih_2) = f'(z_0)h,$$

identifying complex numbers with pairs of reals. Finally, by the Cauchy–Riemann equations,

$$\det J_F(x_0,y_0) = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial v}{\partial x}\frac{\partial u}{\partial y} = \Big(\frac{\partial u}{\partial x}\Big)^2 + \Big(\frac{\partial u}{\partial y}\Big)^2 = \Big|2\frac{\partial u}{\partial z}\Big|^2 = |f'(z_0)|^2. \tag{4}$$

■

The converse:

Theorem 2.4

Suppose $f = u + iv$ is a complex-valued function defined on an open set Ω. If u and v are continuously differentiable and satisfy the Cauchy–Riemann equations on Ω, then f is holomorphic on Ω and $f'(z) = \partial f/\partial z$.

Proof. Write

$$u(x+h_1,y+h_2)-u(x,y) = \frac{\partial u}{\partial x}h_1+\frac{\partial u}{\partial y}h_2+|h|\psi_1(h), \qquad v(x+h_1,y+h_2)-v(x,y) = \frac{\partial v}{\partial x}h_1+\frac{\partial v}{\partial y}h_2+|h|\psi_2(h),$$

with $\psi_j(h) \to 0$ as $|h| \to 0$, $h = h_1 + ih_2$ (differentiability follows from continuity of the partials). Using the Cauchy–Riemann equations,

$$f(z+h) - f(z) = \Big(\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}\Big)(h_1 + ih_2) + |h|\psi(h),$$

with $\psi = \psi_1 + i\psi_2 \to 0$. So f is holomorphic and $f'(z) = 2\partial u/\partial z = \partial f/\partial z$. ■

2.3 Power series

The complex exponential is $e^z = \sum_{n=0}^{\infty} z^n/n!$. For real z this is the usual exponential, and the series converges absolutely for every $z \in \mathbb{C}$, since $|z^n/n!| = |z|^n/n!$ and $\sum |z|^n/n! = e^{|z|} < \infty$; the estimate shows uniform convergence on every disc. We will see that e^z is entire and its derivative is obtained termwise: $(e^z)' = \sum_{n=0}^{\infty} nz^{n-1}/n! = \sum_{m=0}^{\infty} z^m/m! = e^z$. The geometric series $\sum z^n$ converges absolutely only in $|z| < 1$, with sum $1/(1-z)$ (holomorphic on $\mathbb{C} - \{1\}$), since $\sum_{n=0}^{N} z^n = \frac{1-z^{N+1}}{1-z}$ and $z^{N+1} \to 0$ for $|z| < 1$.

A **power series** is an expansion $\sum_{n=0}^{\infty} a_n z^n$ with $a_n \in \mathbb{C}$ (5). If it converges absolutely at z_0, it converges for all $|z| \le |z_0|$.

Theorem 2.5

Given a power series $\sum_{n=0}^{\infty} a_n z^n$, there exists $0 \le R \le \infty$ such that: (i) If $|z| < R$ the series converges absolutely. (ii) If $|z| > R$ the series diverges. Moreover, if we use the convention that $1/0 = \infty$ and $1/\infty = 0$, then R is given by Hadamard's formula

$$1/R = \limsup |a_n|^{1/n}.$$

R is the **radius of convergence** and $|z| < R$ the **disc of convergence**; $R = \infty$ for e^z and $R = 1$ for the geometric series.

Proof. Let $L = 1/R$ and suppose $L \neq 0, \infty$ (these cases are easy). If $|z| < R$, choose $\epsilon > 0$ with $(L+\epsilon)|z| = r < 1$. By definition of L, $|a_n|^{1/n} \le L + \epsilon$ for large n, so $|a_n||z|^n \le \{(L+\epsilon)|z|\}^n = r^n$, and comparison with $\sum r^n$ gives convergence. If $|z| > R$, a similar argument gives a sequence of terms whose absolute values tend to infinity, so the series diverges. ■

Remark

On the boundary $|z| = R$ either convergence or divergence can occur.

Further entire power series are

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \qquad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$

agreeing with the usual functions for real z, and related to the exponential by the **Euler formulas**

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Theorem 2.6

The power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function in its disc of convergence. The derivative of f is also a power series obtained by differentiating term by term the series for f, that is, $f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$. Moreover, f' has the same radius of convergence as f.

Proof. Since $\lim n^{1/n} = 1$, $\limsup |a_n|^{1/n} = \limsup |n a_n|^{1/n}$, so $\sum a_n z^n$ and $\sum n a_n z^n$, hence $\sum n a_n z^{n-1}$, have the same radius of convergence (Hadamard's formula).

Let $g(z) = \sum n a_n z^{n-1}$, let R be the radius of convergence, and $|z_0| < r < R$. Write $f(z) = S_N(z) + E_N(z)$ with $S_N(z) = \sum_{n=0}^{N} a_n z^n$, $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. If $|z_0 + h| < r$,

$$\frac{f(z_0+h) - f(z_0)}{h} - g(z_0) = \left(\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\right) + (S_N'(z_0) - g(z_0)) + \frac{E_N(z_0+h) - E_N(z_0)}{h}.$$

Since $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$,

$$\left|\frac{E_N(z_0+h) - E_N(z_0)}{h}\right| \le \sum_{n=N+1}^{\infty} |a_n| \left|\frac{(z_0+h)^n - z_0^n}{h}\right| \le \sum_{n=N+1}^{\infty} |a_n| n r^{n-1},$$

the tail of a convergent series (as g converges absolutely on $|z| < R$). Given $\epsilon > 0$ choose N_1 so that $N > N_1$ makes this $< \epsilon$, and N_2 so that $N > N_2$ gives $|S_N'(z_0) - g(z_0)| < \epsilon$. Fix $N > \max(N_1, N_2)$; since the derivative of a polynomial is obtained termwise, there is $\delta > 0$ with $\left|\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\right| < \epsilon$ for $|h| < \delta$. Then $\left|\frac{f(z_0+h) - f(z_0)}{h} - g(z_0)\right| < 3\epsilon$ for $|h| < \delta$. ∎

Corollary 2.7

A power series is infinitely complex differentiable in its disc of convergence, and the higher derivatives are also power series obtained by termwise differentiation.

A power series centered at z_0 is $f(z) = \sum a_n (z - z_0)^n$; with $g(w) = \sum a_n w^n$, $f(z) = g(z - z_0)$, so everything about g transfers, e.g. $f'(z) = \sum n a_n (z - z_0)^{n-1}$.

Definition (analytic function)

f defined on an open set Ω is **analytic** (has a power series expansion) at $z_0 \in \Omega$ if there is a power series $\sum a_n (z - z_0)^n$ with positive radius of convergence such that $f(z) = \sum a_n (z - z_0)^n$ for z near z_0. If this holds at every point of Ω, f is analytic on Ω.

By Theorem 2.6, analytic functions are holomorphic; the converse (Chapter 2) justifies using the two terms interchangeably.

3 Integration along curves

A **parametrized curve** is a function $z(t)$ from a closed interval $[a,b] \subset \mathbb{R}$ to $\mathbb{C}$. It is **smooth** if $z'(t)$ exists and is continuous on $[a,b]$ and $z'(t) \neq 0$; at $t = a, b$, z' means the one-sided derivatives

$$z'(a) = \lim_{h \to 0,\, h>0} \frac{z(a+h) - z(a)}{h}, \qquad z'(b) = \lim_{h \to 0,\, h<0} \frac{z(b+h) - z(b)}{h}.$$

It is **piecewise-smooth** if z is continuous on $[a,b]$ and smooth on intervals $[a_k, a_{k+1}]$ for some $a = a_0 < a_1 < \cdots < a_n = b$ (one-sided derivatives at the a_k may differ). Two parametrizations $z : [a,b] \to \mathbb{C}$, $\tilde{z} : [c,d] \to \mathbb{C}$ are **equivalent** if there is a continuously differentiable bijection $s \mapsto t(s)$ of $[c,d]$ onto $[a,b]$ with $t'(s) > 0$ and $\tilde{z}(s) = z(t(s))$; $t' > 0$ means orientation is preserved. The equivalence class determines a smooth curve $\gamma \subset \mathbb{C}$, the image of $[a,b]$ with the orientation given by z. γ^- is γ with orientation reversed, parametrized e.g. by $z^-(t) = z(b+a-t)$. The **end-points** $z(a)$, $z(b)$ do not depend on the parametrization; γ begins at $z(a)$ and ends at $z(b)$. γ is **closed** if $z(a) = z(b)$, and **simple** if $z(t) \neq z(s)$ unless $s = t$ (for closed curves: unless $s = t$, or $s = a$ and $t = b$). For brevity, a *curve* means a piecewise-smooth curve.

The circle $C_r(z_0)$ has the **positive orientation** (counterclockwise) given by $z(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and the negative orientation given by $z(t) = z_0 + re^{-it}$. C denotes a general positively oriented circle.

Definition (integral along a curve; length)

For a smooth curve γ parametrized by $z : [a,b] \to \mathbb{C}$ and f continuous on γ,

$$\int_\gamma f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt.$$

For piecewise-smooth γ, $\int_\gamma f(z)\,dz = \sum_{k=0}^{n-1} \int_{a_k}^{a_{k+1}} f(z(t))z'(t)\,dt$. The **length** of a smooth curve is $\text{length}(\gamma) = \int_a^b |z'(t)|\,dt$, and that of a piecewise-smooth curve is the sum of the lengths of its smooth parts.

These are independent of the parametrization: if $\tilde{z}$ is equivalent to z, by change of variables and the chain rule

$$\int_a^b f(z(t))z'(t)\,dt = \int_c^d f(z(t(s)))z'(t(s))t'(s)\,ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s)\,ds.$$

Proposition 3.1

Integration of continuous functions over curves satisfies the following properties: (i) It is linear: if $\alpha, \beta \in \mathbb{C}$, then $\int_\gamma (\alpha f(z) + \beta g(z))\,dz = \alpha \int_\gamma f(z)\,dz + \beta \int_\gamma g(z)\,dz$. (ii) If γ^- is γ with the reverse orientation, then $\int_\gamma f(z)\,dz = -\int_{\gamma^-} f(z)\,dz$. (iii) One has the inequality $\left|\int_\gamma f(z)\,dz\right| \leq \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$.

Proof. (i) follows from linearity of the Riemann integral. (ii) (supplied): with $z^-(t) = z(a+b-t)$, $\int_a^b f(z^-(t))(z^-)'(t)\,dt = -\int_a^b f(z(a+b-t))z'(a+b-t)\,dt = -\int_a^b f(z(s))z'(s)\,ds$. (iii): $\left|\int_\gamma f(z)\,dz\right| \leq \sup_{t\in[a,b]}|f(z(t))| \int_a^b |z'(t)|\,dt \leq \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$. ■

A **primitive** for f on Ω is a function F holomorphic on Ω with $F'(z) = f(z)$ for all $z \in \Omega$.

Theorem 3.2

If a continuous function f has a primitive F in Ω, and γ is a curve in Ω that begins at w_1 and ends at w_2, then $\int_\gamma f(z)\,dz = F(w_2) - F(w_1)$.

Proof. If γ is smooth with parametrization $z : [a, b] \to \mathbb{C}$, $z(a) = w_1$, $z(b) = w_2$, then by the chain rule and the fundamental theorem of calculus

$$\int_\gamma f(z)\,dz = \int_a^b F'(z(t))z'(t)\,dt = \int_a^b \frac{d}{dt}F(z(t))\,dt = F(z(b)) - F(z(a)).$$

If γ is piecewise-smooth we get a telescoping sum: $\sum_{k=0}^{n-1}[F(z(a_{k+1})) - F(z(a_k))] = F(z(b)) - F(z(a))$. ■

Corollary 3.3

If γ is a closed curve in an open set Ω, and f is continuous and has a primitive in Ω, then $\oint_\gamma f(z)\,dz = 0$.

This is immediate since the end-points of a closed curve coincide. For example, $1/z$ has no primitive in $\mathbb{C} - \{0\}$: for the unit circle C, $z(t) = e^{it}$,

$$\oint_C f(z)\,dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}\,dt = 2\pi i \neq 0.$$

This innocent calculation lies at the heart of the theory.

Corollary 3.4

If f is holomorphic in a region Ω and $f' = 0$, then f is constant.

Proof. Fix $w_0 \in \Omega$; it suffices to show $f(w) = f(w_0)$ for all $w \in \Omega$. Since Ω is connected, there is a curve γ joining w_0 to w. f is a primitive of f', so $\int_\gamma f'(z)\,dz = f(w) - f(w_0)$ by Theorem 3.2. But $f' = 0$, so the integral is 0 and $f(w) = f(w_0)$. ■

Chapter 2. Cauchy's Theorem and Its Applications

Overview

Loosely, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also in Ω, then

$$\int_\gamma f(z)\,dz = 0. \tag{1}$$

A precise general formulation requires defining the "interior" of a curve unambiguously, which is not easy; at this stage we restrict to regions bounded by "toy contours", whose interiors are obvious. The starting point: it suffices that f have a primitive (Corollary 3.3), and primitives exist by *Goursat's theorem*: $\int_T f(z)\,dz = 0$ for a triangle T whose interior lies in Ω. This leads to the *Cauchy integral formula*

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad (z \text{ inside } C),$$

and hence to the regularity of holomorphic functions (existence of derivatives of all orders—decisively false for real variables), power series expansions, analytic continuation, Liouville's theorem and the fundamental theorem of algebra, and Morera's theorem.

1 Goursat's theorem

By Corollary 3.3, if f has a primitive in Ω then $\int_\gamma f = 0$ for closed $\gamma \subset \Omega$. Conversely, if the integral vanishes for certain curves, a primitive will exist. Goursat's theorem is the starting point from which most other results of this chapter are deduced. (Its interest is that its proof needs only the existence of f' at each point, not its continuity.)

Theorem 1.1

If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then

$$\int_T f(z)\,dz = 0$$

whenever f is holomorphic in Ω.

Proof. Let $T^{(0)}$ be the original triangle (positively oriented), with diameter $d^{(0)}$ and perimeter $p^{(0)}$. Bisect each side and connect the midpoints, creating four similar triangles $T_1^{(1)}, \dots, T_4^{(1)}$, oriented consistently with $T^{(0)}$. After cancellation of the integrals over interior sides traversed in opposite directions,

$$\int_{T^{(0)}} f(z)\,dz = \int_{T_1^{(1)}} f + \int_{T_2^{(1)}} f + \int_{T_3^{(1)}} f + \int_{T_4^{(1)}} f. \tag{2}$$

For some j, $\left|\int_{T^{(0)}} f\right| \le 4\left|\int_{T_j^{(1)}} f\right|$, for otherwise (2) would be contradicted. Call this triangle $T^{(1)}$; its diameter and perimeter are $d^{(1)} = \frac{1}{2}d^{(0)}$, $p^{(1)} = \frac{1}{2}p^{(0)}$. Repeating, we obtain triangles $T^{(0)}, T^{(1)}, \dots, T^{(n)}, \dots$ with

$$\left|\int_{T^{(0)}} f(z)\,dz\right| \le 4^n \left|\int_{T^{(n)}} f(z)\,dz\right|, \qquad d^{(n)} = 2^{-n}d^{(0)}, \quad p^{(n)} = 2^{-n}p^{(0)}.$$

Let $\mathcal{T}^{(n)}$ be the solid closed triangle with boundary $T^{(n)}$. These form a nested sequence of compact sets with diameters tending to 0, so by Proposition 1.4 there is a unique point z_0 in all of them.

Since f is holomorphic at z_0,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0), \qquad \psi(z) \to 0 \text{ as } z \to z_0.$$

The constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, so by Corollary 3.3

$$\int_{T^{(n)}} f(z)\,dz = \int_{T^{(n)}} \psi(z)(z - z_0)\,dz. \tag{3}$$

z_0 is in the solid triangle $\mathcal{T}^{(n)}$ and z on its boundary, so $|z - z_0| \le d^{(n)}$, and by Proposition 3.1(iii)

$$\Big|\int_{T^{(n)}} f(z)\,dz\Big| \le \epsilon_n d^{(n)} p^{(n)} = \epsilon_n 4^{-n} d^{(0)} p^{(0)}, \qquad \epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \to 0.$$

Hence $\left|\int_{T^{(0)}} f\right| \le 4^n \left|\int_{T^{(n)}} f\right| \le \epsilon_n d^{(0)} p^{(0)}$, and letting $n \to \infty$ concludes the proof. ∎

Corollary 1.2

If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then $\int_R f(z)\,dz = 0$.

Proof. Split R by a diagonal into two triangles T_1, T_2, oriented consistently; then $\int_R f = \int_{T_1} f + \int_{T_2} f = 0$. ∎

2 Local existence of primitives and Cauchy's theorem in a disc

Theorem 2.1

A holomorphic function in an open disc has a primitive in that disc.

Proof. After a translation we may assume the disc D is centered at 0. For $z \in D$ let γ_z be the polygonal line (at most two segments) from 0 horizontally to $\tilde{z} = \mathrm{Re}(z)$ and then vertically to z, oriented from 0 to z, and define

$$F(z) = \int_{\gamma_z} f(w)\,dw.$$

Fix $z \in D$ and h so small that $z + h \in D$. In $F(z + h) - F(z) = \int_{\gamma_{z+h}} f - \int_{\gamma_z} f$, the integrals over the common segment from the origin cancel; completing a square and a triangle and applying Goursat's theorem for triangles and rectangles, we are left with

$$F(z + h) - F(z) = \int_{\eta} f(w)\,dw,$$

η the segment from z to $z + h$. Since f is continuous at z, $f(w) = f(z) + \psi(w)$ with $\psi(w) \to 0$ as $w \to z$. Hence

$$F(z + h) - F(z) = f(z) \int_{\eta} dw + \int_{\eta} \psi(w)\,dw. \tag{4}$$

The constant 1 has primitive w, so the first integral is h (Theorem 3.2); and $\left|\int_\eta \psi(w)\,dw\right| \le \sup_{w \in \eta} |\psi(w)|\,|h|$. The supremum tends to 0 as $h \to 0$, so (4) gives $\lim_{h \to 0} \frac{F(z+h) - F(z)}{h} = f(z)$. ∎

So every holomorphic function has a primitive *locally*. The theorem holds not only for discs but for other sets as well ("toy contours" below).

Theorem 2.2 (Cauchy's theorem for a disc)

If f is holomorphic in a disc, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in that disc.

Proof. f has a primitive (Theorem 2.1); apply Corollary 3.3. ∎

Corollary 2.3

Suppose f is holomorphic in an open set containing the circle C and its interior. Then $\int_C f(z)\,dz = 0$.

Proof. Let D be the disc bounded by C. There is a slightly larger disc $D' \supset \overline{D}$ on which f is holomorphic; apply Cauchy's theorem in D'. ∎

Definition (toy contour)

A **toy contour** is any closed curve whose interior is obvious and for which a construction as in Theorem 2.1 is possible in a neighborhood of the curve and its interior. Its **positive orientation** is the one for which the interior is to the left as we travel along it.

Circles, triangles and rectangles are toy contours. Another important one is the **keyhole** Γ: two almost complete circles, one large and one small, connected by a narrow corridor. If f is holomorphic in a neighborhood of Γ and its interior, it is holomorphic inside a slightly larger keyhole Γ' whose interior contains $\overline{\operatorname{int}\Gamma}$. Fix $z_0 \in \operatorname{int}\Gamma'$; for $z \in \operatorname{int}\Gamma'$ let γ_z be a curve in $\operatorname{int}\Gamma'$ from z_0 to z made of finitely many horizontal or vertical segments. For any other such curve $\tilde{\gamma}_z$, Corollary 1.2 gives $\int_{\gamma_z} f = \int_{\tilde{\gamma}_z} f$, so $F(z) = \int_{\gamma_z} f$ is well defined, and as above F is a primitive of f; hence $\int_\Gamma f = 0$. In general, for a toy contour γ,

$$\int_\gamma f(z)\,dz = 0$$

whenever f is holomorphic in an open set containing γ and its interior. Further examples used later: the rectangular keyhole, the multiple keyhole, the semicircle, the indented semicircle, the sector and the parallelogram. For general curves, Appendix B proves the Jordan curve theorem for piecewise-smooth curves: a simple closed piecewise-smooth curve has a well-defined "simply connected" interior, and Cauchy's theorem holds in that setting.

3 Evaluation of some integrals

We show how some integrals may be evaluated using Cauchy's theorem; a systematic approach via residues is in Chapter 3.

Example 1

If $\xi \in \mathbb{R}$, then

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\,dx, \tag{5}$$

i.e. $e^{-\pi x^2}$ is its own Fourier transform. For $\xi = 0$ this is the known integral $1 = \int_{-\infty}^{\infty} e^{-\pi x^2}\,dx$. Suppose $\xi > 0$ and let $f(z) = e^{-\pi z^2}$, entire. Let γ_R be the rectangle with vertices R, $R + i\xi$, $-R + i\xi$, $-R$, positively oriented. By Cauchy's theorem

$$\int_{\gamma_R} f(z)\,dz = 0. \tag{6}$$

The integral over the real segment is $\int_{-R}^{R} e^{-\pi x^2}\,dx \to 1$. The integral over the right vertical side is $I(R) = \int_0^\xi f(R+iy)\,i\,dy = \int_0^\xi e^{-\pi(R^2+2iRy-y^2)}\,i\,dy$, and $|I(R)| \le Ce^{-\pi R^2} \to 0$ (ξ fixed); similarly for the left side. The top side contributes

$$\int_R^{-R} e^{-\pi(x+i\xi)^2}\,dx = -e^{\pi\xi^2}\int_{-R}^{R} e^{-\pi x^2} e^{-2\pi i x\xi}\,dx.$$

Letting $R \to \infty$, (6) gives $0 = 1 - e^{\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx$. If $\xi < 0$ use the symmetric rectangle in the lower half-plane. (This *shifting of the contour* has many applications.)

Example 2

$$\int_0^\infty \frac{1-\cos x}{x^2}\, dx = \frac{\pi}{2}.$$

Let $f(z) = (1-e^{iz})/z^2$ and integrate over the indented semicircle in the upper half-plane: the segments $[-R,-\epsilon]$ and $[\epsilon,R]$, the small semicircle γ_ϵ^+ of radius ϵ (negatively oriented) and the large one γ_R^+ (positively oriented). By Cauchy's theorem

$$\int_{-R}^{-\epsilon} \frac{1-e^{ix}}{x^2}\, dx + \int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\, dz + \int_\epsilon^R \frac{1-e^{ix}}{x^2}\, dx + \int_{\gamma_R^+} \frac{1-e^{iz}}{z^2}\, dz = 0.$$

Since $|(1-e^{iz})/z^2| \le 2/|z|^2$ in the upper half-plane, the integral over γ_R^+ tends to 0 as $R \to \infty$, so $\int_{|x|\ge\epsilon} \frac{1-e^{ix}}{x^2}\, dx = -\int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\, dz$. Now $f(z) = \frac{-iz}{z^2} + E(z)$ with E bounded near 0, and on γ_ϵ^+, $z = \epsilon e^{i\theta}$, $dz = i\epsilon e^{i\theta} d\theta$, so

$$\int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\, dz \to \int_\pi^0 (-i)(i)\, d\theta = -\pi \quad \text{as } \epsilon \to 0.$$

Taking real parts, $\int_{-\infty}^{\infty} \frac{1-\cos x}{x^2}\, dx = \pi$; the integrand is even.

4 Cauchy's integral formulas

Integral representations recover a function on a large set from its values on a smaller set (cf. the Poisson integral formula for the steady-state heat equation in the disc).

Theorem 4.1

Suppose f is holomorphic in an open set containing the closure of a disc D. If C denotes the boundary circle of this disc with the positive orientation, then

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \text{for any point } z \in D.$$

Proof. Fix $z \in D$ and consider the keyhole $\Gamma_{\delta,\epsilon}$ omitting z: the circle C and a small circle of radius ϵ about z, joined by a corridor of width δ. $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from $\zeta = z$, so $\int_{\Gamma_{\delta,\epsilon}} F(\zeta)\, d\zeta = 0$ by Cauchy's theorem for this toy contour. Letting $\delta \to 0$, continuity of F makes the integrals over the two sides of the corridor cancel, leaving C (positively oriented) and the small circle C_ϵ about z (negatively oriented). Write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}. \tag{8}$$

The first term is bounded (as f is holomorphic), so its integral over C_ϵ tends to 0 as $\epsilon \to 0$. And

$$\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\, d\zeta = -f(z) \int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\, dt = -f(z) 2\pi i.$$

So in the limit $0 = \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta - 2\pi i f(z)$. ∎

Remark

The same holds for other toy contours: if f is holomorphic in an open set containing a positively oriented rectangle R and its interior, then $f(z) = \frac{1}{2\pi i}\int_R \frac{f(\zeta)}{\zeta - z}\,d\zeta$ for z in the interior of R (use a rectangular keyhole). The integral vanishes for z outside R, since then $f(\zeta)/(\zeta - z)$ is holomorphic inside R.

Corollary 4.2

If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta$$

for all z in the interior of C.

Proof. Induction on n; $n = 0$ is the Cauchy integral formula. Suppose f has up to $n-1$ complex derivatives with $f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^n}\,d\zeta$. For small h,

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{h}\Big[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n}\Big]d\zeta. \tag{9}$$

Since $A^n - B^n = (A-B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}]$, with $A = 1/(\zeta - z - h)$, $B = 1/(\zeta - z)$ the bracket in (9) is

$$\frac{h}{(\zeta - z - h)(\zeta - z)}[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

For small h, $z + h$ and z stay at a finite distance from C, so as $h \to 0$ the quotient converges to

$$\frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{(\zeta - z)^2}\frac{n}{(\zeta - z)^{n-1}}\,d\zeta = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta.$$

■

The formulas of Theorem 4.1 and Corollary 4.2 are the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities)

If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then

$$|f^{(n)}(z_0)| \le \frac{n!\,\|f\|_C}{R^n},$$

where $\|f\|_C = \sup_{z \in C}|f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. By the Cauchy integral formula,

$$|f^{(n)}(z_0)| = \Big|\frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta\Big| = \frac{n!}{2\pi}\Big|\int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\,d\theta\Big| \le \frac{n!}{2\pi}\frac{\|f\|_C}{R^n}2\pi.$$

■

Theorem 4.4

Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure

is contained in Ω, then f has a power series expansion at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by $a_n = \dfrac{f^{(n)}(z_0)}{n!}$ for all $n \geq 0$.

Proof. Fix $z \in D$. By the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta, \tag{10}$$

C the boundary of D. Write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}. \tag{11}$$

Since $\zeta \in C$ and $z \in D$ is fixed, there is $0 < r < 1$ with $\left|\frac{z - z_0}{\zeta - z_0}\right| < r$, so

$$\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)} = \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0}\right)^n, \tag{12}$$

uniformly for $\zeta \in C$. Interchanging sum and integral in (10)–(12),

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta\right)(z - z_0)^n.$$

The formula for a_n follows from the Cauchy integral formulas for derivatives (or by differentiating the series). ∎

This gives another proof that holomorphic functions are indefinitely differentiable. The expansion at z_0 converges in *any* disc centered at z_0 whose closure lies in Ω; in particular an entire function has a power series expansion about 0 converging on all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem)

If f is entire and bounded, then f is constant.

Proof. It suffices to show $f' = 0$ ($\mathbb{C}$ is connected; Corollary 3.4). For each $z_0 \in \mathbb{C}$ and $R > 0$ the Cauchy inequalities give $|f'(z_0)| \leq B/R$, B a bound for f. Let $R \to \infty$. ∎

Corollary 4.6

Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$.

Proof. If P has no roots, $1/P$ is a bounded entire function. Indeed, with $a_n \neq 0$, $\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}\right)$ for $z \neq 0$; the parenthesis tends to 0 as $|z| \to \infty$, so with $c = |a_n|/2$ there is R with $|P(z)| \geq c|z|^n$ for $|z| > R$. P is continuous with no roots in $|z| \leq R$, so it is bounded below there as well. By Liouville's theorem $1/P$ is constant, contradicting that P is non-constant. ∎

Corollary 4.7

Every polynomial $P(z) = a_n z^n + \cdots + a_0$ of degree $n \geq 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \ldots, w_n$, then P can be factored as $P(z) = a_n(z - w_1)(z - w_2)\cdots(z -$

w_n).

Proof. P has a root w_1. Writing $z = (z - w_1) + w_1$ and expanding, $P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0$ with $b_n = a_n$; $P(w_1) = 0$ forces $b_0 = 0$, so $P(z) = (z - w_1)Q(z)$ with Q of degree $n - 1$. By induction $P(z) = c(z - w_1)\cdots(z - w_n)$, and comparing coefficients of z^n, $c = a_n$. ■

Finally, *analytic continuation*: the "genetic code" of a holomorphic function is determined by its values on arbitrarily small sets. Here Ω must be connected.

Theorem 4.8

Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically 0.

Proof. Let $z_0 \in \Omega$ be a limit point of $\{w_k\}$ with $f(w_k) = 0$. First, $f \equiv 0$ near z_0: on a disc $D \subset \Omega$ centered at z_0, $f(z) = \sum a_n(z - z_0)^n$. If $f \not\equiv 0$ on D, let m be the smallest integer with $a_m \neq 0$; then $f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$ with $g(z - z_0) \to 0$ as $z \to z_0$. Taking $z = w_k \neq z_0$ close to z_0, $a_m(w_k - z_0)^m \neq 0$ and $1 + g(w_k - z_0) \neq 0$, contradicting $f(w_k) = 0$.

Now let U be the interior of $\{z \in \Omega : f(z) = 0\}$; U is open and non-empty. U is closed in Ω: if $z_n \in U$ and $z_n \to z \in \Omega$, then $f(z) = 0$ by continuity, and f vanishes near z by the argument just given, so $z \in U$. With $V = \Omega - U$, U and V are disjoint open sets with $\Omega = U \cup V$; Ω is connected and $z_0 \in U$, so $U = \Omega$. ■

Corollary 4.9

Suppose f and g are holomorphic in a region Ω and $f(z) = g(z)$ for all z in some non-empty open subset of Ω (or more generally for z in some sequence of distinct points with limit point in Ω). Then $f(z) = g(z)$ throughout Ω.

If f, F are analytic in regions $\Omega \subset \Omega'$ and agree on Ω, F is an **analytic continuation** of f into Ω'; by the corollary it is unique.

5 Further applications

5.1 Morera's theorem

Theorem 5.1 (Morera)

Suppose f is a continuous function in the open disc D such that for any triangle T contained in D, $\int_T f(z)\,dz = 0$. Then f is holomorphic.

Proof. By the proof of Theorem 2.1, f has a primitive F in D with $F' = f$. By regularity (Corollary 4.2), F is indefinitely (hence twice) complex differentiable, so f is holomorphic. ■

5.2 Sequences of holomorphic functions

Theorem 5.2

If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.

Proof. Let D be a disc with $\overline{D} \subset \Omega$ and T a triangle in D. By Goursat, $\int_T f_n = 0$ for all n. Since $f_n \to f$ uniformly on $\overline{D}$, f is continuous and $\int_T f_n \to \int_T f$, so $\int_T f = 0$, and by Morera's theorem f is holomorphic in D. This holds for every such D. ∎

This fails for real variables: by Weierstrass's theorem every continuous function on $[0,1]$ is a uniform limit of polynomials, yet not every continuous function is differentiable.

Theorem 5.3

Under the hypotheses of the previous theorem, the sequence of derivatives $\{f_n'\}_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.

Proof. We may assume the convergence is uniform on all of Ω. For $\delta > 0$ let $\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}$, the points at distance $> \delta$ from the boundary. It suffices to show $f_n' \to f'$ uniformly on each Ω_δ, which follows by applying to $F = f_n - f$ the inequality

$$\sup_{z\in\Omega_\delta} |F'(z)| \le \frac{1}{\delta} \sup_{\zeta\in\Omega} |F(\zeta)|, \tag{13}$$

valid for F holomorphic in Ω. For $z \in \Omega_\delta$, $\overline{D_\delta(z)} \subset \Omega$ and by the Cauchy integral formula

$$|F'(z)| = \Big|\frac{1}{2\pi i}\int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2}\, d\zeta\Big| \le \frac{1}{2\pi} \sup_{\zeta\in\Omega}|F(\zeta)| \frac{1}{\delta^2} 2\pi\delta = \frac{1}{\delta}\sup_{\zeta\in\Omega}|F(\zeta)|.$$

∎

Likewise $f_n^{(k)} \to f^{(k)}$ uniformly on compact subsets for every $k \ge 0$. In practice Theorem 5.2 constructs holomorphic functions as series $F(z) = \sum_{n=1}^{\infty} f_n(z)$ (14) converging uniformly on compact subsets of Ω, e.g. the zeta function (Chapter 6).

5.3 Holomorphic functions defined in terms of integrals

After a linear change of variables we take the interval $[0,1]$.

Theorem 5.4

Let $F(z,s)$ be defined for $(z,s) \in \Omega \times [0,1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties: (i) $F(z,s)$ is holomorphic in z for each s. (ii) F is continuous on $\Omega \times [0,1]$. Then the function f defined on Ω by $f(z) = \int_0^1 F(z,s)\, ds$ is holomorphic.

Proof. Instead of justifying an interchange of integrals in Morera's theorem, view the integral as a uniform limit of Riemann sums. Let $f_n(z) = \frac{1}{n}\sum_{k=1}^{n} F(z, k/n)$, holomorphic in Ω by (i). On a disc D with $\overline{D} \subset \Omega$, F is uniformly continuous on $\overline{D} \times [0,1]$, so given $\epsilon > 0$ there is $\delta > 0$ with $\sup_{z\in\overline{D}} |F(z,s_1) - F(z,s_2)| < \epsilon$ when $|s_1 - s_2| < \delta$. For $n > 1/\delta$ and $z \in D$,

$$|f_n(z) - f(z)| = \Big|\sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} [F(z,k/n) - F(z,s)]\, ds\Big| \le \sum_{k=1}^{n} \frac{\epsilon}{n} = \epsilon.$$

So $f_n \to f$ uniformly on D, and f is holomorphic in D by Theorem 5.2; hence in Ω. ∎

5.4 Schwarz reflection principle

Holomorphic functions are rigid: there are holomorphic functions in a disc, continuous on its closure, that cannot be continued beyond it; and they vanish identically if they vanish on a line segment. Yet there is a simple and useful extension phenomenon. Let Ω be open and symmetric with respect to the real line ($z \in \Omega \iff \overline{z} \in \Omega$), Ω^+ its part in the upper half-plane, Ω^- the part in the lower half-plane, and $I = \Omega \cap \mathbb{R}$; then $\Omega = \Omega^+ \cup I \cup \Omega^-$.

Theorem 5.5 (Symmetry principle)

If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and $f^+(x) = f^-(x)$ for all $x \in I$, then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & z \in \Omega^+, \\ f^+(z) = f^-(z) & z \in I, \\ f^-(z) & z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. f is continuous on Ω; the issue is holomorphy at points of I. Let $D \subset \Omega$ be a disc centered at a point of I; we apply Morera's theorem. If a triangle $T \subset D$ does not meet I, $\int_T f = 0$ since f is holomorphic in the half-discs. If one side or vertex of T lies on I and the rest is, say, in the upper half-disc, let T_ϵ be obtained by raising that edge or vertex by ϵ; $\int_{T_\epsilon} f = 0$ as T_ϵ lies in the upper half-disc, and letting $\epsilon \to 0$, by continuity $\int_T f = 0$. If the interior of T meets I, write T as a union of triangles each with an edge or vertex on I. By Morera's theorem f is holomorphic in D. ■

Theorem 5.6 (Schwarz reflection principle)

Suppose that f is a holomorphic function in Ω^+ that extends continuously to I and such that f is real-valued on I. Then there exists a function F holomorphic in all of Ω such that $F = f$ on Ω^+.

Proof. Define $F(z) = \overline{f(\overline{z})}$ for $z \in \Omega^-$. If $z, z_0 \in \Omega^-$ then $\overline{z}, \overline{z_0} \in \Omega^+$ and $f(\overline{z}) = \sum a_n(\overline{z} - \overline{z_0})^n$ near $\overline{z_0}$, so $F(z) = \sum \overline{a_n}(z - z_0)^n$ and F is holomorphic in Ω^-. Since f is real on I, $\overline{f(x)} = f(x)$ for $x \in I$, so F extends continuously up to I and agrees with f there. Apply the symmetry principle. ■

5.5 Runge's approximation theorem

By Weierstrass's theorem, continuous functions on a compact interval are uniform limits of polynomials. What conditions on a compact $K \subset \mathbb{C}$ ensure that every function holomorphic in a neighborhood of K is a uniform limit of polynomials on K? For a disc this follows from power series. Some condition is necessary: for $f(z) = 1/z$ on the unit circle $K = C$, $\int_C f = 2\pi i$ while $\int_C p = 0$ for every polynomial p, so uniform approximation is impossible. The right condition is that K^c be connected (a variant of this example shows it is also necessary).

Theorem 5.7 (Runge)

Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c. If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

The second part follows from the first by "pushing the singularities to infinity". The key is an integral representation from the Cauchy formula for squares:

Lemma 5.8

Suppose f is holomorphic in an open set Ω, and $K \subset \Omega$ is compact. Then there exist finitely

many segments $\gamma_1, \dots, \gamma_N$ in $\Omega - K$ such that

$$f(z) = \sum_{n=1}^{N} \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \text{for all } z \in K. \tag{15}$$

Proof. Let $d = c \cdot d(K, \Omega^c)$ with $c < 1/\sqrt{2}$, and consider a grid of closed squares with sides of length d parallel to the axes. Let $\mathcal{Q} = \{Q_1, \dots, Q_M\}$ be the squares of the grid meeting K, with positively oriented boundaries ∂Q_m, and let $\gamma_1, \dots, \gamma_N$ be the sides of squares in $\mathcal{Q}$ that do not belong to two adjacent squares of $\mathcal{Q}$. By the choice of d, each γ_n lies in Ω and does not meet K (if it did, it would belong to two adjacent squares of $\mathcal{Q}$). For $z \in K$ not on the boundary of a square of $\mathcal{Q}$, $z \in Q_j$ for some j, and by Cauchy's theorem and formula

$$\frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \begin{cases} f(z) & m = j, \\ 0 & m \neq j, \end{cases} \qquad \text{so} \qquad f(z) = \sum_{m=1}^{M} \frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

The integrals over a common side of two adjacent squares are taken once in each direction and cancel; this gives (15) for such z. Since the γ_n lie in K^c, continuity extends (15) to all $z \in K$. ∎

Lemma 5.9

For any line segment γ entirely contained in $\Omega - K$, there exists a sequence of rational functions with singularities on γ that approximate the integral $\int_\gamma f(\zeta)/(\zeta - z)\, d\zeta$ uniformly on K.

Proof. With a parametrization $\gamma(t)$, $t \in [0,1]$, $\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta = \int_0^1 \frac{f(\gamma(t))\gamma'(t)}{\gamma(t) - z}\, dt$. Since γ does not meet K, the integrand $F(z,t)$ is jointly continuous on the compact set $K \times [0,1]$, hence uniformly continuous, and as in the proof of Theorem 5.4 the Riemann sums of $\int_0^1 F(z,t)\, dt$ approximate it uniformly on K. Each Riemann sum is a rational function with singularities on γ. ∎

Lemmas 5.8 and 5.9 prove the first part of Theorem 5.7. A rational function whose only singularity is z_0 is a polynomial in $1/(z - z_0)$, so the second part follows from:

Lemma 5.10

If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.

Proof. Choose z_1 outside a large open disc D centered at 0 containing K. Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \frac{1}{1 - z/z_1} = -\sum_{n=0}^{\infty} \frac{z^n}{z_1^{n+1}},$$

uniformly for $z \in K$; the partial sums are polynomials approximating $1/(z - z_1)$ uniformly on K, and hence any power $1/(z - z_1)^k$ is a uniform limit of polynomials on K. It now suffices to approximate $1/(z - z_0)$ uniformly on K by polynomials in $1/(z - z_1)$. Since K^c is connected, there is a curve $\gamma \subset K^c$, $\gamma(0) = z_0$, $\gamma(1) = z_1$. Let $\rho = \frac{1}{2} d(K, \gamma) > 0$ (both are compact), and choose points $w_0 = z_0, w_1, \dots, w_\ell = z_1$ on γ with $|w_j - w_{j+1}| < \rho$.

Claim: if $w, w' \in \gamma$ with $|w - w'| < \rho$, then $1/(z - w)$ is a uniform limit on K of polynomials in $1/(z - w')$. Indeed

$$\frac{1}{z - w} = \frac{1}{z - w'} \cdot \frac{1}{1 - \frac{w - w'}{z - w'}} = \sum_{n=0}^{\infty} \frac{(w - w')^n}{(z - w')^{n+1}},$$

and the series converges uniformly for $z \in K$ since $|z - w'| \geq d(K, \gamma) = 2\rho$ gives $|w - w'|/|z - w'| < \rho/(2\rho) = 1/2$. Applying the claim successively, $1/(z - w_0)$ is a uniform limit of polynomials in $1/(z - w_1)$, hence (as powers of $1/(z - w_1)$ are limits of polynomials in $1/(z - w_2)$) of polynomials in $1/(z - w_2)$, and so on, up to polynomials in $1/(z - w_\ell) = 1/(z - z_1)$. ∎

Chapter 3. Meromorphic Functions and the Logarithm

Overview

A general principle, implicit in Riemann's work: analytic functions are essentially characterized by their singularities—globally, holomorphic functions are "effectively" determined by their zeros, and meromorphic functions by their zeros and poles. We study isolated singularities, in order of increasing severity: *removable singularities*, *poles*, and *essential singularities*. Poles lead to the *calculus of residues*, whose key ingredient is $\int_C dz/z = 2\pi i$. Indefinite integrals of functions with singularities, like $\int dz/z = \log z$, may fail to be single-valued; exploiting this gives the *argument principle*, which counts zeros inside a curve and yields the open mapping theorem and the maximum principle. To define the logarithm properly we introduce homotopies and simply connected domains.

1 Zeros and poles

A **point singularity** (or **isolated singularity**) of f is a point z_0 such that f is defined in a neighborhood of z_0 but not at z_0. For $f(z) = z$ on the punctured plane, 0 is removable (set $f(0) = 0$). $g(z) = 1/z$ cannot be extended continuously to 0; it grows to infinity, and 0 is a pole. $h(z) = e^{1/z}$ grows indefinitely as $z \to 0$ along the positive reals, tends to 0 along the negative reals, and oscillates rapidly but stays bounded along the imaginary axis: removable singularities and poles do not tell the whole story.

z_0 is a **zero** of f if $f(z_0) = 0$. By analytic continuation, the zeros of a non-trivial holomorphic function are isolated.

Theorem 1.1

Suppose that f is holomorphic in a connected open set Ω, has a zero at a point $z_0 \in \Omega$, and does not vanish identically in Ω. Then there exists a neighborhood $U \subset \Omega$ of z_0, a non-vanishing holomorphic function g on U, and a unique positive integer n such that $f(z) = (z - z_0)^n g(z)$ for all $z \in U$.

Proof. Since Ω is connected and $f \not\equiv 0$, f is not identically zero near z_0 (Theorem 4.8). Near z_0, $f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$; let n be the smallest integer with $a_n \neq 0$. Then $f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z)$, with g holomorphic and non-vanishing near z_0 (as $a_n \neq 0$). Uniqueness: if also $f(z) = (z - z_0)^m h(z)$ with $h(z_0) \neq 0$ and $m > n$, then $g(z) = (z - z_0)^{m-n} h(z)$ and letting $z \to z_0$ gives $g(z_0) = 0$, a contradiction; $m < n$ similarly forces $h(z_0) = 0$. So $m = n$ and $h = g$. ∎

f has a **zero of order** n (multiplicity n) at z_0; a zero of order 1 is **simple**. A **deleted neighborhood** of z_0 is $\{z : 0 < |z - z_0| < r\}$. A function f defined in a deleted neighborhood of z_0 has a **pole** at z_0 if $1/f$, defined to be zero at z_0, is holomorphic in a full neighborhood of z_0.

Theorem 1.2

If f has a pole at $z_0 \in \Omega$, then in a neighborhood of that point there exist a non-vanishing holomorphic function h and a unique positive integer n such that $f(z) = (z - z_0)^{-n} h(z)$.

Proof. By Theorem 1.1, $1/f(z) = (z - z_0)^n g(z)$ with g holomorphic and non-vanishing near z_0; take $h = 1/g$. ∎

n is the **order** of the pole (it describes the rate of growth); a pole of order 1 is **simple**.

Theorem 1.3

If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z-z_0} + G(z), \tag{1}$$

where G is a holomorphic function in a neighborhood of z_0.

Proof. Expand $h(z) = A_0 + A_1(z-z_0) + \cdots$ in Theorem 1.2; then $f(z) = (z-z_0)^{-n}(A_0 + A_1(z-z_0) + \cdots)$ has the stated form. ∎

The sum $\frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-1}}{z-z_0}$ is the **principal part** of f at z_0, and a_{-1} the **residue**, $\operatorname{res}_{z_0} f = a_{-1}$. All other terms of the principal part have primitives in a deleted neighborhood of z_0, so for the principal part P and any circle C centered at z_0, $\frac{1}{2\pi i}\int_C P(z)\,dz = a_{-1}$. For a simple pole, $\operatorname{res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f(z)$.

Theorem 1.4

If f has a pole of order n at z_0, then

$$\operatorname{res}_{z_0} f = \lim_{z\to z_0} \frac{1}{(n-1)!}\Big(\frac{d}{dz}\Big)^{n-1}(z-z_0)^n f(z).$$

Proof. By (1), $(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{n-1} + G(z)(z-z_0)^n$; differentiate $n-1$ times and let $z \to z_0$. ∎

2 The residue formula

Theorem 2.1

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then

$$\int_C f(z)\,dz = 2\pi i \operatorname{res}_{z_0} f.$$

Proof. Using a keyhole contour avoiding the pole and letting the corridor width tend to 0, $\int_C f = \int_{C_\epsilon} f$, C_ϵ the small circle of radius ϵ about z_0. By the Cauchy integral formula applied to the constant a_{-1}, $\frac{1}{2\pi i}\int_{C_\epsilon} \frac{a_{-1}}{z-z_0}\,dz = a_{-1}$; by the formulas for derivatives (Corollary 4.2), $\frac{1}{2\pi i}\int_{C_\epsilon} \frac{a_{-k}}{(z-z_0)^k}\,dz = 0$ for $k > 1$; and $\int_{C_\epsilon} G = 0$ by Cauchy's theorem. By (1), $\int_{C_\epsilon} f = 2\pi i a_{-1}$. ∎

Corollary 2.2

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1, \ldots, z_N$ inside C. Then

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Proof. Use a multiple keyhole with a loop around each pole; letting the corridor widths go to zero, the integral over C becomes a sum of integrals over small circles, to which Theorem 2.1 applies. ∎

Corollary 2.3

Suppose that f is holomorphic in an open set containing a toy contour γ and its interior, except for poles at the points $z_1, \dots, z_N$ inside γ. Then

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Here γ has positive orientation.

(Choose a keyhole appropriate to the toy contour and reduce to small circles around the poles.) This identity is the **residue formula**.

2.1 Examples

To compute $\int_{-\infty}^{\infty} f(x)\,dx$, extend f to $\mathbb{C}$ and choose toy contours γ_R with $\lim_{R\to\infty} \int_{\gamma_R} f(z)\,dz = \int_{-\infty}^{\infty} f(x)\,dx$; the residues give $\int_{\gamma_R} f$. Choosing γ_R is the challenge, often guided by the decay of f.

Example 1

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi. \tag{2}$$

(Changing variables $x \mapsto x/y$, this says $\int P_y(x)\,dx = 1$ for the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$.) $f(z) = 1/(1+z^2) = 1/((z-i)(z+i))$ has simple poles at $\pm i$, and $\operatorname{res}_i f = 1/2i$. Let γ_R be the segment $[-R, R]$ together with the upper half-circle C_R^+ of radius R. For large R, $\int_{\gamma_R} f = 2\pi i \cdot \frac{1}{2i} = \pi$. Since $|f(z)| \le B/|z|^2$ on C_R^+, $\left|\int_{C_R^+} f\right| \le \frac{B}{R^2}\pi R \to 0$. Hence $\int_{-\infty}^{\infty} f(x)\,dx = \pi$. (The lower semicircle, with the other pole, works equally well.)

Example 2

An integral that plays an important role in Chapter 6:

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x}\,dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1.$$

Let $f(z) = e^{az}/(1+e^z)$ and γ_R the rectangle with sides on $\mathbb{R}$ and on $\operatorname{Im}(z) = 2\pi$, vertices $\pm R$, $\pm R + 2\pi i$. The only zero of the denominator inside is $z = \pi i$. There

$$(z-\pi i) f(z) = e^{az}\frac{z-\pi i}{e^z - e^{\pi i}}, \qquad \lim_{z\to \pi i} \frac{e^z - e^{\pi i}}{z - \pi i} = e^{\pi i} = -1,$$

so f has a simple pole at πi with residue $-e^{a\pi i}$, and

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

Let $I_R = \int_{-R}^{R} f(x)\,dx \to I$. The top side (right to left) contributes $-e^{2\pi i a} I_R$. On the right vertical side $A_R = \{R + it : 0 \le t \le 2\pi\}$, $\left|\int_{A_R} f\right| \le \int_0^{2\pi} \left|\frac{e^{a(R+it)}}{1+e^{R+it}}\right| dt \le Ce^{(a-1)R} \to 0$ since $a < 1$; the left side is bounded by $Ce^{-aR} \to 0$ since $a > 0$. So (3) gives $I - e^{2\pi i a} I = -2\pi i e^{a\pi i}$, whence

$$I = -2\pi i \frac{e^{a\pi i}}{1 - e^{2\pi i a}} = \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} = \frac{\pi}{\sin \pi a}.$$

Example 3

$1/\cosh \pi x$ is its own Fourier transform: with $\cosh z = \frac{e^z+e^{-z}}{2}$,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{\cosh \pi x}\, dx = \frac{1}{\cosh \pi \xi}.$$

Let $f(z) = e^{-2\pi i z\xi}/\cosh \pi z$ and γ_R the rectangle with vertices $\pm R$, $\pm R+2i$. The denominator vanishes when $e^{2\pi z} = -1$; inside the rectangle the poles are $\alpha = i/2$ and $\beta = 3i/2$. Since

$$(z-\alpha)f(z) = 2e^{-2\pi i z\xi} e^{\pi z} \frac{z-\alpha}{e^{2\pi z} - e^{2\pi \alpha}}, \qquad \lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i \alpha\xi} e^{\pi\alpha} \frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

f has a simple pole at α with residue $e^{\pi\xi}/(\pi i)$; similarly at β with residue $-e^{3\pi\xi}/(\pi i)$. On $z = R + iy$, $0 \le y \le 2$: $|e^{-2\pi i z\xi}| \le e^{4\pi|\xi|}$ and $|\cosh \pi z| \ge \frac{1}{2}(e^{\pi R} - e^{-\pi R}) \to \infty$, so the vertical sides contribute 0 in the limit. Since $\cosh \pi z$ is periodic with period $2i$, the top side contributes $-e^{4\pi\xi} I$. The residue formula gives

$$I - e^{4\pi\xi} I = 2\pi i\Big(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i}\Big) = -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}),$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$,

$$I = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}.$$

A similar argument gives, for $0 < a < 1$, with $\sinh z = (e^z - e^{-z})/2$,

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi} \frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\, dx = \frac{2\sinh 2\pi a\xi}{\sinh 2\pi\xi};$$

$a = 1/2$ is the case above. This yields the Poisson kernel for the strip, and is used in the two-squares theorem (Chapter 10).

3 Singularities and meromorphic functions

Let f be holomorphic in Ω except possibly at $z_0 \in \Omega$. If f can be defined at z_0 so that it becomes holomorphic in all of Ω, z_0 is a **removable singularity**.

Theorem 3.1 (Riemann's theorem on removable singularities)

Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω. If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.

Proof. Take a small disc D centered at z_0 with $\overline{D} \subset \Omega$, boundary circle C positively oriented. We show that for $z \in D$, $z \ne z_0$,

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta. \tag{4}$$

By Theorem 5.4 the right side is holomorphic in all of D and agrees with f for $z \ne z_0$, giving the extension. To prove (4), fix z and use a multiple keyhole avoiding z and z_0; letting the corridors shrink,

$$\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = 0,$$

where γ_ϵ, γ'_ϵ are small negatively oriented circles of radius ϵ about z and z_0. As in the proof of the Cauchy integral formula, $\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = -2\pi i f(z)$. Since f is bounded and $\zeta \in \gamma'_\epsilon$ stays away from z, $\left| \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta \right| \le C\epsilon$. Let $\epsilon \to 0$. ■

Corollary 3.2

Suppose that f has an isolated singularity at the point z_0. Then z_0 is a pole of f if and only if $|f(z)| \to \infty$ as $z \to z_0$.

Proof. If z_0 is a pole, $1/f$ has a zero at z_0, so $|f(z)| \to \infty$. Conversely, if $|f(z)| \to \infty$, then $1/f$ is bounded near z_0 and $1/|f(z)| \to 0$; so $1/f$ has a removable singularity at z_0 and vanishes there, i.e. z_0 is a pole. ■

Isolated singularities fall into three categories: **removable** (f bounded near z_0), **poles** ($|f(z)| \to \infty$ as $z \to z_0$), and **essential singularities** (all others), e.g. $e^{1/z}$ at 0. Near an essential singularity a function typically behaves erratically:

Theorem 3.3 (Casorati–Weierstrass)

Suppose f is holomorphic in the punctured disc $D_r(z_0) - \{z_0\}$ and has an essential singularity at z_0. Then, the image of $D_r(z_0) - \{z_0\}$ under f is dense in the complex plane.

Proof. If not, there are $w \in \mathbb{C}$ and $\delta > 0$ with $|f(z) - w| > \delta$ on the punctured disc. Then $g(z) = 1/(f(z) - w)$ is holomorphic on the punctured disc and bounded by $1/\delta$, so it has a removable singularity at z_0 (Theorem 3.1). If $g(z_0) \neq 0$, $f(z) - w = 1/g(z)$ is holomorphic at z_0; if $g(z_0) = 0$, $f(z) - w$ has a pole at z_0. Both contradict that z_0 is essential. ■

Picard proved much more: under these hypotheses f takes every complex value infinitely many times with at most one exception.

Definition (meromorphic function)

A function f on an open set Ω is **meromorphic** if there is a sequence $\{z_0, z_1, z_2, \dots\}$ with no limit points in Ω such that (i) f is holomorphic in $\Omega - \{z_0, z_1, \dots\}$, and (ii) f has poles at the points $z_0, z_1, \dots$.

If f is holomorphic for all large z, let $F(z) = f(1/z)$, holomorphic in a deleted neighborhood of 0. f has a **pole at infinity** if F has a pole at 0; similarly an **essential singularity at infinity**, or a **removable singularity** (holomorphic) at infinity. A meromorphic function in $\mathbb{C}$ that is holomorphic or has a pole at infinity is **meromorphic in the extended complex plane**. The principle stated at the outset appears here in its simplest form:

Theorem 3.4

The meromorphic functions in the extended complex plane are the rational functions.

Proof. $f(1/z)$ has a pole or removable singularity at 0, hence is holomorphic in a deleted neighborhood of 0; so f has only finitely many poles in the plane, say $z_1, \dots, z_n$. Near each z_k, $f(z) = f_k(z) + g_k(z)$ with f_k the principal part (a polynomial in $1/(z - z_k)$) and g_k holomorphic near z_k. Similarly $f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$ with $\tilde{g}_\infty$ holomorphic near 0 and $\tilde{f}_\infty$ a polynomial in $1/z$; let $f_\infty(z) = \tilde{f}_\infty(1/z)$, a polynomial. Then $H = f - f_\infty - \sum_{k=1}^{n} f_k$ is entire (removable singularities at the z_k) and bounded ($H(1/z)$ is bounded near 0, as the principal part at ∞ was subtracted). By Liouville, H is constant, so f is rational. ■

Consequently, a rational function is determined up to a multiplicative constant by the locations and multiplicities of its zeros and poles.

The Riemann sphere. In $\mathbb{R}^3$ with coordinates (X, Y, Z), identify the XY-plane with $\mathbb{C}$, and let $\mathbb{S}$ be the sphere of radius $1/2$ centered at $(0, 0, 1/2)$, with north pole $N = (0, 0, 1)$. For $W = (X, Y, Z) \in \mathbb{S} - \{N\}$, the line through N and W meets the XY-plane in a single point $w = x + iy$, the **stereographic projection** of W; this is a bijection between $\mathbb{S} - \{N\}$ and $\mathbb{C}$, given by

$$x = \frac{X}{1-Z}, \quad y = \frac{Y}{1-Z}; \qquad X = \frac{x}{x^2+y^2+1}, \quad Y = \frac{y}{x^2+y^2+1}, \quad Z = \frac{x^2+y^2}{x^2+y^2+1}.$$

As $|w| \to \infty$, $W \to N$, so N is the natural "point at infinity". The extended complex plane is thus the sphere $\mathbb{S}$, the **Riemann sphere** (the *one-point compactification* of $\mathbb{C}$). The point at infinity is on equal footing with the others, and a meromorphic function on the extended plane is a map of $\mathbb{S}$ to itself, sending poles to N.

4 The argument principle and applications

$\log f(z)$ is in general multiple-valued, but it must equal $\log|f(z)| + i \arg f(z)$, and its derivative $f'(z)/f(z)$ is single-valued; $\int_\gamma \frac{f'(z)}{f(z)}\,dz$ is the change in the argument of f along γ, determined (for closed γ) by the zeros and poles of f inside γ. While $\log(f_1 f_2) = \log f_1 + \log f_2$ fails in general, additivity holds for logarithmic derivatives:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}, \qquad \frac{(\prod_{k=1}^N f_k)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

If f has a zero of order n at z_0, $f(z) = (z - z_0)^n g(z)$ with g non-vanishing near z_0, so $\frac{f'(z)}{f(z)} = \frac{n}{z-z_0} + G(z)$, $G = g'/g$ holomorphic: f'/f has a simple pole with residue n. If f has a pole of order n, $f(z) = (z - z_0)^{-n} h(z)$ and $\frac{f'(z)}{f(z)} = \frac{-n}{z-z_0} + H(z)$. The residue formula gives:

Theorem 4.1 (Argument principle)

Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C, then

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,dz = (\text{number of zeros of } f \text{ inside } C) - (\text{number of poles of } f \text{ inside } C),$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2

The above theorem holds for toy contours.

Three applications: Rouché's theorem (a continuity statement), the open mapping theorem, and the maximum modulus principle.

Theorem 4.3 (Rouché's theorem)

Suppose that f and g are holomorphic in an open set containing a circle C and its interior. If $|f(z)| > |g(z)|$ for all $z \in C$, then f and $f + g$ have the same number of zeros inside the circle C.

Proof. For $t \in [0, 1]$ let $f_t(z) = f(z) + tg(z)$, so $f_0 = f$, $f_1 = f + g$, and let n_t be the number of zeros of f_t inside C (with multiplicities). $|f| > |g|$ on C implies f_t has no zeros on C, and by the argument principle $n_t = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\,dz$. $f_t'(z)/f_t(z)$ is jointly continuous in $t \in [0, 1]$, $z \in C$

(numerator and denominator are, and the denominator does not vanish on C), so n_t is continuous in t. An integer-valued continuous function on $[0,1]$ is constant (intermediate value theorem), so $n_0 = n_1$. ■

A mapping is **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem)

If f is holomorphic and non-constant in a region Ω, then f is open.

Proof. Let $w_0 = f(z_0)$; we show all w near w_0 are in the image. Write $g(z) = f(z) - w = (f(z) - w_0) + (w_0 - w) = F(z) + G(z)$. Choose $\delta > 0$ with $\{|z - z_0| \leq \delta\} \subset \Omega$ and $f(z) \neq w_0$ on $|z - z_0| = \delta$ (zeros of $f - w_0$ are isolated), then $\epsilon > 0$ with $|f(z) - w_0| \geq \epsilon$ on $|z - z_0| = \delta$. If $|w - w_0| < \epsilon$, then $|F| > |G|$ on that circle, and by Rouché's theorem $g = F + G$ has a zero inside, since F has one (z_0). ■

The maximum of f in Ω means the maximum of $|f|$ in Ω.

Theorem 4.5 (Maximum modulus principle)

If f is a non-constant holomorphic function in a region Ω, then f cannot attain a maximum in Ω.

Proof. If f attained a maximum at z_0, then since f is open, the image $f(D)$ of a small disc D centered at z_0 is open and contains $f(z_0)$; so there are $z \in D$ with $|f(z)| > |f(z_0)|$. ■

Corollary 4.6

Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \overline{\Omega} - \Omega} |f(z)|.$$

Proof. $|f|$ attains its maximum on the compact set $\overline{\Omega}$, and not in Ω if f is non-constant; if f is constant the conclusion is trivial. ■

Remark

Compactness (boundedness) of $\overline{\Omega}$ is essential. Let Ω be the open first quadrant and $F(z) = e^{-iz^2}$. F is entire, $|F| = 1$ on both boundary half-lines $z = x$, $z = iy$, yet F is unbounded in Ω: $F(z) = e^{r^2}$ for $z = r\sqrt{i} = re^{i\pi/4}$.

5 Homotopies and simply connected domains

The key to the general Cauchy theorem and to multiple-valued functions is knowing in what regions a holomorphic function has a primitive—a global question (relevant to the logarithm, the primitive of $1/z$).

Definition (homotopic curves)

Let $\gamma_0, \gamma_1 \subset \Omega$ be curves with parametrizations on $[a,b]$ and common end-points $\gamma_0(a) = \gamma_1(a) = \alpha$, $\gamma_0(b) = \gamma_1(b) = \beta$. They are **homotopic in** Ω if for each $0 \leq s \leq 1$ there is a curve $\gamma_s \subset \Omega$, parametrized by $\gamma_s(t)$ on $[a,b]$, with $\gamma_s(a) = \alpha$, $\gamma_s(b) = \beta$ for every s, $\gamma_s(t)|_{s=0} = \gamma_0(t)$, $\gamma_s(t)|_{s=1} = \gamma_1(t)$, and $\gamma_s(t)$ jointly continuous in $s \in [0,1]$, $t \in [a,b]$.

Loosely, one curve can be deformed into the other continuously without leaving Ω.

Theorem 5.1

If f is holomorphic in Ω, then $\int_{\gamma_0} f(z)\,dz = \int_{\gamma_1} f(z)\,dz$ whenever the two curves γ_0 and γ_1 are homotopic in Ω.

Proof. The key is: if two curves are close and have the same end-points, the integrals agree. $F(s,t) = \gamma_s(t)$ is continuous on $[0,1]\times[a,b]$; its image K is compact. There is $\epsilon > 0$ such that every disc of radius 3ϵ centered at a point of K lies in Ω: otherwise for every $\ell \geq 1$ there are $z_\ell \in K$, $w_\ell \in \Omega^c$ with $|z_\ell - w_\ell| < 1/\ell$; a subsequence $z_{\ell_k} \to z \in K$, then $w_{\ell_k} \to z$ and $z \in \Omega^c$ (Ω^c is closed), a contradiction. By uniform continuity of F, choose δ with $\sup_t |\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon$ whenever $|s_1 - s_2| < \delta$.

Fix such s_1, s_2. Choose discs $D_0, \ldots, D_n$ of radius 2ϵ and consecutive points $z_0, \ldots, z_{n+1}$ on γ_{s_1}, $w_0, \ldots, w_{n+1}$ on γ_{s_2}, such that the discs cover both curves and $z_i, z_{i+1}, w_i, w_{i+1} \in D_i$, with $z_0 = w_0$ the common beginning point and $z_{n+1} = w_{n+1}$ the common end-point. On each D_i let F_i be a primitive of f (Theorem 2.1). On $D_i \cap D_{i+1}$, F_i and F_{i+1} differ by a constant, so $F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1})$, i.e.

$$F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}). \tag{5}$$

Hence

$$\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f = \sum_{i=0}^{n} [F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n} [F_i(w_{i+1}) - F_i(w_i)] = \sum_{i=0}^{n} \left[F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i))\right],$$

which telescopes by (5) to $F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) = 0$. Subdividing $[0,1]$ into intervals of length $< \delta$, finitely many applications take us from γ_0 to γ_1. ∎

Definition (simply connected)

A region Ω is **simply connected** if any two curves in Ω with the same end-points are homotopic.

Example 1

A disc D is simply connected: $\gamma_s(t) = (1-s)\gamma_0(t) + s\gamma_1(t)$ lies on the segment joining $\gamma_0(t)$ and $\gamma_1(t)$, hence in D. The same works for rectangles and any open convex set.

Example 2

The slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ is simply connected: write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ with $r_j > 0$ and θ_j continuous, $|\theta_j(t)| < \pi$, and let $\gamma_s(t) = r_s(t)e^{i\theta_s(t)}$ with $r_s = (1-s)r_0 + sr_1$, $\theta_s = (1-s)\theta_0 + s\theta_1$; then $\gamma_s(t) \in \Omega$.

Example 3

With some effort one shows that the interior of a toy contour is simply connected, by dividing it into several subregions.

Example 4

The punctured plane $\mathbb{C} - \{0\}$ is not simply connected (proved after Corollary 5.3).

Theorem 5.2

Any holomorphic function in a simply connected domain has a primitive.

Proof. Fix $z_0 \in \Omega$ and define $F(z) = \int_\gamma f(w)\,dw$, the integral over any curve $\gamma \subset \Omega$ joining z_0 to z; this is independent of γ by Theorem 5.1, since Ω is simply connected. Then $F(z+h) - F(z) = \int_\eta f(w)\,dw$ with η the segment from z to $z+h$, and as in the proof of Theorem 2.1, $\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = f(z)$. ■

Corollary 5.3

If f is holomorphic in the simply connected region Ω, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in Ω.

That $\mathbb{C} - \{0\}$ is not simply connected now follows rigorously since $\int_C dz/z = 2\pi i \neq 0$ over the unit circle.

6 The complex logarithm

For $z = re^{i\theta}$ it is natural to set $\log z = \log r + i\theta$ ($\log r$ the standard logarithm of $r > 0$), but θ is unique only up to a multiple of 2π. Locally a continuous choice of θ is determined, but not globally: winding once around the origin changes the logarithm by $2\pi i$. To obtain a single-valued function we must restrict the domain—the choice of a **branch** or **sheet** of the logarithm.

Theorem 6.1

Suppose that Ω is simply connected with $1 \in \Omega$, and $0 \notin \Omega$. Then in Ω there is a branch of the logarithm $F(z) = \log_\Omega(z)$ so that (i) F is holomorphic in Ω, (ii) $e^{F(z)} = z$ for all $z \in \Omega$, (iii) $F(r) = \log r$ whenever r is a real number and near 1.

So each branch $\log_\Omega(z)$ extends the standard logarithm.

Proof. $1/z$ is holomorphic in Ω; define $\log_\Omega(z) = F(z) = \int_\gamma \frac{dw}{w}$, γ any curve in Ω from 1 to z (path-independent as Ω is simply connected). As in Theorem 5.2, F is holomorphic with $F'(z) = 1/z$: (i). For (ii), $\frac{d}{dz}(ze^{-F(z)}) = e^{-F(z)} - zF'(z)e^{-F(z)} = (1 - zF'(z))e^{-F(z)} = 0$, so $ze^{-F(z)}$ is constant on the connected Ω (Corollary 3.4); at $z = 1$, $F(1) = 0$, so the constant is 1. For (iii), for real r near 1 integrate along the real segment: $F(r) = \int_1^r \frac{dx}{x} = \log r$. ■

In the slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ we have the **principal branch** $\log z = \log r + i\theta$ for $z = re^{i\theta}$, $|\theta| < \pi$: integrate along the segment from 1 to r and the arc from r to z,

$$\log z = \int_1^r \frac{dx}{x} + \int_{\text{arc}} \frac{dw}{w} = \log r + \int_0^\theta \frac{ire^{it}}{re^{it}}\,dt = \log r + i\theta.$$

In general $\log(z_1 z_2) \neq \log z_1 + \log z_2$: for $z_1 = z_2 = e^{2\pi i/3}$, $\log z_1 = \log z_2 = 2\pi i/3$, while $z_1 z_2 = e^{-2\pi i/3}$ and $\log(z_1 z_2) = -2\pi i/3$. For the principal branch,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = -\sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} \qquad \text{for } |z| < 1, \tag{6}$$

since both sides have derivative $1/(1+z)$ and vanish at $z = 0$.

For Ω simply connected with $1 \in \Omega$, $0 \notin \Omega$, and $\alpha \in \mathbb{C}$, define $z^\alpha = e^{\alpha \log z}$ (branch with $\log 1 = 0$). Then $1^\alpha = 1$, and for $\alpha = 1/n$, $(z^{1/n})^n = \prod_{k=1}^n e^{\frac{1}{n}\log z} = e^{\log z} = z$.

Theorem 6.2

If f is a nowhere vanishing holomorphic function in a simply connected region Ω, then there exists a holomorphic function g on Ω such that $f(z) = e^{g(z)}$.

g may be denoted $\log f(z)$; it determines a "branch" of that logarithm.

Proof. Fix $z_0 \in \Omega$ and let $g(z) = \int_\gamma \frac{f'(w)}{f(w)}\,dw + c_0$, γ any path in Ω from z_0 to z, with $e^{c_0} = f(z_0)$ (path-independent as Ω is simply connected). As in Theorem 2.1, g is holomorphic with $g' = f'/f$, and $\frac{d}{dz}(f(z)e^{-g(z)}) = 0$, so $f(z)e^{-g(z)}$ is constant; at z_0 it is $f(z_0)e^{-c_0} = 1$. ∎

7 Fourier series and harmonic functions

There is a simple and direct relation between Fourier series on the circle and power series of holomorphic functions in the disc. Suppose f is holomorphic in a disc $D_R(z_0)$, so $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ there.

Theorem 7.1

The coefficients of the power series expansion of f are given by

$$a_n = \frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula over the circle γ of radius r about z_0 gives, with $\zeta = z_0 + re^{i\theta}$,

$$a_n = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta = \frac{1}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(re^{i\theta})^{n+1}} rie^{i\theta}\,d\theta = \frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,d\theta.$$

For $n < 0$ the same computation gives $\frac{1}{2\pi r^n}\int_0^{2\pi} f(z_0 + re^{i\theta})e^{-in\theta}\,d\theta = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta$, and since $-n > 0$, $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc; by Cauchy's theorem the integral vanishes. ∎

So the Fourier coefficients of the restriction of f to a circle vanish for $n < 0$, and for $n \geq 0$ equal (up to r^n) the power series coefficients—another special characteristic of holomorphic functions. Since $a_0 = f(z_0)$:

Corollary 7.2 (Mean-value property)

If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{i\theta})\,d\theta \qquad \text{for any } 0 < r < R.$$

Corollary 7.3

If f is holomorphic in a disc $D_R(z_0)$, and $u = \mathrm{Re}(f)$, then

$$u(z_0) = \frac{1}{2\pi}\int_0^{2\pi} u(z_0 + re^{i\theta})\,d\theta \qquad \text{for any } 0 < r < R.$$

This follows by taking real parts. Every harmonic function u in a disc is the real part of a holomorphic function there, so the mean-value property holds for all harmonic functions in a disc.

Chapter 4. The Fourier Transform

Overview

For f on $\mathbb{R}$ with suitable regularity and decay, the Fourier transform and its inversion are

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx, \qquad f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

The theme of this chapter: the possibility of extending f to a holomorphic function is closely related to very rapid (e.g. exponential) decay of $\hat{f}$ at infinity. First, if f continues analytically to a horizontal strip and has moderate decrease, then $\hat{f}$ decays exponentially, and the inversion and Poisson summation formulas follow by contour integration. Second, assuming only Fourier inversion, the *Paley–Wiener theorem* says $\hat{f}$ is supported in $[-M, M]$ iff f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ (*exponential type*). The decisive technique is shifting the contour (the real line) within a horizontal strip, exploiting that $e^{-2\pi i z\xi}$ decays or grows exponentially when $\mathrm{Im}(z) \neq 0$, according to the sign of $\xi\,\mathrm{Im}(z)$.

A function f is of **moderate decrease** if it is continuous and $|f(x)| \le A/(1+x^2)$ for some $A > 0$.

1 The class $\mathfrak{F}$

Book I proved Fourier inversion and Poisson summation when $|f(x)| \le \frac{A}{1+x^2}$ and $|\hat{f}(\xi)| \le \frac{A'}{1+\xi^2}$, a class containing e.g. the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$, $\hat{P}_y(\xi) = e^{-2\pi y|\xi|}$. Here we introduce a class suited to complex analysis.

Definition (the classes $\mathfrak{F}_a$ and $\mathfrak{F}$)

For $a > 0$, $\mathfrak{F}_a$ is the class of functions f such that: (i) f is holomorphic in the horizontal strip $S_a = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < a\}$; (ii) there is $A > 0$ with $|f(x+iy)| \le \dfrac{A}{1+x^2}$ for all $x \in \mathbb{R}$ and $|y| < a$. $\mathfrak{F}$ is the class of functions belonging to $\mathfrak{F}_a$ for some a.

So $\mathfrak{F}_a$ consists of holomorphic functions on S_a of moderate decay on each horizontal line, uniformly in $-a < y < a$. Examples: $e^{-\pi z^2} \in \mathfrak{F}_a$ for all a; $\frac{1}{\pi}\frac{c}{c^2+z^2}$ (simple poles at $\pm ci$) belongs to $\mathfrak{F}_a$ for $0 < a < c$; $1/\cosh \pi z \in \mathfrak{F}_a$ for $|a| < 1/2$. By the Cauchy integral formulas, if $f \in \mathfrak{F}_a$ then $f^{(n)} \in \mathfrak{F}_b$ for $0 < b < a$. (The decrease $A/(1+x^2)$ may be weakened to $A/(1+|x|^{1+\epsilon})$ in most results below.)

2 Action of the Fourier transform on $\mathfrak{F}$

All three theorems are proved by contour integration.

Theorem 2.1

If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \le Be^{-2\pi b|\xi|}$ for any $0 \le b < a$.

Proof. $b = 0$ says $\hat{f}$ is bounded, clear from moderate decrease. Let $0 < b < a$ and $\xi > 0$. Shift the real line down by b: let $g(z) = f(z)e^{-2\pi i z\xi}$ and consider the rectangle with vertices $-R, R, R - ib, -R - ib$. The integral over the left vertical side is bounded by

$$\Big|\int_{-R-ib}^{-R} g(z)\, dz\Big| \le \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\, dt \le \int_0^b \frac{A}{R^2} e^{-2\pi t\xi}\, dt = O(1/R^2),$$

and similarly for the right side. By Cauchy's theorem, letting $R \to \infty$,

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - ib) e^{-2\pi i(x-ib)\xi}\, dx, \tag{1}$$

so $|\hat{f}(\xi)| \le \int_{-\infty}^{\infty} \frac{A}{1+x^2} e^{-2\pi b\xi}\, dx \le B e^{-2\pi b\xi}$. For $\xi < 0$ shift the real line up by b. ■

The further f extends (larger a), the better the decay of $\hat{f}$.

Theorem 2.2

If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi \qquad \text{for all } x \in \mathbb{R}.$$

Lemma 2.3

If A is positive and B is real, then $\int_0^{\infty} e^{-(A+iB)\xi}\, d\xi = \dfrac{1}{A+iB}$.

Proof. $|e^{-(A+iB)\xi}| = e^{-A\xi}$, so the integral converges, and $\int_0^R e^{-(A+iB)\xi}\, d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB}\right]_0^R \to \frac{1}{A+iB}$. ■

Proof of Theorem 2.2. Split $\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi} d\xi = \int_{-\infty}^{0} + \int_0^{\infty}$. Say $f \in \mathfrak{F}_a$, $0 < b < a$. By (1) and the lemma (the integrals converge absolutely),

$$\begin{aligned}\int_0^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi &= \int_0^{\infty}\int_{-\infty}^{\infty} f(u-ib) e^{-2\pi i(u-ib)\xi} e^{2\pi i x\xi}\, du\, d\xi = \int_{-\infty}^{\infty} f(u-ib) \int_0^{\infty} e^{-2\pi i(u-ib-x)\xi}\, d\xi\, du \\ &= \int_{-\infty}^{\infty} f(u-ib) \frac{1}{2\pi b + 2\pi i(u-x)}\, du = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{f(u-ib)}{u-ib-x}\, du = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\, d\zeta,\end{aligned}$$

$L_1 = \{u - ib : u \in \mathbb{R}\}$ traversed left to right. Similarly $\int_{-\infty}^{0} \hat{f}(\xi) e^{2\pi i x\xi} d\xi = -\frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\, d\zeta$, L_2 the real line shifted up by b, left to right. Now for $x \in \mathbb{R}$ let γ_R be the rectangle with vertices $-R \pm ib$, $R \pm ib$. $f(\zeta)/(\zeta - x)$ has a simple pole at x with residue $f(x)$, so $f(x) = \frac{1}{2\pi i}\int_{\gamma_R} \frac{f(\zeta)}{\zeta - x}\, d\zeta$. As $R \to \infty$ the vertical sides contribute 0, and

$$f(x) = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\, d\zeta - \frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\, d\zeta = \int_0^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi + \int_{-\infty}^{0} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

■

Theorem 2.4 (Poisson summation formula)

If $f \in \mathfrak{F}$, then $\displaystyle\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)$.

Proof. Say $f \in \mathfrak{F}_a$, $0 < b < a$. $1/(e^{2\pi iz} - 1)$ has simple poles with residue $1/(2\pi i)$ at the integers, so $f(z)/(e^{2\pi iz} - 1)$ has simple poles at the integers n with residues $f(n)/2\pi i$. Apply the residue formula to the rectangle γ_N with vertices $\pm(N + \frac{1}{2}) \pm ib$:

$$\sum_{|n|\le N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi iz} - 1}\, dz.$$

Let $N \to \infty$; by moderate decrease the vertical sides contribute 0, and

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi iz} - 1}\, dz - \int_{L_2} \frac{f(z)}{e^{2\pi iz} - 1}\, dz, \tag{2}$$

L_1, L_2 the real line shifted down and up by b. If $|w| > 1$, $\frac{1}{w-1} = w^{-1}\sum_{n=0}^{\infty} w^{-n}$, so on L_1 (where $|e^{2\pi iz}| > 1$) $\frac{1}{e^{2\pi iz}-1} = e^{-2\pi iz}\sum_{n=0}^{\infty} e^{-2\pi inz}$; if $|w| < 1$, $\frac{1}{w-1} = -\sum_{n=0}^{\infty} w^n$, so on L_2 $\frac{1}{e^{2\pi iz}-1} = -\sum_{n=0}^{\infty} e^{2\pi inz}$. Substituting in (2) and shifting L_1, L_2 back to the real line (by (1) and its analogue),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n=0}^{\infty}\int_{L_1} f(z)e^{-2\pi i(n+1)z}\,dz + \sum_{n=0}^{\infty}\int_{L_2} f(z)e^{2\pi inz}\,dz = \sum_{n=0}^{\infty}\hat{f}(n+1) + \sum_{n=0}^{\infty}\hat{f}(-n) = \sum_{n\in\mathbb{Z}}\hat{f}(n).$$

■

Consequences. Since $e^{-\pi x^2}$ is its own Fourier transform (Example 1), for $t > 0$ and $a \in \mathbb{R}$ the change of variables $x \mapsto t^{1/2}(x+a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi ia\xi}$. Poisson summation gives

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi ina}. \tag{3}$$

For $a = 0$ this is the transformation law of the theta function $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ $(t > 0)$:

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0, \tag{4}$$

used in Chapter 6 for the functional equation of ζ; the general case is used in Chapter 10 for the Jacobi theta function. Likewise, since $1/\cosh \pi x$ is its own Fourier transform (Example 3), $f(x) = e^{-2\pi iax}/\cosh(\pi x/t)$ has $\hat{f}(\xi) = t/\cosh(\pi(\xi+a)t)$, and

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi ian}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n+a)t)}, \tag{5}$$

used in Chapter 10 for the two-squares theorem.

3 Paley–Wiener theorem

Now we assume no analyticity, but the validity of Fourier inversion when $|f(x)| \le A/(1+x^2)$ and $|\hat{f}(\xi)| \le A'/(1+\xi^2)$ (proved in Book I). A partial converse to Theorem 2.1:

Theorem 3.1

Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \le Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.

Proof. Let $f_n(z) = \int_{-n}^{n} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$, entire by Theorem 5.4. For $z \in S_b$ define $f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; the integral converges absolutely, being majorized by $A\int e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi < \infty$ as $b < a$. Moreover for $z \in S_b$, $|f(z) - f_n(z)| \le A\int_{|\xi|\ge n} e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi \to 0$, so $f_n \to f$ uniformly in S_b, and f is holomorphic there by Theorem 5.2. ■

Corollary 3.2

If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.

f is analytic in a region containing $\mathbb{R}$, so this follows from Theorem 4.8. In particular f and $\hat{f}$ cannot both have compact support unless $f = 0$.

Theorem 3.3 (Paley–Wiener)

Suppose f is continuous and of moderate decrease on $\mathbb{R}$. Then, f has an extension to the complex plane that is entire with $|f(z)| \le Ae^{2\pi M|z|}$ for some $A > 0$, if and only if $\hat{f}$ is supported in the interval $[-M, M]$.

Proof. *($\Leftarrow$)* If $\hat{f}$ is supported in $[-M, M]$, both f and $\hat{f}$ have moderate decrease and $f(x) = \int_{-M}^{M} \hat{f}(\xi) e^{2\pi i \xi x}\, d\xi$. Define $g(z) = \int_{-M}^{M} \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$; $g = f$ on $\mathbb{R}$, g is entire by Theorem 5.4, and for $z = x + iy$, $|g(z)| \le \int_{-M}^{M} |\hat{f}(\xi)| e^{-2\pi \xi y}\, d\xi \le Ae^{2\pi M|z|}$.

($\Rightarrow$) If $\hat{f}$ were supported in $[-M, M]$, the argument above would give the stronger bound $Ae^{2\pi M|y|}$. We reduce to that situation, adding also decay in x, and then remove the extra assumptions step by step.

Step 1. Assume f is entire with

$$|f(x + iy)| \le A \frac{e^{2\pi M|y|}}{1 + x^2}. \tag{6}$$

Then $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shifting the real line down by $y > 0$ (as in (1)),

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy) e^{-2\pi i (x - iy)\xi}\, dx, \qquad |\hat{f}(\xi)| \le \int_{-\infty}^{\infty} A \frac{e^{2\pi M y - 2\pi y \xi}}{1 + x^2}\, dx \le Ce^{-2\pi y(\xi - M)},$$

and letting $y \to \infty$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ shift up.

Step 2. Assume only

$$|f(x + iy)| \le Ae^{2\pi M|y|}. \tag{7}$$

Let $\xi > M$ and for $\epsilon > 0$ set $f_\epsilon(z) = \dfrac{f(z)}{(1 + i\epsilon z)^2}$. The factor $1/(1 + i\epsilon z)^2$ has absolute value ≤ 1 in the closed lower half-plane and tends to 1 as $\epsilon \to 0$; hence $\hat{f}_\epsilon(\xi) \to \hat{f}(\xi)$, since $|\hat{f}_\epsilon(\xi) - \hat{f}(\xi)| \le \int |f(x)| \left| \frac{1}{(1 + i\epsilon x)^2} - 1 \right| dx \to 0$ by moderate decrease. Each f_ϵ satisfies $|f_\epsilon(x + iy)| \le A' \frac{e^{2\pi M|y|}}{1 + x^2}$ in the lower half-plane, which is where Step 1 shifts the contour for $\xi > M$; so $\hat{f}_\epsilon(\xi) = 0$, and hence $\hat{f}(\xi) = 0$. For $\xi < -M$ argue in the upper half-plane with $1/(1 - i\epsilon z)^2$.

Step 3. It remains to show the hypotheses imply (7). Dividing by a constant, it suffices: if $|f(x)| \le 1$ for real x and $|f(z)| \le e^{2\pi M|z|}$ for all z, then $|f(x + iy)| \le e^{2\pi M|y|}$. This follows from the Phragmén–Lindelöf theorem below, rotated to the first quadrant $Q = \{x > 0, y > 0\}$: $F(z) = f(z) e^{2\pi i M z}$ is bounded by 1 on the positive real and imaginary axes and $|F(z)| \le Ce^{c|z|}$ in Q, so $|F(z)| \le 1$ in Q, i.e. $|f(z)| \le e^{2\pi M y}$. The other quadrants are similar. ∎

The idea of Phragmén and Lindelöf adapts the maximum modulus principle to unbounded regions:

Theorem 3.4 (Phragmén–Lindelöf)

Suppose F is a holomorphic function in the sector $S = \{z : -\pi/4 < \arg z < \pi/4\}$ that is continuous on the closure of S. Assume $|F(z)| \le 1$ on the boundary of the sector, and that there are constants $C, c > 0$ such that $|F(z)| \le Ce^{c|z|}$ for all z in the sector. Then $|F(z)| \le 1$ for all $z \in S$.

Some growth restriction is needed: $F(z) = e^{z^2}$ is bounded by 1 on ∂S but unbounded on $\mathbb{R}_+$.

Proof. Subdue the "enemy" e^{z^2}: replace it by e^{z^α} with $\alpha < 2$; take $\alpha = 3/2$. For $\epsilon > 0$ let $F_\epsilon(z) = F(z) e^{-\epsilon z^{3/2}}$, with the principal branch, so $z^{3/2} = r^{3/2} e^{3i\theta/2}$ for $z = re^{i\theta}$, $-\pi < \theta < \pi$. F_ϵ is holomorphic in S and continuous up to the boundary, and $|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2} \cos(3\theta/2)}$. In the sector $-\pi/4 < \theta < \pi/4$ we have $-\frac{\pi}{2} < -\frac{3\pi}{8} < \frac{3\theta}{2} < \frac{3\pi}{8} < \frac{\pi}{2}$, so $\cos(3\theta/2)$ is bounded below by a positive constant. With $|F(z)| \le Ce^{c|z|}$, F_ϵ decreases rapidly as $|z| \to \infty$ in the closed sector and is bounded. Let $M = \sup_{z \in \overline{S}} |F_\epsilon(z)|$; if $F \not\equiv 0$, take w_j with $|F_\epsilon(w_j)| \to M$. Since $M \neq 0$ and F_ϵ

decays at infinity, the w_j accumulate at some $w \in \overline{S}$. By the maximum principle $w \notin S$, so $w \in \partial S$, where $|F| \le 1$ and $|e^{-\epsilon z^{3/2}}| \le 1$; so $M \le 1$. Thus $|F_\epsilon| \le 1$ on $\overline{S}$, and letting $\epsilon \to 0$ gives $|F| \le 1$. ■

Another version of the Paley–Wiener idea characterizes Fourier transforms vanishing for $\xi < 0$:

Theorem 3.5

Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f can be extended to a continuous and bounded function in the closed upper half-plane $\{z = x + iy : y \ge 0\}$ with f holomorphic in the interior.

Proof. If $\hat{f}(\xi) = 0$ for $\xi < 0$, then $f(x) = \int_0^\infty \hat{f}(\xi) e^{2\pi i \xi x}\, d\xi$, and we extend f for $y \ge 0$ by $f(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$. The integral converges, and $|f(z)| \le A \int_0^\infty \frac{d\xi}{1+\xi^2} < \infty$, so f is bounded. The functions $f_n(z) = \int_0^n \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$ converge to f uniformly in the closed half-plane, which gives continuity there and holomorphy in the interior.

Conversely, in the spirit of Theorem 3.3, for $\epsilon, \delta > 0$ set $f_{\epsilon,\delta}(z) = \dfrac{f(z + i\delta)}{(1 - i\epsilon z)^2}$, holomorphic in a region containing the closed upper half-plane. As before, by Cauchy's theorem (shifting the contour up), $\hat{f}_{\epsilon,\delta}(\xi) = 0$ for all $\xi < 0$. Passing to the limit successively, $\hat{f}_{\epsilon,0}(\xi) = 0$ for $\xi < 0$, and finally $\hat{f}(\xi) = \hat{f}_{0,0}(\xi) = 0$ for $\xi < 0$. ■

Remark

Compare Theorem 7.1: there, for a function holomorphic in a disc, the Fourier coefficients vanish for $n < 0$; here, for a function holomorphic in the upper half-plane, the Fourier transform vanishes for $\xi < 0$.

Chapter 5. Entire Functions

Overview

Entire functions are holomorphic in the whole complex plane. The chapter is organized around three questions.

1. *Where can they vanish?* The obvious necessary condition is sufficient: for any sequence $\{z_n\}$ with no limit point in $\mathbb{C}$ there is an entire function vanishing exactly there. The construction is inspired by Euler's product for $\sin \pi z$, refined by the *Weierstrass canonical factors*.
2. *How do they grow at infinity?* The larger a function is, the more zeros it can have; for a polynomial of degree d, $\sup_{|z|=R}|P(z)| \approx R^d$ and P has d zeros. A precise version is *Jensen's formula*, linking the number of zeros in a disc to the logarithmic average of $|f|$ on the circle.
3. *To what extent are they determined by their zeros?* If f has finite order, it is determined by its zeros up to a simple factor: the *Hadamard factorization theorem*.

1 Jensen's formula

D_R and C_R denote the open disc and circle of radius R centered at 0. We exclude the function that vanishes identically.

Theorem 1.1 (Jensen's formula)

Let Ω be an open set that contains the closure of a disc D_R and suppose that f is holomorphic in Ω, $f(0) \neq 0$, and f vanishes nowhere on the circle C_R. If $z_1, \dots, z_N$ denote the zeros of f inside the disc (counted with multiplicities), then

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta. \tag{1}$$

Proof.

Step 1. If f_1, f_2 satisfy the hypotheses and (1), so does $f_1 f_2$, since $\log xy = \log x + \log y$ for $x, y > 0$ and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2.

Step 2. $g(z) = f(z)/((z-z_1)\cdots(z-z_N))$, defined on $\Omega - \{z_1, \dots, z_N\}$, is bounded near each z_j, so each z_j is a removable singularity: $f(z) = (z-z_1)\cdots(z-z_N)g(z)$ with g holomorphic in Ω and nowhere vanishing on $\overline{D_R}$. By Step 1 it suffices to prove (1) for such g and for functions $z - z_j$.

Step 3. For g nowhere vanishing on $\overline{D_R}$ we must show $\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta$. In a slightly larger disc, $g = e^h$ with h holomorphic (discs are simply connected; Theorem 6.2). Then $|g| = e^{\mathrm{Re}(h)}$, so $\log|g| = \mathrm{Re}(h)$, and the mean value property (Corollary 7.3) gives the formula.

Step 4. For $f(z) = z - w$, $w \in D_R$, we must show $\log|w| = \log\frac{|w|}{R} + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta$. Since $\log(|w|/R) = \log|w| - \log R$ and $\log|Re^{i\theta} - w| = \log R + \log|e^{i\theta} - w/R|$, it suffices to show $\int_0^{2\pi} \log|e^{i\theta} - a|\, d\theta = 0$ for $|a| < 1$, equivalently (via $\theta \mapsto -\theta$, $|e^{i\theta} - a| = |1 - ae^{-i\theta}|$) $\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0$. $F(z) = 1 - az$ vanishes nowhere on the closed unit disc, so $F = e^G$ in a disc of radius > 1 and $\log|F| = \mathrm{Re}(G)$ is harmonic. Since $\log|F(0)| = 0$, the mean value property gives the result. ∎

For f holomorphic on $\overline{D_R}$, let $n(r) = n_f(r)$ be the number of zeros of f (with multiplicities) in D_r, $0 < r < R$; $n(r)$ is non-decreasing. If $f(0) \neq 0$ and f does not vanish on C_R, then

$$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta - \log|f(0)|, \tag{2}$$

which follows from Jensen's formula and:

Lemma 1.2

If $z_1, \dots, z_N$ are the zeros of f inside the disc D_R, then

$$\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|.$$

Proof. $\sum_{k=1}^{N} \log|\frac{R}{z_k}| = \sum_{k=1}^{N} \int_{|z_k|}^{R} \frac{dr}{r}$. With $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, $\sum_k \eta_k(r) = n(r)$, so $\sum_k \int_{|z_k|}^{R} \frac{dr}{r} = \sum_k \int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$. ∎

2 Functions of finite order

Definition (order of growth)

An entire function f has an **order of growth** $\le \rho$ if $|f(z)| \le Ae^{B|z|^\rho}$ for all $z \in \mathbb{C}$, for some constants $A, B > 0$. The **order of growth** of f is $\rho_f = \inf \rho$ over all such $\rho > 0$.

For example, e^{z^2} has order of growth 2.

Theorem 2.1

If f is an entire function that has an order of growth $\le \rho$, then: (i) $n(r) \le Cr^\rho$ for some $C > 0$ and all sufficiently large r. (ii) If $z_1, z_2, \dots$ denote the zeros of f, with $z_k \neq 0$, then for all $s > \rho$ we have $\displaystyle\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. It suffices to treat $f(0) \neq 0$: otherwise use $F(z) = f(z)/z^\ell$, ℓ the order of the zero at 0; n_f and n_F differ by a constant and F also has order of growth $\le \rho$. By (2) with $R = 2r$,

$$\int_r^{2r} n(x)\frac{dx}{x} \le \int_0^{2r} n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{i\theta})|\,d\theta - \log|f(0)|.$$

Since n is increasing, the left side is $\ge n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$; and by the growth condition $\int_0^{2\pi}\log|f(2re^{i\theta})|\,d\theta \le \int_0^{2\pi}\log|Ae^{B(2r)^\rho}|\,d\theta \le C'r^\rho$ for large r. Hence $n(r) \le Cr^\rho$. For (ii),

$$\sum_{|z_k|\ge 1} |z_k|^{-s} = \sum_{j=0}^{\infty} \sum_{2^j \le |z_k| < 2^{j+1}} |z_k|^{-s} \le \sum_{j=0}^{\infty} 2^{-js} n(2^{j+1}) \le c\sum_{j=0}^{\infty} 2^{-js}2^{(j+1)\rho} \le c'\sum_{j=0}^{\infty}(2^{\rho-s})^j < \infty,$$

since $s > \rho$. ∎

The condition $s > \rho$ cannot be improved:

Example 1

$f(z) = \sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$ satisfies $|f(z)| \le e^{\pi|z|}$, and taking $z = ix$ shows the order of growth is exactly 1. f vanishes to order 1 at each $n \in \mathbb{Z}$, and $\sum_{n\neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

Example 2

$f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty}(-1)^n \frac{z^n}{(2n)!}$ is entire with $|f(z)| \le e^{|z|^{1/2}}$, of order $1/2$. It vanishes at $z_n = ((n+1/2)\pi)^2$, and $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

Given $z_1, z_2, \ldots$, is there an entire f with zeros exactly there? Necessarily $\lim_{k\to\infty}|z_k| = \infty$ (Theorem 4.8). Weierstrass showed this suffices, by inserting factors into $(z-z_1)(z-z_2)\cdots$ that guarantee convergence without introducing new zeros.

3 Infinite products

3.1 Generalities

For complex a_n, the product $\prod_{n=1}^{\infty}(1+a_n)$ **converges** if $\lim_{N\to\infty}\prod_{n=1}^{N}(1+a_n)$ exists.

Proposition 3.1

If $\sum|a_n| < \infty$, then the product $\prod_{n=1}^{\infty}(1+a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.

Proof. For large n, $|a_n| < 1/2$; discarding finitely many terms, assume this for all n. Define $\log(1+a_n)$ by the power series (6) of Chapter 3, so $1+z = e^{\log(1+z)}$ for $|z| < 1$. Then $\prod_{n=1}^{N}(1+a_n) = \prod_{n=1}^{N} e^{\log(1+a_n)} = e^{B_N}$ with $B_N = \sum_{n=1}^{N} b_n$, $b_n = \log(1+a_n)$. From the power series, $|\log(1+z)| \le 2|z|$ for $|z| < 1/2$, so $|b_n| \le 2|a_n|$ and $B_N \to B$. By continuity of the exponential, $e^{B_N} \to e^B$. If $1+a_n \neq 0$ for all n, the limit is $e^B \neq 0$. ∎

Proposition 3.2

Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω. If there exist constants $c_n > 0$ such that $\sum c_n < \infty$ and $|F_n(z) - 1| \le c_n$ for all $z \in \Omega$, then: (i) The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$. (ii) If $F_n(z)$ does not vanish for any n, then

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F_n'(z)}{F_n(z)}.$$

Proof. (i) Write $F_n(z) = 1 + a_n(z)$ with $|a_n(z)| \le c_n$ and argue as in Proposition 3.1; the estimates are uniform in z since the c_n are constants. The uniform limit is holomorphic (Theorem 5.2). (ii) Let $K \subset \Omega$ be compact and $G_N = \prod_{n=1}^{N} F_n$. Since $G_N \to F$ uniformly, Theorem 5.3 gives $G_N' \to F'$ uniformly on K. G_N is uniformly bounded below on K, so $G_N'/G_N \to F'/F$ uniformly on K; and $G_N'/G_N = \sum_{n=1}^{N} F_n'/F_n$ (Section 4). ∎

3.2 Example: the product formula for the sine function

$$\sin \pi z = \pi z \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right) \tag{3}$$

for all $z \in \mathbb{C}$; it is derived from the sum formula for the cotangent

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\le N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \tag{4}$$

valid for $z \notin \mathbb{Z}$. The halves of $\sum 1/(z+n)$ with $n > 0$ and $n < 0$ diverge separately; only the symmetric interpretation converges.

Proof of (4). $F(z) = \pi\cot\pi z$ has the properties: (i) $F(z+1) = F(z)$ for $z \notin \mathbb{Z}$; (ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0; (iii) F has simple poles at the integers and no other singularities. The function $\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ has the same three properties: for (i),

$$\sum_{|n|\le N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\le N} \frac{1}{z+n},$$

and let $N \to \infty$; (ii), (iii) are evident from $\frac{1}{z} + \sum_{n\ge 1} \frac{2z}{z^2-n^2}$. Hence $\Delta(z) = F(z) - \sum \frac{1}{z+n}$ is periodic with period 1, has a removable singularity at 0 by (ii), hence at all integers by periodicity, and is entire. It suffices to show Δ is bounded, hence (by periodicity) bounded in the strip $|\mathrm{Re}(z)| \le 1/2$. It is bounded in the rectangle $|\mathrm{Im}(z)| \le 1$. For $z = x + iy$ with $y > 1$,

$$\pi\cot\pi z = \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$$

is bounded, and

$$\Big|\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}\Big| = \Big|\frac{1}{x+iy} + \sum_{n=1}^{\infty} \frac{2(x+iy)}{x^2-y^2-n^2+2ixy}\Big| \le C + C\sum_{n=1}^{\infty} \frac{y}{y^2+n^2}.$$

The last sum is majorized by $\int_0^\infty \frac{y}{y^2+x^2}\,dx$ (the integrand is decreasing in x), which is independent of y (change variables $x \mapsto yx$). Similarly for $y < -1$. So Δ is bounded on $\mathbb{C}$, hence constant by Liouville; Δ is odd, so the constant is 0. ∎

Proof of (3). Let $G(z) = \sin\pi z$ and $P(z) = \pi z\prod_{n=1}^{\infty}(1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 3.2 shows P converges and, away from the integers, $\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$. Since $G'/G = \pi\cot\pi z$, (4) gives

$$\Big(\frac{P}{G}\Big)' = \frac{P}{G}\Big[\frac{P'}{P} - \frac{G'}{G}\Big] = 0,$$

so $P = cG$. Dividing by z and letting $z \to 0$, $c = 1$. ∎

4 Weierstrass infinite products

Theorem 4.1 (Weierstrass)

Given any sequence $\{a_n\}$ of complex numbers with $|a_n| \to \infty$ as $n \to \infty$, there exists an entire function f that vanishes at all $z = a_n$ and nowhere else. Any other such entire function is of the form $f(z)e^{g(z)}$, where g is entire.

Repetitions are allowed in $\{a_n\}$, so this gives entire functions with prescribed zeros *and multiplicities* (the multiplicity of a zero a is the m with $f(z) = (z-a)^m g(z)$, $g(a) \ne 0$).

Proof. If f_1, f_2 both vanish exactly at the a_n (with multiplicities), f_1/f_2 has removable singularities there, so it is entire and nowhere vanishing; hence $f_1/f_2 = e^g$ for an entire g (Theorem 6.2). This proves the last statement.

To construct f: the naive product $\prod(1 - z/a_n)$ converges only for suitable $\{a_n\}$, so we insert exponential factors. Define the **canonical factors** of degree k:

$$E_0(z) = 1 - z, \qquad E_k(z) = (1-z)e^{z+z^2/2+\cdots+z^k/k} \quad (k \ge 1).$$

Suppose f is to have a zero of order m at the origin and $a_1, a_2, \ldots$ are non-zero. The **Weierstrass product** is

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Fix $R > 0$ and $|z| < R$. Factors with $|a_n| \le 2R$ are finitely many, and their product vanishes exactly at the a_n with $|a_n| < R$. If $|a_n| > 2R$ then $|z/a_n| \le 1/2$, and by Lemma 4.2 (with c independent of n)

$$|1 - E_n(z/a_n)| \le c\Big|\frac{z}{a_n}\Big|^{n+1} \le \frac{c}{2^{n+1}}.$$

By Section 3, $\prod_{|a_n|>2R} E_n(z/a_n)$ is holomorphic and non-vanishing in $|z| < R$. As R is arbitrary, f has the required properties. ■

Lemma 4.2

If $|z| \le 1/2$, then $|1 - E_k(z)| \le c|z|^{k+1}$ for some $c > 0$.

Proof. For $|z| \le 1/2$, $1 - z = e^{\log(1-z)}$ (power series), so $E_k(z) = e^{\log(1-z)+z+z^2/2+\cdots+z^k/k} = e^w$ with $w = -\sum_{n=k+1}^{\infty} z^n/n$. Since $|z| \le 1/2$,

$$|w| \le |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \le |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \le 2|z|^{k+1}.$$

In particular $|w| \le 1$, so $|1 - E_k(z)| = |1 - e^w| \le c'|w| \le c|z|^{k+1}$. ■

Remark

The constant c is independent of k: one may take $c' = e$, $c = 2e$.

5 Hadamard's factorization theorem

Weierstrass's theorem writes a function with zeros $a_1, a_2, \ldots$ as $e^{g(z)} z^m \prod E_n(z/a_n)$. Hadamard showed that for functions of finite order the degree of the canonical factors can be taken constant, and g is then a polynomial. Recall that if f has order of growth $\le \rho$, then $n(r) \le Cr^\rho$ for large r and $\sum |a_n|^{-s} < \infty$ for $s > \rho$ (Theorem 2.1).

Theorem 5.1 (Hadamard)

Suppose f is entire and has growth order ρ_0. Let k be the integer so that $k \le \rho_0 < k + 1$. If $a_1, a_2, \ldots$ denote the (non-zero) zeros of f, then

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\le k$, and m is the order of the zero of f at $z = 0$.

Main lemmas.

Lemma 5.2

The canonical products satisfy

$$|E_k(z)| \ge e^{-c|z|^{k+1}} \text{ if } |z| \le 1/2 \qquad \text{and} \qquad |E_k(z)| \ge |1 - z|e^{-c'|z|^k} \text{ if } |z| \ge 1/2.$$

Proof. If $|z| \le 1/2$, $E_k(z) = e^{\log(1-z)+\sum_{n=1}^{k} z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w$, and $|e^w| \ge e^{-|w|}$ with $|w| \le c|z|^{k+1}$. If $|z| \ge 1/2$, $|E_k(z)| = |1 - z||e^{z+z^2/2+\cdots+z^k/k}|$, and $|e^{z+\cdots+z^k/k}| \ge e^{-|z+\cdots+z^k/k|} \ge e^{-c'|z|^k}$ (as $|z| \ge 1/2$). ■

The key is a lower bound for the product of canonical factors away from the zeros.

Lemma 5.3

For any s with $\rho_0 < s < k+1$, we have

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \geq e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \ldots$.

Proof. Write $\prod_n E_k(z/a_n) = \prod_{|a_n|\leq 2|z|} E_k(z/a_n) \prod_{|a_n|>2|z|} E_k(z/a_n)$. For the second product, with no restriction on z, Lemma 5.2 gives

$$\Big|\prod_{|a_n|>2|z|} E_k(z/a_n)\Big| \geq \prod_{|a_n|>2|z|} e^{-c|z/a_n|^{k+1}} = e^{-c|z|^{k+1}\sum_{|a_n|>2|z|}|a_n|^{-k-1}},$$

and since $|a_n| > 2|z|$ and $s < k+1$, $|a_n|^{-k-1} = |a_n|^{-s}|a_n|^{s-k-1} \leq C|a_n|^{-s}|z|^{s-k-1}$; as $\sum|a_n|^{-s} < \infty$, this product is $\geq e^{-c|z|^s}$. For the first product, by the second part of Lemma 5.2,

$$\Big|\prod_{|a_n|\leq 2|z|} E_k(z/a_n)\Big| \geq \prod_{|a_n|\leq 2|z|} \Big|1 - \frac{z}{a_n}\Big| \prod_{|a_n|\leq 2|z|} e^{-c'|z/a_n|^k}. \tag{5}$$

Here $\prod_{|a_n|\leq 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k \sum_{|a_n|\leq 2|z|}|a_n|^{-k}}$ and $|a_n|^{-k} = |a_n|^{-s}|a_n|^{s-k} \leq C|a_n|^{-s}|z|^{s-k}$, so this is $\geq e^{-c|z|^s}$. For the first factor of (5) we need the restriction on z: if z is not in the disc of radius $|a_n|^{-k-1}$ about a_n, then $|a_n - z| \geq |a_n|^{-k-1}$, so

$$\prod_{|a_n|\leq 2|z|} \Big|1 - \frac{z}{a_n}\Big| = \prod_{|a_n|\leq 2|z|} \Big|\frac{a_n - z}{a_n}\Big| \geq \prod_{|a_n|\leq 2|z|} |a_n|^{-k-1}|a_n|^{-1} = \prod_{|a_n|\leq 2|z|} |a_n|^{-k-2}.$$

Finally $\log \prod_{|a_n|\leq 2|z|} |a_n|^{k+2} = (k+2)\sum_{|a_n|\leq 2|z|} \log|a_n| \leq (k+2)n(2|z|)\log 2|z| \leq c|z|^s \log 2|z| \leq c'|z|^{s'}$ for any $s' > s$, using $n(2|z|) \leq c|z|^s$ (Theorem 2.1). Since $s > \rho_0$ was arbitrary, starting with s close enough to ρ_0 gives the lemma (with s' in place of s). ∎

Corollary 5.4

There exists a sequence of radii $r_1, r_2, \ldots$ with $r_m \to \infty$, such that

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \geq e^{-c|z|^s} \qquad \text{for } |z| = r_m.$$

Proof. Since $\sum|a_n|^{-k-1} < \infty$, there is N with $\sum_{n\geq N}|a_n|^{-k-1} < 1/10$. Given consecutive large integers L, $L+1$, there is $r \in [L, L+1]$ such that the circle of radius r about 0 meets none of the forbidden discs of Lemma 5.3: otherwise the intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$, of length $2|a_n|^{-k-1}$, would cover $[L, L+1]$, giving $2\sum_{n\geq N}|a_n|^{-k-1} \geq 1$, a contradiction. Apply Lemma 5.3 with $|z| = r$. ∎

Proof of Hadamard's theorem. Let $E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n)$. E is entire, as in the proof of Theorem 4.1: by Lemma 4.2, $|1 - E_k(z/a_n)| \leq c|z/a_n|^{k+1}$ for large n, and $\sum|a_n|^{-k-1}$ converges ($\rho_0 < s < k+1$). E has the zeros of f, so f/E is entire and nowhere vanishing: $f(z)/E(z) = e^{g(z)}$ for an entire g. Since f has growth order ρ_0 and by Corollary 5.4,

$$e^{\mathrm{Re}(g(z))} = \Big|\frac{f(z)}{E(z)}\Big| \leq c'e^{c|z|^s} \qquad \text{for } |z| = r_m,$$

so $\mathrm{Re}(g(z)) \leq C|z|^s$ for $|z| = r_m$. Lemma 5.5 gives that g is a polynomial of degree $\leq s$, hence $\leq k$ (as $s < k+1$). ∎

Lemma 5.5

Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies $u(z) \le Cr^s$ whenever $|z| = r$, for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\le s$.

Proof. Write $g(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 7.1,

$$\frac{1}{2\pi}\int_0^{2\pi} g(re^{i\theta})e^{-in\theta}\,d\theta = \begin{cases} a_n r^n & n \ge 0, \\ 0 & n < 0. \end{cases} \tag{6}$$

Taking complex conjugates, $\frac{1}{2\pi}\int_0^{2\pi} \overline{g(re^{i\theta})}e^{-in\theta}\,d\theta = 0$ for $n > 0$ (7). Since $2u = g + \overline{g}$, adding (6) and (7),

$$a_n r^n = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})e^{-in\theta}\,d\theta \quad (n > 0), \qquad 2\operatorname{Re}(a_0) = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})\,d\theta.$$

(Supplied completion.) Since $\int_0^{2\pi} e^{-in\theta}d\theta = 0$ for $n > 0$, $a_n r^n = -\frac{1}{\pi}\int_0^{2\pi}[Cr^s - u(re^{i\theta})]e^{-in\theta}\,d\theta$, and $Cr^s - u \ge 0$ on the circles in question, so

$$|a_n| r^n \le \frac{1}{\pi}\int_0^{2\pi}[Cr^s - u(re^{i\theta})]\,d\theta = 2Cr^s - 2\operatorname{Re}(a_0).$$

Letting $r \to \infty$ along the sequence, $a_n = 0$ for $n > s$. ∎

Chapter 6. The Gamma and Zeta Functions

Overview

The gamma and zeta functions are among the most important non-elementary functions. The gamma function is ubiquitous; its basic structural property essentially characterizes it: $1/\Gamma(s)$ is the (simplest) entire function with zeros exactly at $s = 0, -1, -2, \ldots$. The zeta function is fundamental in analytic number theory, through Euler's identity

$$\prod_p \frac{1}{1-p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$$

over all primes. ζ is easily analytic for $\mathrm{Re}(s) > 1$; Riemann realized that the study of primes is bound up with its meromorphic continuation to $\mathbb{C}$. We prove this and the functional equation (a symmetry about $\mathrm{Re}(s) = 1/2$, based on the theta function identity), and estimate ζ near $\mathrm{Re}(s) = 1$ for the prime number theorem (Chapter 7). Following convention, the variable is s.

1 The gamma function

For $s > 0$ the **gamma function** is

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1}\, dt. \tag{1}$$

The integral converges: t^{s-1} is integrable near 0, and e^{-t} decays at infinity.

Proposition 1.1

The gamma function extends to an analytic function in the half-plane $\mathrm{Re}(s) > 0$, and is still given there by the integral formula (1).

Proof. It suffices to show the integral is holomorphic in each strip $S_{\delta,M} = \{\delta < \mathrm{Re}(s) < M\}$, $0 < \delta < M < \infty$. With $\sigma = \mathrm{Re}(s)$, $|e^{-t}t^{s-1}| = e^{-t}t^{\sigma-1}$, so $\Gamma(s) = \lim_{\epsilon\to 0} \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$ converges for $s \in S_{\delta,M}$. Let $F_\epsilon(s) = \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$, holomorphic in $S_{\delta,M}$ by Theorem 5.4. By Theorem 5.2 it suffices that $F_\epsilon \to \Gamma$ uniformly on the strip. Now

$$|\Gamma(s) - F_\epsilon(s)| \le \int_0^\epsilon e^{-t}t^{\sigma-1}\, dt + \int_{1/\epsilon}^\infty e^{-t}t^{\sigma-1}\, dt.$$

The first integral is $\le \epsilon^\delta/\delta$ for $0 < \epsilon < 1$, and the second is $\le \int_{1/\epsilon}^\infty e^{-t}t^{M-1}\, dt \le C\int_{1/\epsilon}^\infty e^{-t/2}\, dt \to 0$, uniformly in the strip. ∎

1.1 Analytic continuation

Although the integral does not converge absolutely elsewhere, there is a meromorphic function on $\mathbb{C}$ equal to Γ for $\mathrm{Re}(s) > 0$, its **analytic continuation**, still denoted Γ (unique, since the complement of the poles of a meromorphic function is connected).

Lemma 1.2

If $\mathrm{Re}(s) > 0$, then

$$\Gamma(s+1) = s\Gamma(s). \tag{2}$$

As a consequence $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \ldots$.

Proof. Integrating by parts,

$$\int_\epsilon^{1/\epsilon} \frac{d}{dt}(e^{-t}t^s)\,dt = -\int_\epsilon^{1/\epsilon} e^{-t}t^s\,dt + s\int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\,dt,$$

and letting $\epsilon \to 0$ the left side vanishes, since $e^{-t}t^s \to 0$ as $t \to 0$ or $t \to \infty$. Also $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$; apply (2) successively. ■

Theorem 1.3

The function $\Gamma(s)$ initially defined for $\mathrm{Re}(s) > 0$ has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at the negative integers $s = 0, -1, \ldots$. The residue of Γ at $s = -n$ is $(-1)^n/n!$.

Proof. It suffices to extend Γ to each half-plane $\mathrm{Re}(s) > -m$, $m \geq 1$. For $\mathrm{Re}(s) > -m$ define

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

F_m is meromorphic in $\mathrm{Re}(s) > -m$ with simple poles at $s = 0, -1, \ldots, -m+1$ and residues

$$\mathrm{res}_{s=-n} F_m(s) = \frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(m-n-1)!}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(-1)^n}{n!}.$$

By Lemma 1.2, $F_m = \Gamma$ for $\mathrm{Re}(s) > 0$, so by uniqueness $F_m = F_k$ $(1 \leq k \leq m)$ on the domain of F_k; this gives the continuation. ■

Remark

By analytic continuation, $\Gamma(s+1) = s\Gamma(s)$ holds whenever $s \neq 0, -1, -2, \ldots$. At $s = -n$, $n \geq 1$, both sides are infinite and $\mathrm{res}_{s=-n}\Gamma(s+1) = -n\,\mathrm{res}_{s=-n}\Gamma(s)$. At $s = 0$, $\Gamma(1) = \lim_{s\to 0} s\Gamma(s)$.

An alternate proof. For $\mathrm{Re}(s) > 0$ split $\Gamma(s) = \int_0^1 e^{-t}t^{s-1}\,dt + \int_1^\infty e^{-t}t^{s-1}\,dt$. The second integral is entire; expanding e^{-t} and integrating termwise, $\int_0^1 e^{-t}t^{s-1}\,dt = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)}$, so

$$\Gamma(s) = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t}t^{s-1}\,dt \qquad \text{for } \mathrm{Re}(s) > 0. \tag{3}$$

The series is meromorphic on $\mathbb{C}$ with poles at the negative integers and residue $(-1)^n/n!$ at $-n$: for fixed R, split it at $N > 2R$; the finite part is meromorphic in $|s| < R$ with the right poles and residues, and for $n > N$, $|s| < R$, $\left|\frac{(-1)^n}{n!(n+s)}\right| \leq \frac{1}{n!R}$, so the tail converges uniformly and is holomorphic there. Hence (3) holds on all of $\mathbb{C}$.

1.2 Further properties of Γ

Theorem 1.4

For all $s \in \mathbb{C}$,

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}. \tag{4}$$

$\Gamma(1-s)$ has simple poles at $s = 1, 2, \ldots$, so $\Gamma(s)\Gamma(1-s)$ is meromorphic with simple poles at all integers, as is $\pi/\sin \pi s$. By analytic continuation it suffices to prove (4) for $0 < s < 1$.

Lemma 1.5

For $0 < a < 1$, $\displaystyle\int_0^\infty \frac{v^{a-1}}{1+v}\,dv = \frac{\pi}{\sin \pi a}$.

Proof. With $v = e^x$, $\int_0^\infty \frac{v^{a-1}}{1+v}\,dv = \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x}\,dx$, which equals $\pi/\sin \pi a$ by Example 2. ∎

Proof of Theorem 1.4. For $0 < s < 1$ and $t > 0$, the change of variables $u = vt$ gives $\Gamma(1-s) = \int_0^\infty e^{-u}u^{-s}\,du = t\int_0^\infty e^{-vt}(vt)^{-s}\,dv$. Hence

$$\begin{aligned}\Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t}t^{s-1}\Gamma(1-s)\,dt = \int_0^\infty e^{-t}t^{s-1}\Big(t\int_0^\infty e^{-vt}(vt)^{-s}\,dv\Big)dt \\ &= \int_0^\infty\int_0^\infty e^{-t[1+v]}v^{-s}\,dt\,dv = \int_0^\infty \frac{v^{-s}}{1+v}\,dv = \frac{\pi}{\sin\pi(1-s)} = \frac{\pi}{\sin \pi s}.\end{aligned}$$

∎

With $s = 1/2$, since $\Gamma(s) > 0$ for $s > 0$, $\Gamma(1/2) = \sqrt{\pi}$.

Theorem 1.6

The function Γ has the following properties: (i) $1/\Gamma(s)$ is an entire function of s with simple zeros at $s = 0, -1, -2, \ldots$ and it vanishes nowhere else. (ii) $1/\Gamma(s)$ has growth $\left|\frac{1}{\Gamma(s)}\right| \le c_1 e^{c_2|s|\log|s|}$. Therefore, $1/\Gamma$ is of order 1 in the sense that for every $\epsilon > 0$, there exists a bound $c(\epsilon)$ so that $\left|\frac{1}{\Gamma(s)}\right| \le c(\epsilon)e^{c_2|s|^{1+\epsilon}}$.

Proof. By Theorem 1.4,

$$\frac{1}{\Gamma(s)} = \Gamma(1-s)\frac{\sin \pi s}{\pi}, \tag{5}$$

so the simple poles of $\Gamma(1-s)$ at $s = 1, 2, \ldots$ are cancelled by zeros of $\sin \pi s$, and $1/\Gamma$ is entire with simple zeros at $0, -1, -2, \ldots$.

For the estimate, first $\int_1^\infty e^{-t}t^\sigma\,dt \le e^{(\sigma+1)\log(\sigma+1)}$ for $\sigma > 0$: choose n with $\sigma \le n \le \sigma + 1$; then $\int_1^\infty e^{-t}t^\sigma\,dt \le \int_0^\infty e^{-t}t^n\,dt = n! \le n^n = e^{n\log n} \le e^{(\sigma+1)\log(\sigma+1)}$. Since (3) holds on $\mathbb{C}$, by (5)

$$\frac{1}{\Gamma(s)} = \Big(\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)}\Big)\frac{\sin \pi s}{\pi} + \Big(\int_1^\infty e^{-t}t^{-s}\,dt\Big)\frac{\sin \pi s}{\pi}.$$

$\left|\int_1^\infty e^{-t}t^{-s}\,dt\right| \le \int_1^\infty e^{-t}t^{|\sigma|}\,dt \le e^{(|s|+1)\log(|s|+1)}$ and $|\sin \pi s| \le e^{\pi|s|}$, so the second term is $\le c_1 e^{c_2|s|\log|s|}$. For the first term: if $|\mathrm{Im}(s)| > 1$ it is $\le ce^{\pi|s|}$. If $|\mathrm{Im}(s)| \le 1$, let k be the integer with $k - 1/2 \le \mathrm{Re}(s) < k + 1/2$. If $k \ge 1$,

$$\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)}\frac{\sin \pi s}{\pi} = (-1)^{k-1}\frac{\sin \pi s}{\pi(k-1)!(k-s)} + \sum_{n\ne k-1}\frac{(-1)^n}{n!(n+1-s)}\frac{\sin \pi s}{\pi};$$

the first term is bounded since $\sin \pi s$ vanishes at $s = k$, and the second is majorized by $c\sum 1/n!$. If $k \le 0$ then $\mathrm{Re}(s) < 1/2$ and the series is again bounded by $c\sum 1/n!$. ∎

Theorem 1.7

For all $s \in \mathbb{C}$,

$$\frac{1}{\Gamma(s)} = e^{\gamma s}s\prod_{n=1}^\infty \Big(1 + \frac{s}{n}\Big)e^{-s/n}.$$

Here γ is **Euler's constant**, $\gamma = \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \log N\right)$. The limit exists: $\sum_{n=1}^{N}\frac{1}{n} - \log N = \sum_{n=1}^{N-1}\int_n^{n+1}\left[\frac{1}{n} - \frac{1}{x}\right]dx + \frac{1}{N}$, and by the mean value theorem $0 \le \frac{1}{n} - \frac{1}{x} \le \frac{1}{n^2}$ for $n \le x \le n+1$; so the sum is $\sum_{n=1}^{N-1} a_n + \frac{1}{N}$ with $0 \le a_n \le 1/n^2$, and $\sum a_n$ converges.

Proof. By Hadamard's theorem, since $1/\Gamma$ is entire of growth order 1 with simple zeros at $0, -1, -2, \dots$,

$$\frac{1}{\Gamma(s)} = e^{As+B}s\prod_{n=1}^{\infty}\left(1+\frac{s}{n}\right)e^{-s/n}.$$

Since $s\Gamma(s) \to 1$ as $s \to 0$, $B = 0$ (or an integer multiple of $2\pi i$, which gives the same result). Putting $s = 1$ and using $\Gamma(1) = 1$,

$$e^{-A} = \prod_{n=1}^{\infty}\left(1+\frac{1}{n}\right)e^{-1/n} = \lim_{N\to\infty} e^{\sum_{n=1}^{N}[\log(1+1/n)-1/n]} = \lim_{N\to\infty} e^{-(\sum_{n=1}^{N}1/n)+\log N+\log(1+1/N)} = e^{-\gamma}.$$

So $A = \gamma + 2\pi ik$; since $\Gamma(s)$ is real for real s, $k = 0$. ∎

Thus $1/\Gamma$ is essentially characterized (up to two normalizing constants) as the entire function with (i) simple zeros at $s = 0, -1, -2, \dots$ and no others, and (ii) order of growth 1. $\sin \pi s$ has a similar characterization (zeros at all integers), and satisfies the stricter $\sin \pi s = O(e^{c|s|})$; this estimate without the logarithm fails for $1/\Gamma$.

2 The zeta function

The **Riemann zeta function** is initially defined for real $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty}\frac{1}{n^s}$.

2.1 Functional equation and analytic continuation

Proposition 2.1

The series defining $\zeta(s)$ converges for $\mathrm{Re}(s) > 1$, and the function ζ is holomorphic in this half-plane.

Proof. For $s = \sigma + it$, $|n^{-s}| = |e^{-s\log n}| = e^{-\sigma\log n} = n^{-\sigma}$. If $\sigma \ge 1 + \delta > 1$ the series is uniformly bounded by $\sum 1/n^{1+\delta} < \infty$; so it converges uniformly on each half-plane $\mathrm{Re}(s) \ge 1 + \delta$ and defines a holomorphic function in $\mathrm{Re}(s) > 1$. ∎

The continuation is subtler than for Γ; it relates ζ to the **theta function** $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$, $t > 0$, which satisfies (by Poisson summation, (4) of Chapter 4) $\vartheta(t) = t^{-1/2}\vartheta(1/t)$. We need

$$\vartheta(t) \le Ct^{-1/2} \quad \text{as } t \to 0, \qquad |\vartheta(t) - 1| \le Ce^{-\pi t} \quad \text{for } t \ge 1:$$

the first follows from the functional equation, the second from $\sum_{n\ge1} e^{-\pi n^2 t} \le \sum_{n\ge1} e^{-\pi nt} \le Ce^{-\pi t}$ for $t \ge 1$.

Theorem 2.2

If $\mathrm{Re}(s) > 1$, then

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^{\infty} u^{(s/2)-1}[\vartheta(u) - 1]\,du.$$

Proof. The key is

$$\int_0^{\infty} e^{-\pi n^2 u}u^{(s/2)-1}\,du = \pi^{-s/2}\Gamma(s/2)n^{-s}, \qquad n \ge 1: \tag{6}$$

with $u = t/\pi n^2$ the left side becomes $\left(\int_0^\infty e^{-t} t^{(s/2)-1}\,dt\right)(\pi n^2)^{-s/2}$. Since $\frac{\vartheta(u)-1}{2} = \sum_{n=1}^\infty e^{-\pi n^2 u}$, and the estimates on ϑ justify interchanging sum and integral,

$$\frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u)-1]\,du = \sum_{n=1}^\infty \int_0^\infty u^{(s/2)-1} e^{-\pi n^2 u}\,du = \pi^{-s/2}\Gamma(s/2)\sum_{n=1}^\infty n^{-s} = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

■

The **xi function**, a more symmetric modification of ζ, is defined for $\mathrm{Re}(s) > 1$ by

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{7}$$

Theorem 2.3

The function ξ is holomorphic for $\mathrm{Re}(s) > 1$ and has an analytic continuation to all of $\mathbb{C}$ as a meromorphic function with simple poles at $s = 0$ and $s = 1$. Moreover, $\xi(s) = \xi(1-s)$ for all $s \in \mathbb{C}$.

Proof. Heuristically: multiply $\sum_n e^{-\pi n^2 u} = u^{-1/2}\sum_n e^{-\pi n^2/u}$ by $u^{(s/2)-1}$ and integrate, using (6) and the change $u \mapsto 1/u$ (discarding the $n = 0$ terms). Precisely: let $\psi(u) = [\vartheta(u)-1]/2$. The functional equation of ϑ gives

$$\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

By Theorem 2.2, for $\mathrm{Re}(s) > 1$,

$$\begin{aligned}
\pi^{-s/2}\Gamma(s/2)\zeta(s) &= \int_0^1 u^{(s/2)-1}\psi(u)\,du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \int_0^1 u^{(s/2)-1}\Big[u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}\Big]du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du
\end{aligned}$$

(substituting $u \mapsto 1/u$ in the first integral). Therefore

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du.$$

Since ψ decays exponentially at infinity, the integral is entire, so ξ continues to $\mathbb{C}$ with simple poles at $s = 0$, $s = 1$. The integral is unchanged under $s \mapsto 1-s$, and so is $\frac{1}{s-1} - \frac{1}{s}$; hence $\xi(s) = \xi(1-s)$. ■

Theorem 2.4

The zeta function has a meromorphic continuation into the entire complex plane, whose only singularity is a simple pole at $s = 1$.

Proof. By (7), $\zeta(s) = \pi^{s/2}\xi(s)/\Gamma(s/2)$. $1/\Gamma(s/2)$ is entire with simple zeros at $0, -2, -4, \ldots$, so the simple pole of ξ at 0 is cancelled, and the only singularity of ζ is a simple pole at $s = 1$. ■

A more elementary approach gives the continuation to $\mathrm{Re}(s) > 0$ and is useful for the growth of ζ near $\mathrm{Re}(s) = 1$: compare $\sum n^{-s}$ with $\int_1^\infty x^{-s}\,dx$.

Proposition 2.5

There is a sequence of entire functions $\{\delta_n(s)\}_{n=1}^\infty$ that satisfy the estimate $|\delta_n(s)| \le |s|/n^{\sigma+1}$,

where $s = \sigma + it$, and such that

$$\sum_{1\le n<N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \delta_n(s), \tag{8}$$

whenever N is an integer > 1.

Proof. Set

$$\delta_n(s) = \int_n^{n+1} \Big[\frac{1}{n^s} - \frac{1}{x^s}\Big] dx. \tag{9}$$

By the mean-value theorem applied to x^{-s}, $\big|\frac{1}{n^s} - \frac{1}{x^s}\big| \le \frac{|s|}{n^{\sigma+1}}$ for $n \le x \le n+1$, so $|\delta_n(s)| \le |s|/n^{\sigma+1}$; and $\int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \int_n^{n+1} \frac{dx}{x^s}$. ∎

Corollary 2.6

For $\mathrm{Re}(s) > 0$ we have

$$\zeta(s) - \frac{1}{s-1} = H(s),$$

where $H(s) = \sum_{n=1}^{\infty} \delta_n(s)$ is holomorphic in the half-plane $\mathrm{Re}(s) > 0$.

Proof. First let $\mathrm{Re}(s) > 1$ and $N \to \infty$ in (8): $\int_1^N x^{-s} dx \to \frac{1}{s-1}$, and by $|\delta_n(s)| \le |s|/n^{\sigma+1}$ the series $\sum \delta_n(s)$ converges uniformly on compact subsets of $\mathrm{Re}(s) \ge \delta > 0$. This proves the identity for $\mathrm{Re}(s) > 1$; the uniform convergence shows H is holomorphic in $\mathrm{Re}(s) > 0$, so ζ extends there and the identity persists. ∎

For $\mathrm{Re}(s) > 1$, $|\zeta(s)| \le \sum n^{-\sigma}$, so ζ is bounded in $\mathrm{Re}(s) \ge 1 + \delta$. The growth near $\mathrm{Re}(s) = 1$ is mild (the following estimates are crude but suffice for Chapter 7).

Proposition 2.7

Suppose $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then for each σ_0, $0 \le \sigma_0 \le 1$, and every $\epsilon > 0$, there exists a constant c_ϵ so that (i) $|\zeta(s)| \le c_\epsilon |t|^{1-\sigma_0+\epsilon}$, if $\sigma_0 \le \sigma$ and $|t| \ge 1$. (ii) $|\zeta'(s)| \le c_\epsilon |t|^{\epsilon}$, if $1 \le \sigma$ and $|t| \ge 1$.

In particular $\zeta(1 + it) = O(|t|^\epsilon)$ as $|t| \to \infty$, and likewise for ζ'.

Proof. Besides $|\delta_n(s)| \le |s|/n^{\sigma+1}$ we have $|\delta_n(s)| \le 2/n^\sigma$, from (9) and $|n^{-s}| = n^{-\sigma}$, $|x^{-s}| \le n^{-\sigma}$ for $x \ge n$. Combining via $A = A^\delta A^{1-\delta}$ $(0 \le \delta \le 1)$,

$$|\delta_n(s)| \le \Big(\frac{|s|}{n^{\sigma_0+1}}\Big)^{\delta} \Big(\frac{2}{n^{\sigma_0}}\Big)^{1-\delta} \le \frac{2|s|^\delta}{n^{\sigma_0+\delta}} \qquad (\sigma \ge \sigma_0).$$

Take $\delta = 1 - \sigma_0 + \epsilon$ and use Corollary 2.6: for $\sigma \ge \sigma_0$,

$$|\zeta(s)| \le \Big|\frac{1}{s-1}\Big| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

which gives (i). For (ii), by the Cauchy integral formula on the circle of radius r about s,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{-i\theta}\, d\theta.$$

Take $r = \epsilon$; this circle lies in $\mathrm{Re} \ge 1 - \epsilon$, and (ii) follows from (i) (with $\sigma_0 = 1 - \epsilon$ and 2ϵ in place of ϵ). ∎

Chapter 7. The Zeta Function and Prime Number Theorem

Overview

Euler's product formula for ζ connects analysis with the arithmetic of primes; an easy consequence is that $\sum_p 1/p$ diverges. To understand the distribution of primes, let

$$\pi(x) = \text{number of primes less than or equal to } x.$$

Legendre and Gauss conjectured

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty, \tag{1}$$

where $f(x) \sim g(x)$ means $f(x)/g(x) \to 1$. Tchebychev proved by elementary methods the weaker $\pi(x) \approx x/\log x$, i.e. $A\frac{x}{\log x} \le \pi(x) \le B\frac{x}{\log x}$ for large x (2). In 1896 Hadamard and de la Vallée Poussin proved (1), the *prime number theorem*, using complex analysis, as we do. At the heart of the proof lies the fact that $\zeta(s)$ does not vanish on the line $\mathrm{Re}(s) = 1$ (the two statements are in fact equivalent).

1 Zeros of the zeta function

Euler's identity. For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Proof. Each factor is a convergent geometric series $1 + p^{-s} + p^{-2s} + \cdots$; formally multiplying them gives the result. Precisely, take first $s > 1$ real and integers $M > N$. By unique factorization, every $n \le N$ is a product of primes $\le N$, each repeated fewer than M times, so

$$\sum_{n=1}^{N} \frac{1}{n^s} \le \prod_{p \le N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}}\right) \le \prod_{p \le N} \frac{1}{1 - p^{-s}} \le \prod_{p} \frac{1}{1 - p^{-s}},$$

and letting $N \to \infty$, $\sum_{n=1}^{\infty} n^{-s} \le \prod_p (1 - p^{-s})^{-1}$. Conversely, again by unique factorization, $\prod_{p \le N}(1 + p^{-s} + \cdots + p^{-Ms}) \le \sum_{n=1}^{\infty} n^{-s}$; letting $M \to \infty$, then $N \to \infty$, gives the reverse inequality. (For complex s with $\mathrm{Re}(s) > 1$ the identity follows by analytic continuation, or by the same argument with absolute convergence.) ∎

By Proposition 3.1, $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$. The functional equation $\xi(s) = \xi(1 - s)$ reads $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta(1-s)$, so

$$\zeta(s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}\zeta(1-s).$$

For $\mathrm{Re}(s) < 0$: (i) $\zeta(1-s)$ has no zeros, as $\mathrm{Re}(1-s) > 1$; (ii) $\Gamma((1-s)/2)$ is zero free; (iii) $1/\Gamma(s/2)$ has zeros at $s = -2, -4, -6, \ldots$. Hence:

Theorem 1.1

The only zeros of ζ outside the strip $0 \le \mathrm{Re}(s) \le 1$ are at the negative even integers, $-2, -4, -6, \ldots$.

The region $0 \le \mathrm{Re}(s) \le 1$ is the **critical strip**; the zeros outside it are the **trivial zeros**. Riemann conjectured the **Riemann hypothesis**: *the zeros of $\zeta(s)$ in the critical strip lie on the line* $\mathrm{Re}(s) = 1/2$—still one of the most famous unsolved problems. We prove:

Theorem 1.2

The zeta function has no zeros on the line $\mathrm{Re}(s) = 1$.

(Since ζ has a pole at 1 there are no zeros near $s = 1$; the point is the deeper fact $\zeta(1 + it) \neq 0$ for all $t \in \mathbb{R}$.) By the functional equation, ζ then has no zeros on $\mathrm{Re}(s) = 0$ either.

Lemma 1.3

If $\mathrm{Re}(s) > 1$, then

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s}$$

for some $c_n \geq 0$.

Proof. For $s > 1$, take the logarithm of Euler's product and use $\log \frac{1}{1-x} = \sum_{m=1}^{\infty} \frac{x^m}{m}$ $(0 \leq x < 1)$:

$$\log \zeta(s) = \sum_{p} \log\Big(\frac{1}{1 - p^{-s}}\Big) = \sum_{p,m} \frac{p^{-ms}}{m}.$$

The double sum converges absolutely, so the order of summation is irrelevant (see the Note at the end of the chapter). The formula holds for $\mathrm{Re}(s) > 1$ by analytic continuation; $\log \zeta$ is well defined in the simply connected half-plane $\mathrm{Re}(s) > 1$ since ζ has no zeros there (Theorem 6.2). Finally $c_n = 1/m$ if $n = p^m$ and $c_n = 0$ otherwise. ∎

Lemma 1.4

If $\theta \in \mathbb{R}$, then $3 + 4\cos\theta + \cos 2\theta \geq 0$.

This follows from $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$.

Corollary 1.5

If $\sigma > 1$ and t is real, then $\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \geq 0$.

Proof. With $s = \sigma + it$, $\mathrm{Re}(n^{-s}) = \mathrm{Re}(e^{-(\sigma+it)\log n}) = n^{-\sigma}\cos(t \log n)$. Hence, with $\theta_n = t \log n$,

$$\begin{aligned}\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| &= 3\,\mathrm{Re}[\log \zeta(\sigma)] + 4\,\mathrm{Re}[\log \zeta(\sigma + it)] + \mathrm{Re}[\log \zeta(\sigma + 2it)] \\ &= \sum c_n n^{-\sigma}(3 + 4\cos\theta_n + \cos 2\theta_n) \geq 0,\end{aligned}$$

by Lemma 1.4 and $c_n \geq 0$. ∎

Proof of Theorem 1.2. Suppose $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. ζ is holomorphic at $1 + it_0$ and vanishes there to order ≥ 1, so $|\zeta(\sigma + it_0)|^4 \leq C(\sigma - 1)^4$ as $\sigma \to 1$. Since $s = 1$ is a simple pole, $|\zeta(\sigma)|^3 \leq C'(\sigma - 1)^{-3}$ as $\sigma \to 1$. And $|\zeta(\sigma + 2it_0)|$ stays bounded as $\sigma \to 1$. Hence $|\zeta^3(\sigma)\zeta^4(\sigma + it_0)\zeta(\sigma + 2it_0)| \to 0$ as $\sigma \to 1^+$, contradicting Corollary 1.5 (the logarithm of a number in $(0, 1)$ is negative). ∎

1.1 Estimates for $1/\zeta(s)$

The proof uses the logarithmic derivative ζ'/ζ near $\mathrm{Re}(s) = 1$, so we need growth estimates for ζ' (Proposition 2.7) and for $1/\zeta$. The following is a quantitative version of Theorem 1.2.

Proposition 1.6

For every $\epsilon > 0$, we have $1/|\zeta(s)| \leq c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \geq 1$, and $|t| \geq 1$.

Proof. From Corollary 1.5, $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 1$ for $\sigma \geq 1$. With the estimate for ζ in Proposition 2.7 and $|\zeta(\sigma)| \leq c/(\sigma-1)$,

$$|\zeta^4(\sigma+it)| \geq c|\zeta^{-3}(\sigma)||t|^{-\epsilon} \geq c'(\sigma-1)^3|t|^{-\epsilon},$$

so

$$|\zeta(\sigma+it)| \geq c'(\sigma-1)^{3/4}|t|^{-\epsilon/4} \qquad (\sigma \geq 1,\ |t| \geq 1). \tag{3}$$

Two cases, with a constant A to be chosen. If $\sigma - 1 \geq A|t|^{-5\epsilon}$, (3) gives $|\zeta(\sigma+it)| \geq A'|t|^{-4\epsilon}$, and replacing 4ϵ by ϵ gives the estimate. If $\sigma - 1 < A|t|^{-5\epsilon}$, choose $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$. Then $|\zeta(\sigma+it)| \geq |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)|$, and by the mean value theorem and the bound on ζ' (Proposition 2.7), $|\zeta(\sigma'+it) - \zeta(\sigma+it)| \leq c''|\sigma'-\sigma||t|^{\epsilon} \leq c''(\sigma'-1)|t|^{\epsilon}$. With (3) at σ',

$$|\zeta(\sigma+it)| \geq c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma'-1)|t|^{\epsilon}.$$

Choose $A = (c'/(2c''))^4$; since $\sigma'-1 = A|t|^{-5\epsilon}$, this makes $c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma'-1)|t|^{\epsilon}$, hence $|\zeta(\sigma+it)| \geq c''(\sigma'-1)|t|^{\epsilon} = A''|t|^{-4\epsilon}$. Replace 4ϵ by ϵ. ∎

2 Reduction to the functions ψ and ψ_1

Tchebychev introduced a function easier to handle than $\pi(x)$ but carrying the same information:

$$\psi(x) = \sum_{p^m \leq x} \log p,$$

summed over integers of the form $p^m \leq x$ (p prime, $m \geq 1$). Equivalently, with $\Lambda(n) = \log p$ if $n = p^m$ for some prime p and $m \geq 1$, and $\Lambda(n) = 0$ otherwise,

$$\psi(x) = \sum_{1 \leq n \leq x} \Lambda(n), \qquad \psi(x) = \sum_{p \leq x} \left[\frac{\log x}{\log p}\right] \log p,$$

$[u]$ the greatest integer $\leq u$ (since $p^m \leq x$ iff $m \leq \log x/\log p$).

Proposition 2.1

If $\psi(x) \sim x$ as $x \to \infty$, then $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. We show $1 \leq \liminf \pi(x)\frac{\log x}{x}$ and $\limsup \pi(x)\frac{\log x}{x} \leq 1$ (4). Crudely, $\psi(x) = \sum_{p\leq x}\left[\frac{\log x}{\log p}\right]\log p \leq \sum_{p\leq x}\frac{\log x}{\log p}\log p = \pi(x)\log x$, so $\frac{\psi(x)}{x} \leq \frac{\pi(x)\log x}{x}$, and $\psi(x) \sim x$ gives the first inequality. For the second, fix $0 < \alpha < 1$:

$$\psi(x) \geq \sum_{p\leq x} \log p \geq \sum_{x^\alpha < p \leq x} \log p \geq (\pi(x) - \pi(x^\alpha))\alpha \log x,$$

so $\psi(x) + \alpha\pi(x^\alpha)\log x \geq \alpha\pi(x)\log x$. Dividing by x, and using $\pi(x^\alpha) \leq x^\alpha$, $\alpha < 1$, $\psi(x) \sim x$, we get $1 \geq \alpha \limsup \pi(x)\frac{\log x}{x}$. As $\alpha < 1$ is arbitrary, the proof is complete. ∎

(The converse also holds.) It is more convenient to work with

$$\psi_1(x) = \int_1^x \psi(u)\,du.$$

Proposition 2.2

If $\psi_1(x) \sim x^2/2$ as $x \to \infty$, then $\psi(x) \sim x$ as $x \to \infty$, and therefore $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. By Proposition 2.1 it suffices to show $\psi(x) \sim x$. Since ψ is increasing, for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x}\int_{\alpha x}^{x} \psi(u)\,du \le \psi(x) \le \frac{1}{(\beta-1)x}\int_{x}^{\beta x} \psi(u)\,du.$$

So $\psi(x) \le \frac{1}{(\beta-1)x}[\psi_1(\beta x) - \psi_1(x)]$, i.e. $\frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{\psi_1(\beta x)}{(\beta x)^2}\beta^2 - \frac{\psi_1(x)}{x^2}\Big]$, whence

$$\limsup_{x\to\infty} \frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{1}{2}\beta^2 - \frac{1}{2}\Big] = \frac{1}{2}(\beta+1).$$

As $\beta > 1$ is arbitrary, $\limsup \psi(x)/x \le 1$; a similar argument with $\alpha < 1$ gives $\liminf \psi(x)/x \ge 1$. ∎

To relate ψ_1 and ζ: differentiating Lemma 1.3, for $\mathrm{Re}(s) > 1$, $\frac{\zeta'(s)}{\zeta(s)} = -\sum_{m,p}(\log p)p^{-ms} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}$, i.e.

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}. \tag{5}$$

Proposition 2.3

For all $c > 1$,

$$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)\,ds. \tag{6}$$

Lemma 2.4

If $c > 0$, then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{a^s}{s(s+1)}\,ds = \begin{cases} 0 & \text{if } 0 < a \le 1,\\ 1 - 1/a & \text{if } 1 \le a.\end{cases}$$

Here, the integral is over the vertical line $\mathrm{Re}(s) = c$.

Proof. $|a^s| = a^c$ on the line, so the integral converges. Suppose $a \ge 1$, $a = e^{\beta}$, $\beta = \log a \ge 0$, and $f(s) = \frac{a^s}{s(s+1)} = \frac{e^{\beta s}}{s(s+1)}$; then $\mathrm{res}_{s=0} f = 1$ and $\mathrm{res}_{s=-1} f = -1/a$. For $T > 0$ let $\Gamma(T)$ be the vertical segment $S(T)$ from $c - iT$ to $c + iT$ together with the half-circle $C(T)$ of radius T centered at c, to the left of the segment, positively oriented (a toy contour). For T large, 0 and -1 are inside, and $\frac{1}{2\pi i}\int_{\Gamma(T)} f = 1 - 1/a$. On $C(T)$, for large T, $|s(s+1)| \ge \frac{1}{2}T^2$ and (as $\sigma \le c$) $|e^{\beta s}| \le e^{\beta c}$, so $\big|\int_{C(T)} f\big| \le \frac{C}{T^2}2\pi T \to 0$. If $0 < a \le 1$, use the half-circle to the right of $\mathrm{Re}(s) = c$; there are no poles inside and the same estimate applies (now $\beta \le 0$ and $\sigma \ge c$). ∎

Proof of Proposition 2.3. $\psi(u) = \sum_{n=1}^{\infty}\Lambda(n)f_n(u)$ with $f_n(u) = 1$ if $n \le u$ and 0 otherwise. Hence

$$\psi_1(x) = \int_0^x \psi(u)\,du = \sum_{n=1}^{\infty}\Lambda(n)\int_0^x f_n(u)\,du = \sum_{n\le x}\Lambda(n)(x-n).$$

By (5) and Lemma 2.4 with $a = x/n$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)ds = x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds = x\sum_{n\le x}\Lambda(n)\Big(1-\frac{n}{x}\Big) = \psi_1(x).$$

∎

2.1 Proof of the asymptotics for ψ_1

We show $\psi_1(x) \sim x^2/2$, which completes the proof of the prime number theorem. The ingredients: the formula (6); the non-vanishing $\zeta(1+it) \neq 0$; and the estimates of Proposition 2.7 and Proposition 1.6. Strategy: move the line of integration from $\mathrm{Re}(s) = c$ to $\mathrm{Re}(s) = 1$, so that x^{s+1} has size x^2 instead of x^{c+1}; the pole of ζ at $s = 1$ contributes exactly the main term $x^2/2$, and the rest must be shown to be essentially smaller.

Fix $c = 2$, $x \geq 2$, and let

$$F(s) = \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big).$$

Deform the line $\mathrm{Re}(s) = c$ to the path $\gamma(T)$ ($T \geq 3$, to be chosen): along $\mathrm{Re}(s) = 1$ for $|t| \geq T$, and around $s = 1$ through the segment $\mathrm{Re}(s) = c$, $|t| \leq T$, joined by horizontal segments at height $\pm T$. Then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)\,ds = \frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds. \tag{7}$$

Indeed, by Proposition 2.7 and Proposition 1.6, $|\zeta'(s)/\zeta(s)| \leq A_\eta |t|^\eta$ for any fixed $\eta > 0$, when $\sigma \geq 1$, $|t| \geq 1$; so $|F(s)| \leq A'|t|^{-2+\eta}$ in the two infinite regions between the line and $\gamma(T)$, where F is regular, and Cauchy's theorem gives (7).

Next pass from $\gamma(T)$ to $\gamma(T,\delta)$, which consists of γ_1: $\mathrm{Re}(s) = 1$, $t \leq -T$; γ_2: horizontal segment at $t = -T$ from 1 to $1-\delta$; γ_3: $\mathrm{Re}(s) = 1-\delta$, $|t| \leq T$; γ_4: horizontal segment at $t = T$ from $1-\delta$ to 1; γ_5: $\mathrm{Re}(s) = 1$, $t \geq T$. For fixed T, choose $\delta > 0$ so small that ζ has no zeros in the box $\{\sigma + it : 1-\delta \leq \sigma \leq 1,\ |t| \leq T\}$ (possible since ζ does not vanish on $\sigma = 1$). F has a simple pole at $s = 1$: by Corollary 2.6, $\zeta(s) = \frac{1}{s-1} + H(s)$ with H regular near 1, so $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic near 1, and $\mathrm{res}_{s=1} F = x^2/2$. Thus

$$\frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds = \frac{x^2}{2} + \frac{1}{2\pi i}\int_{\gamma(T,\delta)} F(s)\,ds.$$

Let $\epsilon > 0$ be given. γ_1, γ_5. For $\sigma = 1$, $|x^{1+s}| = x^2$, and by Proposition 1.6 (with Proposition 2.7) $|\zeta'/\zeta| \leq A|t|^{1/2}$, so $\big|\int_{\gamma_5} F\big| \leq Cx^2 \int_T^\infty \frac{|t|^{1/2}}{t^2}\,dt$; the integral converges, so we can choose T with $\big|\int_{\gamma_1} F\big|, \big|\int_{\gamma_5} F\big| \leq \epsilon x^2$. γ_3. With T fixed, $|x^{1+s}| = x^{2-\delta}$ on γ_3, so $\big|\int_{\gamma_3} F\big| \leq C_T x^{2-\delta}$. γ_2, γ_4. $\big|\int_{\gamma_2} F\big| \leq C'_T \int_{1-\delta}^1 x^{1+\sigma}\,d\sigma \leq C'_T \frac{x^2}{\log x}$.

Hence there are constants C_T, C'_T with

$$\Big|\psi_1(x) - \frac{x^2}{2}\Big| \leq \epsilon x^2 + C_T x^{2-\delta} + C'_T \frac{x^2}{\log x}, \qquad \Big|\frac{2\psi_1(x)}{x^2} - 1\Big| \leq 2\epsilon + 2C_T x^{-\delta} + \frac{2C'_T}{\log x},$$

and for all large x, $|\frac{2\psi_1(x)}{x^2} - 1| \leq 4\epsilon$. So $\psi_1(x) \sim x^2/2$, and the prime number theorem is proved.

Note on interchanging double sums

Proposition

Let $\{a_{k\ell}\}_{1 \leq k,\ell < \infty}$ be complex numbers with

$$\sum_{k=1}^{\infty}\Big(\sum_{\ell=1}^{\infty}|a_{k\ell}|\Big) < \infty. \tag{8}$$

Then: (i) The double sum $A = \sum_{k=1}^\infty \sum_{\ell=1}^\infty a_{k\ell}$ converges, and the order of summation may be interchanged: $A = \sum_{\ell=1}^\infty \sum_{k=1}^\infty a_{k\ell}$. (ii) Given $\epsilon > 0$, there is N so that for all $K, L > N$,

$|A - \sum_{k=1}^{K}\sum_{\ell=1}^{L} a_{k\ell}| < \epsilon$. (iii) If $m \mapsto (k(m), \ell(m))$ is a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$ and $c_m = a_{k(m)\ell(m)}$, then $A = \sum_{m=1}^{\infty} c_m$.

Proof. (i) Let $b_k = \sum_\ell |a_{k\ell}|$, and let $S = \{x_0, x_1, \dots\}$ with $x_n \to x_0$. On S define $f_k(x_0) = \sum_{\ell=1}^{\infty} a_{k\ell}$, $f_k(x_n) = \sum_{\ell=1}^{n} a_{k\ell}$, and $g(x) = \sum_k f_k(x)$. Each f_k is continuous at x_0; $|f_k(x)| \le b_k$ with $\sum b_k < \infty$, so the series for g converges uniformly on S and g is continuous at x_0. Hence

$$\sum_k \sum_\ell a_{k\ell} = g(x_0) = \lim_n g(x_n) = \lim_n \sum_k \sum_{\ell=1}^{n} a_{k\ell} = \lim_n \sum_{\ell=1}^{n} \sum_k a_{k\ell} = \sum_\ell \sum_k a_{k\ell}.$$

(ii) $|A - \sum_{k\le K}\sum_{\ell\le L} a_{k\ell}| \le \sum_{k\le K}\sum_{\ell>L}|a_{k\ell}| + \sum_{k>K}\sum_\ell |a_{k\ell}|$. Choose N with $\sum_{k>N} b_k < \epsilon/2$; the first sum is $\le \sum_{k\le N}\sum_{\ell>L}|a_{k\ell}| + \sum_{k>N} b_k$, and for L large the finitely many tails $\sum_{\ell>L}|a_{k\ell}|$, $k \le N$, are small (supplied). (iii) (Supplied.) Given ϵ, take N as in (ii) (for the absolute values as well), and M so large that $\{c_1, \dots, c_M\}$ contains all $a_{k\ell}$ with $k, \ell \le N$. For $M' \ge M$, $\sum_{m\le M'} c_m - \sum_{k,\ell\le N} a_{k\ell}$ is a sum of terms $a_{k\ell}$ with $k > N$ or $\ell > N$, of total absolute value $< \epsilon$; so $|\sum_{m\le M'} c_m - A| < 2\epsilon$. ∎

Chapter 8. Conformal Mappings

Overview

This chapter is geometric and "global": we study mapping properties of holomorphic functions, motivated by the question

> *Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?*

(The inverse will automatically be holomorphic.) The most fruitful choice is $V = \mathbb{D}$: *what conditions on Ω guarantee a holomorphic bijection $\Omega \to \mathbb{D}$?* (For $V = \mathbb{C}$ only $U = \mathbb{C}$ is possible.) We first give explicit examples (the upper half-plane maps to the disc by a fractional linear transformation; compositions of elementary maps) and apply them to the Dirichlet problem. Then the Schwarz lemma determines the automorphisms of the disc and of the upper half-plane. The heart of the chapter is the *Riemann mapping theorem*: Ω is conformally equivalent to $\mathbb{D}$ whenever it is simply connected and not all of $\mathbb{C}$, with no assumption on $\partial\Omega$. For polygons the map is given by the *Schwarz–Christoffel formula*; for rectangles it involves elliptic integrals, which lead to the doubly periodic functions of Chapter 9.

1 Conformal equivalence and examples

A bijective holomorphic $f : U \to V$ is a **conformal map** or **biholomorphism**; then U and V are **conformally equivalent** (**biholomorphic**).

Proposition 1.1

If $f : U \to V$ is holomorphic and injective, then $f'(z) \neq 0$ for all $z \in U$. In particular, the inverse of f defined on its range is holomorphic, and thus the inverse of a conformal map is also holomorphic.

Proof. Suppose $f'(z_0) = 0$. Then $f(z) - f(z_0) = a(z - z_0)^k + G(z)$ near z_0, with $a \neq 0$, $k \geq 2$, and G vanishing to order $k+1$ at z_0. For small w write $f(z) - f(z_0) - w = F(z) + G(z)$ with $F(z) = a(z - z_0)^k - w$. On a small circle about z_0, $|G| < |F|$, and F has at least two zeros inside, so by Rouché's theorem $f(z) - f(z_0) - w$ has at least two zeros there. Since $f'(z) \neq 0$ for $z \neq z_0$ near z_0 (zeros of f' are isolated), these roots are distinct, so f is not injective.

Let $g = f^{-1}$ on the range V, $w_0 = f(z_0)$, $w = f(z)$ close to w_0, $w \neq w_0$. Then

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w - w_0}{g(w) - g(w_0)}} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}.$$

Since $f'(z_0) \neq 0$, letting $z \to z_0$ shows g is holomorphic at w_0 with $g'(w_0) = 1/f'(g(w_0))$. ∎

So U, V are conformally equivalent iff there are holomorphic $f : U \to V$, $g : V \to U$ with $g(f(z)) = z$ and $f(g(w)) = w$. (Some authors call f conformal when merely $f' \neq 0$, e.g. z^2 on $\mathbb{C} - \{0\}$, which is not injective. The condition $f'(z) \neq 0$ means f is a local bijection, and such a map preserves angles between curves.)

1.1 The disc and upper half-plane

The **upper half-plane** is $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im}(z) > 0\}$. Let

$$F(z) = \frac{i - z}{i + z} \qquad \text{and} \qquad G(w) = i\frac{1 - w}{1 + w}.$$

Theorem 1.2

The map $F : \mathbb{H} \to \mathbb{D}$ is a conformal map with inverse $G : \mathbb{D} \to \mathbb{H}$.

Proof. Both maps are holomorphic in their domains. Points of $\mathbb{H}$ are closer to i than to $-i$, so $|F(z)| < 1$. For $w = u + iv \in \mathbb{D}$,

$$\operatorname{Im}(G(w)) = \operatorname{Re}\Big(\frac{1-u-iv}{1+u+iv}\Big) = \operatorname{Re}\frac{(1-u-iv)(1+u-iv)}{(1+u)^2+v^2} = \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0.$$

Finally

$$F(G(w)) = \frac{i - i\frac{1-w}{1+w}}{i + i\frac{1-w}{1+w}} = \frac{1+w-1+w}{1+w+1-w} = w,$$

and similarly $G(F(z)) = z$. ∎

F is holomorphic on $\mathbb{C}$ except at $-i$, hence continuous on $\mathbb{R} = \partial\mathbb{H}$, and $|F(x)| = 1$ for real x. With $x = \tan t$, $t \in (-\pi/2, \pi/2)$,

$$F(x) = \frac{1-x^2}{1+x^2} + i\frac{2x}{1+x^2} = \cos 2t + i \sin 2t = e^{2it},$$

so F maps $\mathbb{R}$ onto the circle minus -1, traversed from -1 first through the lower half; -1 corresponds to the point at infinity of $\mathbb{H}$.

Remark

Maps $z \mapsto \frac{az+b}{cz+d}$ ($a, b, c, d \in \mathbb{C}$, denominator not a multiple of the numerator) are **fractional linear transformations**; they reappear as the automorphisms of the disc and of $\mathbb{H}$ (Theorems 2.2, 2.4).

1.2 Further examples

Example 1

Translations and dilations. For $h \in \mathbb{C}$, $z \mapsto z+h$ is a conformal map of $\mathbb{C}$ (inverse $w \mapsto w-h$); for real h it is also a conformal map of $\mathbb{H}$. For $c \neq 0$, $z \mapsto cz$ is conformal on $\mathbb{C}$ (inverse $w \mapsto c^{-1}w$): a rotation if $c = e^{i\varphi}$, a dilation if $c > 0$, a dilation by $|c|$ followed by rotation by π if $c < 0$.

Example 2

For a positive integer n, $z \mapsto z^n$ is conformal from the sector $\{0 < \arg z < \pi/n\}$ to $\mathbb{H}$, with inverse $w \mapsto w^{1/n}$ (principal branch). More generally, for $0 < \alpha < 2$, $f(z) = z^\alpha$ maps $\mathbb{H}$ to the sector $S = \{0 < \arg w < \alpha\pi\}$: with the branch obtained by deleting the positive real axis and $z = re^{i\theta}$, $0 < \theta < \pi$, $f(z) = |z|^\alpha e^{i\alpha\theta}$; the inverse is $w^{1/\alpha}$ with $0 < \arg w < \alpha\pi$. Composing with Example 1, $\mathbb{H}$ maps conformally onto any infinite sector. On the boundary: as x goes from $-\infty$ to 0, $f(x)$ goes from $\infty e^{i\alpha\pi}$ to 0 along $\arg z = \alpha\pi$; as x goes from 0 to ∞, $f(x)$ goes from 0 to ∞ along $\mathbb{R}$.

Example 3

$f(z) = \frac{1+z}{1-z}$ maps the upper half-disc $\{|z| < 1,\ y > 0\}$ conformally onto the first quadrant

$\{u > 0, v > 0\}$: for $z = x + iy$,

$$f(z) = \frac{1 - (x^2 + y^2)}{(1-x)^2 + y^2} + i\frac{2y}{(1-x)^2 + y^2}.$$

The inverse $g(w) = \frac{w-1}{w+1}$ is holomorphic in the first quadrant, maps it into $\mathbb{D}$ (the distance from w to -1 exceeds that to 1) with $\operatorname{Im} g(w) > 0$. On the boundary, for $z = e^{i\theta}$, $f(z) = \frac{e^{-i\theta/2}+e^{i\theta/2}}{e^{-i\theta/2}-e^{i\theta/2}} = \frac{i}{\tan(\theta/2)}$, so as θ goes from 0 to π, $f(e^{i\theta})$ travels down the imaginary axis from ∞ to 0; f maps $(-1, 1)$ increasingly onto $(0, \infty)$, $f(0) = 1$.

Example 4

$z \mapsto \log z$, the branch obtained by deleting the negative imaginary axis, maps $\mathbb{H}$ onto the strip $\{w = u + iv : 0 < v < \pi\}$: for $z = re^{i\theta}$, $-\pi/2 < \theta < 3\pi/2$, $\log z = \log r + i\theta$. The inverse is $w \mapsto e^w$. As x goes from $-\infty$ to 0, $f(x)$ goes from $\infty + i\pi$ to $-\infty + i\pi$; as x goes from 0 to ∞, $f(x)$ goes from $-\infty$ to ∞ along $\mathbb{R}$.

Example 5

$z \mapsto \log z$ also maps the half-disc $\{|z| < 1, y > 0\}$ onto the half-strip $\{u < 0,\ 0 < v < \pi\}$: $(0, 1)$ goes to $(-\infty, 0)$, the upper half-circle from 1 to -1 to the segment from 0 to $i\pi$, and $(-1, 0)$ to the half-line from $i\pi$ to $-\infty + i\pi$.

Example 6

$f(z) = e^{iz}$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ onto the half-disc $\{|w| < 1,\ u > 0\}$, since $e^{iz} = e^{-y}e^{ix}$. The boundary goes: from $\pi/2 + i\infty$ to $\pi/2$, f goes from 0 to i; from $\pi/2$ to $-\pi/2$, f goes from i to $-i$ along the half-circle; from $-\pi/2$ to $-\pi/2 + i\infty$, from $-i$ back to 0. (This is closely related to the inverse of Example 5.)

Example 7

$f(z) = -\frac{1}{2}(z+1/z)$ maps the half-disc $\{|z| < 1, y > 0\}$ conformally onto $\mathbb{H}$. On the boundary: as x goes from 0 to 1, $f(x)$ goes from $-\infty$ to -1; on the half-circle $z = e^{i\theta}$, $f(z) = -\cos\theta$ runs from -1 to 1 as θ goes from 0 to π; and as x goes from -1 to 0, $f(x)$ goes from 1 to ∞.

Example 8

$f(z) = \sin z$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ conformally onto $\mathbb{H}$. With $\zeta = e^{iz}$,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = -\frac{1}{2}\Big(i\zeta + \frac{1}{i\zeta}\Big),$$

so f is the map of Example 6, followed by multiplication by i (rotation by $\pi/2$, giving the upper half-disc), followed by Example 7. As z goes from $-\pi/2 + i\infty$ to $-\pi/2$, f goes from $-\infty$ to -1; on $[-\pi/2, \pi/2]$, f runs over $[-1, 1]$; from $\pi/2$ to $\pi/2 + i\infty$, f goes from 1 to ∞.

1.3 The Dirichlet problem in a strip

The **Dirichlet problem** in Ω is

$$\triangle u = 0 \text{ in } \Omega, \qquad u = f \text{ on } \partial\Omega, \tag{1}$$

$\triangle = \partial^2/\partial x^2 + \partial^2/\partial y^2$, f given on $\partial\Omega$. For the disc and $\mathbb{H}$ it is solved by convolution with Poisson kernels. We solve it for a strip using conformal maps and the solution in the disc.

Lemma 1.3

Let V and U be open sets in $\mathbb{C}$ and $F : V \to U$ a holomorphic function. If $u : U \to \mathbb{C}$ is a harmonic function, then $u \circ F$ is harmonic on V.

Proof. The question is local, so take U a disc. There is G holomorphic on U with $\operatorname{Re} G = u$ (every harmonic function in a disc is the real part of a holomorphic function). Then $u \circ F = \operatorname{Re}(G \circ F)$ with $G \circ F$ holomorphic, so $u \circ F$ is harmonic. ∎

Let $\Omega = \{x + iy : x \in \mathbb{R},\ 0 < y < 1\}$, with boundary data $u(x,0) = f_0(x)$, $u(x,1) = f_1(x)$, f_0, f_1 continuous and vanishing at infinity. Transfer the problem to $\mathbb{D}$ by

$$F(w) = \frac{1}{\pi} \log\Big(i\frac{1-w}{1+w}\Big) : \mathbb{D} \to \Omega, \qquad G(z) = \frac{i - e^{\pi z}}{i + e^{\pi z}} : \Omega \to \mathbb{D},$$

compositions of maps above, conformal and mutually inverse. F maps the lower half-circle onto $i+\mathbb{R}$ and the upper half-circle onto $\mathbb{R}$: as φ goes from $-\pi$ to 0, $F(e^{i\varphi})$ goes from $i+\infty$ to $i-\infty$, and as φ goes from 0 to π, $F(e^{i\varphi})$ goes from $-\infty$ to ∞. Define $\tilde f_1(\varphi) = f_1(F(e^{i\varphi}) - i)$ for $-\pi < \varphi < 0$ and $\tilde f_0(\varphi) = f_0(F(e^{i\varphi}))$ for $0 < \varphi < \pi$; since f_0, f_1 vanish at infinity, $\tilde f$ (equal to $\tilde f_1$, $\tilde f_0$ on the two half-circles and 0 at $\varphi = \pm\pi, 0$) is continuous on the circle. The solution in the disc is the Poisson integral, for $w = re^{i\theta}$,

$$\tilde u(w) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta-\varphi)\tilde f(\varphi)\,d\varphi = \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde f_1(\varphi)\,d\varphi + \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta-\varphi)\tilde f_0(\varphi)\,d\varphi,$$

$P_r(\gamma) = \frac{1-r^2}{1-2r\cos\gamma+r^2}$ the Poisson kernel. By Lemma 1.3, $u(z) = \tilde u(G(z))$ is harmonic in the strip, with the correct boundary values. For $z = iy$, $0 < y < 1$, with $re^{i\theta} = G(iy)$, a change of variables gives

$$\frac{1}{2\pi}\int_0^{\pi} P_r(\theta-\varphi)\tilde f_0(\varphi)\,d\varphi = \frac{\sin\pi y}{2}\int_{-\infty}^{\infty}\frac{f_0(t)}{\cosh\pi t - \cos\pi y}\,dt, \qquad \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde f_1(\varphi)\,d\varphi = \frac{\sin\pi y}{2}\int_{-\infty}^{\infty}\frac{f_1(t}{\cosh\pi t +}$$

A solution vanishing at infinity is unique, so translating the boundary data by x translates the solution, and

$$u(x,y) = \frac{\sin\pi y}{2}\Big(\int_{-\infty}^{\infty}\frac{f_0(x-t)}{\cosh\pi t - \cos\pi y}\,dt + \int_{-\infty}^{\infty}\frac{f_1(x-t)}{\cosh\pi t + \cos\pi y}\,dt\Big),$$

convolutions with f_0 and f_1. At $y = 1/2$ the kernel is $1/\cosh\pi t$, its own Fourier transform (Example 3).

Remarks about the Dirichlet problem. If $F : \mathbb{D} \to \Omega$ is conformal and extends to a continuous bijection $\partial\mathbb{D} \to \partial\Omega$, then for f on $\partial\Omega$, $\tilde f = f \circ F$ is defined on the circle, the Poisson integral $\tilde u(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta-\varphi)\tilde f(e^{i\varphi})\,d\varphi$ solves the problem in the disc, and $u = \tilde u \circ F^{-1}$ is expected to solve (1). This requires: (i) existence of a conformal map $\Phi = F^{-1} : \Omega \to \mathbb{D}$, settled in general by the Riemann mapping theorem; (ii) its extension to a continuous bijection of the boundaries, which needs some regularity of $\partial\Omega$ (treated for polygons in Section 4). Riemann reversed the implication: to find Φ with $\Phi(z_0) = 0$, write $\Phi(z) = (z - z_0)e^{H(z)}$; $|\Phi| = 1$ on $\partial\Omega$ means $u = \operatorname{Re} H$ solves the Dirichlet problem with $u(z) = \log(1/|z - z_0|)$ on $\partial\Omega$ (u is the Green's function with source z_0). This approach requires showing Φ is a bijection, some boundary regularity, and solving the Dirichlet problem (Riemann proposed the "Dirichlet principle", whose implementation has difficulties). We prove the existence in general by different methods in Section 3.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

A **rotation** is a map $z \mapsto cz$ with $|c| = 1$, $c = e^{i\varphi}$, φ the angle of rotation (defined modulo 2π).

Lemma 2.1 (Schwarz lemma)

Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then (i) $|f(z)| \le |z|$ for all $z \in \mathbb{D}$. (ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation. (iii) $|f'(0)| \le 1$, and if equality holds, then f is a rotation.

Proof. Expand $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots$ in $\mathbb{D}$; $f(0) = 0$ gives $a_0 = 0$, so $f(z)/z$ has a removable singularity at 0 and is holomorphic in $\mathbb{D}$. If $|z| = r < 1$, since $|f(z)| \le 1$, $|f(z)/z| \le 1/r$, and by the maximum modulus principle this holds for $|z| \le r$. Letting $r \to 1$ gives (i). For (ii), $f(z)/z$ attains its maximum in the interior of $\mathbb{D}$, so it is a constant c, and $|c| = 1$ (evaluate at z_0). For (iii), $g(z) = f(z)/z$ satisfies $|g| \le 1$ on $\mathbb{D}$ and $g(0) = \lim_{z\to 0} \frac{f(z)-f(0)}{z} = f'(0)$; if $|f'(0)| = 1$, then $|g(0)| = 1$, so g is constant by the maximum principle and $f(z) = cz$, $|c| = 1$. ∎

2.1 Automorphisms of the disc

A conformal map of Ω to itself is an **automorphism** of Ω; they form a group $\mathrm{Aut}(\Omega)$ under composition, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. Rotations $r_\theta : z \mapsto e^{i\theta} z$ are automorphisms of $\mathbb{D}$ (inverse $r_{-\theta}$). More interesting are

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}, \qquad \alpha \in \mathbb{C},\ |\alpha| < 1.$$

ψ_α is holomorphic in $\mathbb{D}$. If $z = e^{i\theta}$, $\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \overline{\alpha})} = -e^{-i\theta}\frac{w}{\overline{w}}$ with $w = \alpha - e^{i\theta}$, so $|\psi_\alpha| = 1$ on the circle and $|\psi_\alpha| < 1$ in $\mathbb{D}$ by the maximum modulus principle. Moreover

$$(\psi_\alpha \circ \psi_\alpha)(z) = \frac{\alpha - \frac{\alpha - z}{1 - \overline{\alpha} z}}{1 - \overline{\alpha}\frac{\alpha - z}{1 - \overline{\alpha} z}} = \frac{\alpha - |\alpha|^2 z - \alpha + z}{1 - \overline{\alpha} z - |\alpha|^2 + \overline{\alpha} z} = \frac{(1 - |\alpha|^2) z}{1 - |\alpha|^2} = z,$$

so ψ_α is its own inverse; and it interchanges 0 and α: $\psi_\alpha(0) = \alpha$, $\psi_\alpha(\alpha) = 0$.

Theorem 2.2

If f is an automorphism of the disc, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \overline{\alpha} z}.$$

Proof. There is a unique $\alpha \in \mathbb{D}$ with $f(\alpha) = 0$. Let $g = f \circ \psi_\alpha$; then $g(0) = 0$ and by the Schwarz lemma

$$|g(z)| \le |z| \qquad \text{for all } z \in \mathbb{D}. \tag{2}$$

Also $g^{-1}(0) = 0$, so $|g^{-1}(w)| \le |w|$; with $w = g(z)$,

$$|z| \le |g(z)| \qquad \text{for all } z \in \mathbb{D}. \tag{3}$$

By (2), (3), $|g(z)| = |z|$, so $g(z) = e^{i\theta} z$ by the Schwarz lemma. Replacing z by $\psi_\alpha(z)$ and using $\psi_\alpha \circ \psi_\alpha = \mathrm{id}$, $f(z) = e^{i\theta}\psi_\alpha(z)$. ∎

Corollary 2.3

The only automorphisms of the unit disc that fix the origin are the rotations.

The group $\mathrm{Aut}(\mathbb{D})$ acts *transitively*: given $\alpha, \beta \in \mathbb{D}$, $\psi_\beta \circ \psi_\alpha$ maps α to β. $\mathrm{Aut}(\mathbb{D})$ is "almost" isomorphic to the matrix group $SU(1,1)$ of 2×2 complex matrices preserving the hermitian form $\langle Z, W \rangle = z_1\overline{w_1} - z_2\overline{w_2}$ on $\mathbb{C}^2 \times \mathbb{C}^2$.

2.2 Automorphisms of the upper half-plane

With $F : \mathbb{H} \to \mathbb{D}$ from Section 1, define $\Gamma : \mathrm{Aut}(\mathbb{D}) \to \mathrm{Aut}(\mathbb{H})$ by "conjugation by F": $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$. Γ is a bijection with inverse $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$, and it preserves composition:

$$\Gamma(\varphi_1 \circ \varphi_2) = F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F = \Gamma(\varphi_1) \circ \Gamma(\varphi_2).$$

So $\mathrm{Aut}(\mathbb{D}) \cong \mathrm{Aut}(\mathbb{H})$. Let $SL_2(\mathbb{R}) = \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R},\ \det(M) = ad - bc = 1 \right\}$, the **special linear group**, and for $M \in SL_2(\mathbb{R})$ let $f_M(z) = \frac{az+b}{cz+d}$.

Theorem 2.4

Every automorphism of $\mathbb{H}$ takes the form f_M for some $M \in SL_2(\mathbb{R})$. Conversely, every map of this form is an automorphism of $\mathbb{H}$.

Proof. Write $G = SL_2(\mathbb{R})$.

Step 1. If $M \in G$, f_M maps $\mathbb{H}$ to itself:

$$\mathrm{Im}(f_M(z)) = \frac{(ad-bc)\,\mathrm{Im}(z)}{|cz+d|^2} = \frac{\mathrm{Im}(z)}{|cz+d|^2} > 0 \qquad (z \in \mathbb{H}). \tag{4}$$

Step 2. $f_M \circ f_{M'} = f_{MM'}$ (a straightforward calculation). Hence each f_M is an automorphism, with holomorphic inverse $f_{M^{-1}}$: $(f_M \circ f_{M^{-1}})(z) = f_I(z) = z$.

Step 3. G acts transitively on $\mathbb{H}$; it suffices to map any $z \in \mathbb{H}$ to i. With $d = 0$ in (4), $\mathrm{Im}(f_M(z)) = \mathrm{Im}(z)/|cz|^2$, and we choose $c \in \mathbb{R}$ making this 1; let $M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$, so $f_{M_1}(z)$ has imaginary part 1. Then translate by $M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $b \in \mathbb{R}$, to bring it to i; $M = M_2 M_1$ works.

Step 4. For $\theta \in \mathbb{R}$, $M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in G$, and $F \circ f_{M_\theta} \circ F^{-1}$ is the rotation of angle -2θ in the disc, since $F \circ f_{M_\theta} = e^{-2i\theta} F(z)$ (easily verified).

Step 5. Let $f \in \mathrm{Aut}(\mathbb{H})$ with $f(\beta) = i$, and $N \in G$ with $f_N(i) = \beta$. Then $g = f \circ f_N$ fixes i, so $F \circ g \circ F^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, hence a rotation; by Step 4, $F \circ g \circ F^{-1} = F \circ f_{M_\theta} \circ F^{-1}$ for some θ. So $g = f_{M_\theta}$ and $f = f_{M_\theta N^{-1}}$. ∎

M and $-M$ give the same f_M; identifying them gives the **projective special linear group** $PSL_2(\mathbb{R})$, which is isomorphic to $\mathrm{Aut}(\mathbb{H})$.

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

If $\Omega = \mathbb{C}$ there is no conformal map $F : \Omega \to \mathbb{D}$ (by Liouville F would be constant); Ω must be connected (as $\mathbb{D}$ is) and simply connected (as $\mathbb{D}$ is). Remarkably, these conditions suffice. A subset of $\mathbb{C}$ is **proper** if it is non-empty and not all of $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem)

Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \to \mathbb{D}$ such that $F(z_0) = 0$ and $F'(z_0) > 0$.

Corollary 3.2

Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.

The corollary follows using $\mathbb{D}$ as an intermediate step. *Uniqueness*: if F, G satisfy the conditions, $H = F \circ G^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, so $H(z) = e^{i\theta}z$; $H'(0) > 0$ forces $e^{i\theta} = 1$, so $F = G$. For existence, consider all injective holomorphic $f : \Omega \to \mathbb{D}$ with $f(z_0) = 0$, and choose one with $|f'(z_0)|$ as large as possible; it will fill out $\mathbb{D}$. We need to extract limits from sequences of functions.

3.2 Montel's theorem

A family $\mathcal{F}$ of holomorphic functions on Ω is **normal** if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on every compact subset of Ω (the limit need not be in $\mathcal{F}$). $\mathcal{F}$ is **uniformly bounded on compact subsets** of Ω if for each compact $K \subset \Omega$ there is $B > 0$ with $|f(z)| \le B$ for $z \in K$, $f \in \mathcal{F}$. $\mathcal{F}$ is **equicontinuous** on a compact K if for every $\epsilon > 0$ there is $\delta > 0$ such that $z, w \in K$, $|z - w| < \delta$ imply $|f(z) - f(w)| < \epsilon$ for all $f \in \mathcal{F}$. (E.g. differentiable functions on $[0, 1]$ with uniformly bounded derivatives are equicontinuous; $f_n(x) = x^n$ on $[0, 1]$ are not.)

Theorem 3.3 (Montel)

Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω. Then: (i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω. (ii) $\mathcal{F}$ is a normal family.

Part (i) uses the Cauchy integral formula and contrasts with the real case ($\sin(nx)$ on $(0, 1)$ is uniformly bounded but not equicontinuous, with no convergent subsequence on any subinterval). Part (ii) uses only uniform boundedness and equicontinuity (the Arzelà–Ascoli theorem). A sequence $\{K_\ell\}_{\ell=1}^{\infty}$ of compact subsets of Ω is an **exhaustion** if (a) K_ℓ is contained in the interior of $K_{\ell+1}$ for all ℓ, and (b) any compact $K \subset \Omega$ is contained in some K_ℓ; in particular $\Omega = \bigcup_\ell K_\ell$.

Lemma 3.4

Any open set Ω in the complex plane has an exhaustion.

Proof. If Ω is bounded, let K_ℓ be the set of points of Ω at distance $\ge 1/\ell$ from $\partial\Omega$. If Ω is unbounded, require also $|z| \le \ell$ for $z \in K_\ell$. ∎

Proof of Montel's theorem. (i) Let $K \subset \Omega$ be compact and $r > 0$ with $3r < d(K, \partial\Omega)$, so $D_{3r}(z) \subset \Omega$ for $z \in K$. Let $z, w \in K$ with $|z - w| < r$ and γ the boundary circle of $D_{2r}(w)$. By Cauchy's integral formula

$$f(z) - f(w) = \frac{1}{2\pi i}\int_\gamma f(\zeta)\Big[\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big]d\zeta, \qquad \Big|\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big| = \frac{|z - w|}{|\zeta - z||\zeta - w|} \le \frac{|z - w|}{r^2},$$

since $\zeta \in \gamma$ and $|z - w| < r$. Hence $|f(z) - f(w)| \le \frac{1}{2\pi} 2\pi(2r)\frac{B|z-w|}{r^2}$, B the uniform bound for $\mathcal{F}$ on the compact set of points at distance $\le 2r$ from K. So $|f(z) - f(w)| < C|z - w|$ for all $f \in \mathcal{F}$: equicontinuity.

(ii) Let $\{f_n\} \subset \mathcal{F}$ and $K \subset \Omega$ compact. Choose a sequence $\{w_j\}$ dense in Ω. Since $\{f_n\}$ is uniformly bounded, there is a subsequence $\{f_{n,1}\}$ with $f_{n,1}(w_1)$ convergent; from it a subsequence $\{f_{n,2}\}$ with $f_{n,2}(w_2)$ convergent; and so on. The diagonal sequence $g_n = f_{n,n}$ converges at each w_j. It converges

uniformly on K: given $\epsilon > 0$, choose δ as in equicontinuity; K is covered by finitely many discs $D_\delta(w_1), \ldots, D_\delta(w_J)$. Choose N with $|g_m(w_j) - g_n(w_j)| < \epsilon$ for $n, m > N$, $j \le J$. For $z \in K$, $z \in D_\delta(w_j)$ for some $j \le J$, and

$$|g_n(z) - g_m(z)| \le |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + |g_m(w_j) - g_m(z)| < 3\epsilon.$$

Finally, with an exhaustion $K_1 \subset K_2 \subset \cdots$, extract $\{g_{n,1}\}$ converging uniformly on K_1, then $\{g_{n,2}\} \subset \{g_{n,1}\}$ on K_2, and so on; $\{g_{n,n}\}$ converges uniformly on every K_ℓ, hence on every compact subset of Ω. ■

Proposition 3.5

If Ω is a connected open subset of $\mathbb{C}$ and $\{f_n\}$ a sequence of injective holomorphic functions on Ω that converges uniformly on every compact subset of Ω to a holomorphic function f, then f is either injective or constant.

Proof. Suppose $f(z_1) = f(z_2)$ with $z_1 \neq z_2$. Let $g_n(z) = f_n(z) - f_n(z_1)$, which has no zero besides z_1, and $g_n \to g = f - f(z_1)$ uniformly on compact sets. If $g \not\equiv 0$, z_2 is an isolated zero of g (Ω is connected), so

$$1 \le \frac{1}{2\pi i} \int_\gamma \frac{g'(\zeta)}{g(\zeta)}\, d\zeta,$$

γ a small circle about z_2 on and inside which g vanishes only at z_2. Then $1/g_n \to 1/g$ and $g_n' \to g'$ uniformly on γ (Theorem 5.3), so $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} \to \frac{1}{2\pi i}\int_\gamma \frac{g'}{g}$. But g_n has no zeros inside γ, so $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} = 0$ for all n, a contradiction. Hence $g \equiv 0$ and f is constant. ■

3.3 Proof of the Riemann mapping theorem

Step 1. Ω is conformally equivalent to an open subset of $\mathbb{D}$ containing 0. Choose $\alpha \notin \Omega$. $z - \alpha$ never vanishes on the simply connected Ω, so there is a holomorphic $f(z) = \log(z - \alpha)$ with $e^{f(z)} = z - \alpha$; in particular f is injective. Fix $w \in \Omega$; then $f(z) \neq f(w) + 2\pi i$ for $z \in \Omega$ (exponentiating would give $z = w$, hence $f(z) = f(w)$). In fact $f(\Omega)$ stays away from $f(w) + 2\pi i$: if $f(z_n) \to f(w) + 2\pi i$, exponentiating gives $z_n \to w$, hence $f(z_n) \to f(w)$, a contradiction. Thus

$$F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$$

is injective and bounded on Ω, so $F : \Omega \to F(\Omega)$ is conformal with $F(\Omega)$ bounded; translating and rescaling gives a conformal map of Ω onto an open subset of $\mathbb{D}$ containing 0.

Step 2. So assume $\Omega \subset \mathbb{D}$ is open with $0 \in \Omega$, and let

$$\mathcal{F} = \{f : \Omega \to \mathbb{D} \text{ holomorphic, injective, and } f(0) = 0\}.$$

$\mathcal{F} \neq \emptyset$ (it contains the identity) and is uniformly bounded. The numbers $|f'(0)|$, $f \in \mathcal{F}$, are uniformly bounded by the Cauchy inequalities on a small disc about 0. Let $s = \sup_{f \in \mathcal{F}} |f'(0)|$ and $f_n \in \mathcal{F}$ with $|f_n'(0)| \to s$. By Montel's theorem a subsequence converges uniformly on compact sets to a holomorphic f on Ω. Since $s \ge 1$ (the identity is in $\mathcal{F}$), f is non-constant, hence injective by Proposition 3.5. By continuity $|f(z)| \le 1$, and by the maximum modulus principle $|f(z)| < 1$; also $f(0) = 0$. So $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3. f is surjective onto $\mathbb{D}$. Otherwise we construct a function in $\mathcal{F}$ with derivative at 0 greater than s. Suppose $\alpha \in \mathbb{D}$ is not in $f(\Omega)$, and let $\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}$. $U = (\psi_\alpha \circ f)(\Omega)$ is simply connected and does not contain 0, so a square root $g(w) = e^{\frac{1}{2}\log w}$ is defined on U. Let

$$F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f.$$

F is holomorphic, maps 0 to 0 ($g(\psi_\alpha(0)) = g(\alpha)$), maps Ω into $\mathbb{D}$, and is injective (each factor is). So $F \in \mathcal{F}$. With $h(w) = w^2$,

$$f = \psi_\alpha^{-1} \circ h \circ g^{-1} \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F.$$

$\Phi = \psi_\alpha \circ h \circ \psi_{g(\alpha)}$ maps $\mathbb{D}$ into $\mathbb{D}$ with $\Phi(0) = 0$, and is not injective (since h is not). By the last part of the Schwarz lemma, $|\Phi'(0)| < 1$. Since $f'(0) = \Phi'(0)F'(0)$, $|f'(0)| < |F'(0)|$, contradicting the maximality of $|f'(0)|$.

Finally multiply f by a constant of absolute value 1 so that $f'(0) > 0$. ■

Remark

Simple connectivity was used only for the logarithm and the square root. It would suffice to assume Ω proper and *holomorphically simply connected*: $\int_\gamma f(z)\,dz = 0$ for every holomorphic f in Ω and every closed curve $\gamma \subset \Omega$. See Appendix B.

4 Conformal mappings onto polygons

The Riemann mapping theorem gives little insight into the exact form of the map; but for polygons there are nice formulas: the *Schwarz–Christoffel formula* for conformal maps from the disc (or $\mathbb{H}$) to polygons.

4.1 Some examples

Example 1

For $0 < \alpha < 2$, the map $f(z) = z^\alpha$ of $\mathbb{H}$ onto the sector $\{0 < \arg z < \alpha\pi\}$ (Section 1) can be written

$$z^\alpha = f(z) = \int_0^z f'(\zeta)\,d\zeta = \alpha \int_0^z \zeta^{-\beta}\,d\zeta, \qquad \alpha + \beta = 1,$$

integrated along any path in the (closed) upper half-plane. $\zeta^{-\beta}$ is integrable near 0 since $\beta < 1$, so $f(0) = 0$. For $z = x > 0$, $f'(x) = \alpha x^{\alpha-1} > 0$ and is not integrable at ∞, so f maps $[0, \infty)$ onto $[0, \infty)$. For $x < 0$, $f'(x) = \alpha|x|^{\alpha-1}e^{i\pi(\alpha-1)} = -\alpha|x|^{\alpha-1}e^{i\pi\alpha}$, so f maps $(-\infty, 0]$ onto $(\infty e^{i\pi\alpha}, 0]$.

Example 2

For $z \in \mathbb{H}$ let $f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}}$, with the branch of $(1-\zeta^2)^{1/2}$ holomorphic in $\mathbb{H}$ and positive for $-1 < \zeta < 1$; then $(1-\zeta^2)^{-1/2} = -i(\zeta^2-1)^{-1/2}$ for $\zeta > 1$ (and the analogous formula for $\zeta < -1$). f maps $\mathbb{R}$ onto the boundary of the half-strip $\{-\pi/2 < \mathrm{Re} < \pi/2,\ \mathrm{Im} > 0\}$: $f(\pm 1) = \pm\pi/2$ and $f'(x) > 0$ on $(-1, 1)$, so $[-1, 1] \mapsto [-\pi/2, \pi/2]$; for $x > 1$, $f(x) = \frac{\pi}{2} + i\int_1^x \frac{dx}{(x^2-1)^{1/2}}$ and $\int_1^\infty \frac{dx}{(x^2-1)^{1/2}} = \infty$, so $[1, \infty)$ goes onto the vertical half-line above $\pi/2$; similarly for $(-\infty, -1]$. In fact f is the inverse of $\sin z$ (Example 8), so f maps $\mathbb{H}$ conformally onto the half-strip.

Example 3

For fixed $0 < k < 1$, let

$$f(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}, \qquad z \in \mathbb{H},$$

the branch positive for $-1 < \zeta < 1$. Such integrals are **elliptic integrals** (variants arise in the arc length of an ellipse). With

$$K = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K' = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}},$$

f maps $\mathbb{R}$ onto the rectangle with vertices $-K$, K, $K+iK'$, $-K+iK'$. Divide $\mathbb{R}$ by $-1/k, -1, 1, 1/k$. $f(\pm 1) = \pm K$ and $f' > 0$ on $(-1,1)$, so $[-1,1] \mapsto [-K,K]$. For $1 < x < 1/k$, $f(x) = K + i\int_1^x \frac{d\xi}{[(\xi^2-1)(1-k^2\xi^2)]^{1/2}}$, so $[1,1/k] \mapsto [K, K+iK']$; similarly $[-1/k,-1] \mapsto [-K+iK', -K]$. For $x > 1/k$, $f'(x) = -[(x^2-1)(k^2x^2-1)]^{-1/2}$, so $f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}}$, and since $\int_{1/k}^\infty \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}} = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}} = K$ (substitute $x = 1/ku$), $[1/k, \infty)$ goes onto $[K+iK', iK')$; similarly $(-\infty, -1/k]$ onto $[-K+iK', iK')$. The point at infinity goes to the midpoint iK' of the upper side.

Two problems arise: generalize these examples (the Schwarz–Christoffel integral maps $\mathbb{R}$ to a polygonal line), and show that the map is conformal from $\mathbb{H}$ onto the interior of the polygon.

4.2 The Schwarz–Christoffel integral

Definition (Schwarz–Christoffel integral)

For real $A_1 < A_2 < \cdots < A_n$ and exponents $\beta_k < 1$ with $1 < \sum_{k=1}^n \beta_k$, let

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}, \tag{5}$$

where $(z-A_k)^{\beta_k}$ is the branch in the plane slit along $\{A_k + iy : y \le 0\}$ that is positive for real $x > A_k$; thus $(x-A_k)^{\beta_k} = |x-A_k|^{\beta_k} e^{i\pi\beta_k}$ for real $x < A_k$.

The plane slit along the rays $\bigcup_k \{A_k + iy : y \le 0\}$ is simply connected, so S is holomorphic there. Since $\beta_k < 1$, the singularities are integrable, and S is continuous up to $\mathbb{R}$, including the A_k; the integral may be taken along any path avoiding the open slits. Since $\prod_k |\zeta - A_k|^{-\beta_k} \le c|\zeta|^{-\sum \beta_k}$ for large $|\zeta|$ and $\sum \beta_k > 1$, the integral converges at infinity, and by Cauchy's theorem $\lim_{r\to\infty} S(re^{i\theta})$ exists and is independent of $0 \le \theta \le \pi$; call it a_∞, and let $a_k = S(A_k)$. (The cases $\sum \beta_k \le 1$ of Examples 1–2 are excluded; a modification handles them, but then S is unbounded in $\mathbb{H}$.)

Proposition 4.1

Suppose $S(z)$ is given by (5). (i) If $\sum_{k=1}^n \beta_k = 2$, and p denotes the polygon whose vertices are given (in order) by $a_1, \ldots, a_n$, then S maps the real axis onto $p - \{a_\infty\}$. The point a_∞ lies on the segment $[a_n, a_1]$ and is the image of the point at infinity. Moreover, the (interior) angle at the vertex a_k is $\alpha_k \pi$ where $\alpha_k = 1 - \beta_k$. (ii) There is a similar conclusion when $1 < \sum_{k=1}^n \beta_k < 2$, except now the image of the extended line is the polygon of $n+1$ sides with vertices $a_1, a_2, \ldots, a_n, a_\infty$. The angle at the vertex a_∞ is $\alpha_\infty \pi$ with $\alpha_\infty = 1 - \beta_\infty$, where $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.

Proof. Let $\sum \beta_k = 2$. For $A_k < x < A_{k+1}$,

$$S'(x) = \prod_{j \le k} (x - A_j)^{-\beta_j} \prod_{j > k} (x - A_j)^{-\beta_j}, \qquad \arg S'(x) = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

constant on (A_k, A_{k+1}). Since $S(x) = S(A_k) + \int_{A_k}^x S'(y)\,dy$, as x goes from A_k to A_{k+1}, $S(x)$ moves along the straight segment $[a_k, a_{k+1}]$ (where $[z,w] = \{(1-t)z + tw : t \in [0,1]\}$), which makes angle

$-\pi\sum_{j>k}\beta_j$ with the real axis. For $x > A_n$, $S'(x) > 0$; for $x < A_1$, $\arg S'(x) = -\pi\sum_k \beta_k = -2\pi$, so $S'(x) > 0$ again. Thus on $[A_n, \infty)$, S moves parallel to the real axis from a_n to a_∞, and on $(-\infty, A_1]$ from a_∞ to a_1; $[a_n, a_\infty) \cup (a_\infty, a_1] = [a_n, a_1] - \{a_\infty\}$. The angle of $[a_k, a_{k+1}]$ exceeds that of $[a_{k-1}, a_k]$ by $\pi\beta_k$, so the interior angle at a_k is $\pi - \pi\beta_k = \alpha_k\pi$. The case $1 < \sum \beta_k < 2$ is similar. ■

This does not settle the problem: (1) for general n and A_k the image polygon need not be simple, nor S conformal on $\mathbb{H}$; (2) it does not show that for a given simply connected region P bounded by a polygon, S (with suitable A_k and simple modifications) is a conformal map of $\mathbb{H}$ onto P. That is the case, as we now prove.

4.3 Boundary behavior

A **polygonal region** P is a bounded simply connected open set whose boundary is a closed polygonal line p (a *polygon*). For a conformal $f : U \to f(U)$,

$$\operatorname{Area}(f(U)) = \iint_U |f'(z)|^2\, dx\, dy,$$

since the Jacobian of $w = f(z)$ is $|f'(z)|^2$ (Proposition 2.3).

Theorem 4.2

If $F : \mathbb{D} \to P$ is a conformal map, then F extends to a continuous bijection from the closure $\overline{\mathbb{D}}$ of the disc to the closure $\overline{P}$ of the polygonal region. In particular, F gives rise to a bijection from the boundary of the disc to the boundary polygon p.

Lemma 4.3

For each $0 < r < 1/2$, let C_r denote the circle centered at z_0 (a point of the unit circle) of radius r. Suppose that for all sufficiently small r we are given two points z_r and z_r' in the unit disc that also lie on C_r. If we let $\rho(r) = |F(z_r) - F(z_r')|$, then there exists a sequence $\{r_n\}$ of radii that tends to zero, and such that $\lim_{n\to\infty} \rho(r_n) = 0$.

Proof. If not, there are $c > 0$, $0 < R < 1/2$ with $c \le \rho(r)$ for $0 < r \le R$. Now $F(z_r) - F(z_r') = \int_\gamma F'(\zeta)\, d\zeta$, γ the arc of C_r in $\mathbb{D}$ joining z_r, z_r'; parametrizing it by $z_0 + re^{i\theta}$, $\theta_1(r) \le \theta \le \theta_2(r)$,

$$\rho(r) \le \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)| r\, d\theta \le \Big(\int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\, d\theta\Big)^{1/2} \Big(\int_{\theta_1(r)}^{\theta_2(r)} r\, d\theta\Big)^{1/2}$$

by Cauchy–Schwarz. Squaring and dividing by r, $\frac{\rho(r)^2}{r} \le 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\, d\theta$. Integrating from 0 to R,

$$c^2 \int_0^R \frac{dr}{r} \le 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\, d\theta\, dr \le 2\pi \iint_{\mathbb{D}} |F'(z)|^2\, dx\, dy = 2\pi \operatorname{Area}(P).$$

The left side is infinite and the right side finite—a contradiction. ■

Lemma 4.4

Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.

Proof. If not, there are sequences $z_n, z_n' \to z_0$ in $\mathbb{D}$ with $F(z_n) \to \zeta$, $F(z_n') \to \zeta'$, distinct points of $\overline{P}$; since F is conformal, they lie on p. Choose disjoint discs D, D' about ζ, ζ' at distance $d > 0$. For large n, $F(z_n) \in D$, $F(z_n') \in D'$, so there are continuous curves $\Lambda \subset D \cap P$, $\Lambda' \subset D' \cap P$ (images of continuous maps of a closed interval) containing $F(z_n)$, $F(z_n')$ for all large n, with end-points ζ,

ζ'. Then $\lambda = F^{-1}(\Lambda)$ and $\lambda' = F^{-1}(\Lambda')$ are continuous curves in $\mathbb{D}$, containing infinitely many z_n, resp. z'_n, which tend to z_0. By continuity, for all small r the circle C_r meets λ and λ' at points z_r, z'_r, and $|F(z_r) - F(z'_r)| > d$, contradicting Lemma 4.3. ■

Lemma 4.5

The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.

Proof. Define $F(z_0) = \lim_{z\to z_0} F(z)$ for $|z_0| = 1$ (Lemma 4.4). Given ϵ, there is δ with $|F(z) - F(z_0)| < \epsilon$ for $z \in \mathbb{D}$, $|z - z_0| < \delta$. If $|z| = 1$ and $|z - z_0| < \delta$, choose $w \in \mathbb{D}$ with $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$; then $|F(z) - F(z_0)| \le |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon$. ■

Proof of Theorem 4.2. F extends continuously to $\overline{\mathbb{D}} \to \overline{P}$. The argument applies to the inverse G of F: the key geometric property used was that for $z_0 \in \partial\mathbb{D}$ and a small circle C about z_0, $C \cap \mathbb{D}$ is an arc; this holds at every boundary point of P. So G extends continuously to $\overline{P} \to \overline{\mathbb{D}}$. If $z \in \overline{\mathbb{D}}$ and $z_k \to z$ in $\mathbb{D}$, then $G(F(z_k)) = z_k$, and by continuity $G(F(z)) = z$; similarly $F(G(w)) = w$ for $w \in \overline{P}$. ■

4.4 The mapping formula

Let P be a polygonal region bounded by p with consecutive vertices $a_1, \dots, a_n$, $n \ge 3$, interior angle $\alpha_k\pi$ at a_k, and exterior angle $\beta_k\pi$ with $\alpha_k + \beta_k = 1$; a simple geometric argument gives $\sum_{k=1}^{n} \beta_k = 2$. The correspondence $w = (i - z)/(i + z)$ between $\mathbb{H}$ and $\mathbb{D}$ sends the point at infinity to $w = -1$, so a conformal map of $\mathbb{H}$ onto P (which exists by the Riemann mapping theorem) extends to a continuous bijection of $\mathbb{R} \cup \{\infty\}$ onto p. Let F be such a map, and first assume no vertex corresponds to ∞: $F(A_k) = a_k$ for real A_k. Since F is continuous and injective and the vertices are consecutive, after relabeling $A_1 < A_2 < \cdots < A_n$, and F maps $[A_k, A_{k+1}]$ onto $[a_k, a_{k+1}]$ and $(-\infty, A_1] \cup [A_n, \infty)$ onto $[a_n, a_1]$.

Theorem 4.6 (Schwarz–Christoffel)

There exist complex numbers c_1 and c_2 so that the conformal map F of $\mathbb{H}$ to P is given by $F(z) = c_1 S(z) + c_2$, where S is the Schwarz–Christoffel integral introduced in Section 4.2.

Proof. Let $1 < k < n$. F maps $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$ onto two segments meeting at $a_k = F(A_k)$ at angle $\alpha_k\pi$. Choosing a branch of the logarithm, define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

on the half-strip in $\mathbb{H}$ between $\mathrm{Re}(z) = A_{k-1}$ and $\mathrm{Re}(z) = A_{k+1}$. h_k is continuous up to (A_{k-1}, A_{k+1}) and maps $[A_{k-1}, A_{k+1}]$ onto a straight segment L_k, with $A_k \mapsto 0$. By the Schwarz reflection principle, h_k continues analytically to the whole strip $A_{k-1} < \mathrm{Re}(z) < A_{k+1}$. h'_k never vanishes there: in the upper half-strip,

$$\frac{F'(z)}{F(z) - F(A_k)} = \frac{1}{\alpha_k}\frac{h'_k(z)}{h_k(z)},$$

and $F'(z) \neq 0$ (Proposition 1.1), so $h'_k(z) \neq 0$; by reflection the same holds in the lower half-strip. For $A_{k-1} < x < A_{k+1}$, the image under h_k of a small half-disc about x in $\mathbb{H}$ lies on one side of L_k; h_k is injective up to L_k (since F is), and by the symmetry of reflection h_k is injective on the whole disc about x, so $h'_k(x) \neq 0$.

Since $F = a_k + h_k^{\alpha_k}$, $F' = \alpha_k h_k^{\alpha_k - 1} h'_k$ and $F'' = \alpha_k(\alpha_k - 1)h_k^{\alpha_k - 2}(h'_k)^2 + \alpha_k h_k^{\alpha_k - 1} h''_k$; as $h'_k \neq 0$ and h_k has a simple zero at A_k,

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

E_k holomorphic in the strip $A_{k-1} < \mathrm{Re}(z) < A_{k+1}$. Similarly $\frac{F''}{F'} = \frac{-\beta_1}{z-A_1} + E_1$ in $-\infty < \mathrm{Re}(z) < A_2$ and $\frac{F''}{F'} = \frac{-\beta_n}{z-A_n} + E_n$ in $A_{n-1} < \mathrm{Re}(z) < \infty$. Moreover F continues across $(-\infty, A_1) \cup (A_n, \infty)$ by reflection (the image is a straight segment), so F continues to $|z| > R$ for large R; since F maps $\mathbb{H}$ into a bounded region, the continuation is bounded, hence holomorphic at infinity: $F(z) = c_0 + \frac{c_1}{z} + \frac{c_2}{z^2} + \cdots$. Differentiating, F''/F' decays like $1/z$ at infinity. Since the strips overlap and cover $\mathbb{C}$,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^{n} \frac{\beta_k}{z - A_k}$$

is entire and vanishes at infinity; by Liouville it is zero. With $Q(z) = (z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$, $\frac{Q'}{Q} = -\sum_k \frac{\beta_k}{z-A_k} = \frac{F''}{F'}$, so $\frac{d}{dz}\left(\frac{F'(z)}{Q(z)}\right) = 0$ and $F' = cQ$. A final integration yields the theorem. ■

Theorem 4.7

If F is a conformal map from the upper half-plane to the polygonal region P and maps the points $A_1, \ldots, A_{n-1}, \infty$ to the vertices of p, then there exist constants C_1 and C_2 such that

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

So the formula is obtained by deleting the last factor of the Schwarz–Christoffel integral.

Proof. After a translation assume $A_j \neq 0$ for $j = 1, \ldots, n-1$. Choose a real $A_n^* > 0$ (supplied: different from the others) and let $\Phi(z) = A_n^* - \frac{1}{z}$, an automorphism of $\mathbb{H}$. Let $A_k^* = \Phi(A_k)$, $k \leq n-1$, and note $A_n^* = \Phi(\infty)$. Then $(F \circ \Phi^{-1})(A_k^*) = a_k$ for $k = 1, \ldots, n$, and by Theorem 4.6

$$(F \circ \Phi^{-1})(z') = C_1' \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2'.$$

Change variables $\zeta = \Phi(w)$, $d\zeta = dw/w^2$; since $\zeta - A_n^* = -1/w$ and $\zeta - A_k^* = (w(A_n^* - A_k^*) - 1)/w$, and $2 = \beta_1 + \cdots + \beta_n$, the integrand becomes $C \prod_{k<n}(w - 1/(A_n^* - A_k^*))^{-\beta_k}$ up to a constant factor. Since $1/(A_n^* - A_k^*) = A_k$, setting $\Phi^{-1}(z') = z$ gives the formula. ■

4.5 Return to elliptic integrals

Consider again $I(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}$, $0 < k < 1$, which maps $\mathbb{R}$ onto the rectangle R with vertices $-K, K, K+iK', -K+iK'$ (Example 3). We show I is a conformal map of $\mathbb{H}$ onto the interior of R. By Theorem 4.6 there is a conformal map F of $\mathbb{H}$ onto R mapping four real points to the vertices; composing with an automorphism of $\mathbb{H}$ we may assume F maps $-1, 0, 1$ to $-K, 0, K$. Let $1/\ell$ $(0 < \ell < 1)$ be the real point mapped to $K + iK'$. Then $F(-1/\ell) = -K + iK'$: $\tilde{F}(z) = -\overline{F(-\bar{z})}$ is, by the symmetry of R, also a conformal map of $\mathbb{H}$ onto R with $\tilde{F}(0) = 0$, $\tilde{F}(\pm 1) = \pm K$; so $F^{-1} \circ \tilde{F}$ is an automorphism of $\mathbb{H}$ fixing $-1, 0, 1$, hence the identity, and $F(-1/\ell) = -\overline{F(1/\ell)} = -K + iK'$. By Theorem 4.6,

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-\ell^2\zeta^2)]^{1/2}} + c_2.$$

$z = 0$ gives $c_2 = 0$; $z = 1$ and $z = 1/\ell$ give $K(k) = c_1 K(\ell)$ and $K'(k) = c_1 K'(\ell)$, where $K(k) = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}$, $K'(k) = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}}$. $K(k)$ is strictly increasing in $k \in (0,1)$, and a change of variables gives $K'(k) = K(\tilde{k})$ with $\tilde{k}^2 = 1 - k^2$, $\tilde{k} > 0$, so K' is strictly decreasing. Hence $K(k)/K'(k)$ is strictly increasing; since $K(k)/K'(k) = K(\ell)/K'(\ell)$, $k = \ell$ and $c_1 = 1$. Thus $I = F$ is conformal.

Passing to the inverse function gives basic insight: let $z \mapsto \mathrm{sn}(z)$ be the inverse of $z \mapsto I(z)$ (Jacobi's notation, by analogy with $\sin z$). It maps the closed rectangle $R_0 = R$ onto the closed upper half-plane. Reflect successively along the lower sides to get rectangles $R_1, R_2, \ldots$: by the reflection

principle, $\mathrm{sn}(z) = \overline{\mathrm{sn}(\overline{z})}$ for $z \in R_1$, then $\mathrm{sn}(z) = \mathrm{sn}(-iK' + \overline{z})$ for $z \in R_2$, and so on; this extends sn to the strip $-K < \mathrm{Re}(z) < K$ with $\mathrm{sn}(z) = \mathrm{sn}(z + 2iK')$. Reflecting also in horizontal rows of rectangles, sn continues to all of $\mathbb{C}$ with $\mathrm{sn}(z) = \mathrm{sn}(z+4K)$. Thus sn is *doubly periodic*, with periods $4K$ and $2iK'$ (and meromorphic, with two simple poles in a period parallelogram)—the subject of Chapter 9.

Chapter 9. An Introduction to Elliptic Functions

Overview

Elliptic functions grew out of the study of *elliptic integrals* $\int R(x, \sqrt{P(x)})\,dx$, R rational and P a polynomial of degree three or four (degree two reduces to the circular functions), which arose in computing the arc length of an ellipse or lemniscate. Their early study centered on transformation properties and an inherent double periodicity (cf. the map of $\mathbb{H}$ onto a rectangle, Section 4). Jacobi began the systematic study of doubly periodic functions (*elliptic functions*), with theta functions playing a decisive role; Weierstrass's approach, based on his function $\wp$, is simpler in its first steps, and we sketch it here, reaching the *Eisenstein series* and their expression via divisor functions, a link with number theory. Theta functions and further number-theoretic applications follow in Chapter 10.

1 Elliptic functions

We consider meromorphic f on $\mathbb{C}$ with two periods: non-zero ω_1, ω_2 with $f(z+\omega_1) = f(z)$ and $f(z+\omega_2) = f(z)$ for all z; f is **doubly periodic**. If $\omega_2/\omega_1 \in \mathbb{R}$, f is either simply periodic (quotient rational) or constant (quotient irrational), so we assume ω_1, ω_2 are linearly independent over $\mathbb{R}$. *Normalization*: let $\tau = \omega_2/\omega_1$; since τ and $1/\tau$ have imaginary parts of opposite signs, after possibly interchanging ω_1, ω_2 we may assume $\operatorname{Im}(\tau) > 0$. f has periods ω_1, ω_2 iff $F(z) = f(\omega_1 z)$ has periods 1 and τ, and F is meromorphic iff f is. So we assume f is meromorphic on $\mathbb{C}$ with periods 1 and τ, $\operatorname{Im}(\tau) > 0$. Then

$$f(z+n+m\tau) = f(z) \qquad \text{for all integers } n, m \text{ and all } z \in \mathbb{C}, \tag{1}$$

and we consider the **lattice** $\Lambda = \{n+m\tau : n, m \in \mathbb{Z}\}$ **generated** by 1 and τ, and the **fundamental parallelogram**

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau,\ 0 \le a < 1,\ 0 \le b < 1\}.$$

z and w are **congruent modulo** Λ, $z \sim w$, if $z - w \in \Lambda$; by (1), $f(z) = f(w)$ when $z \sim w$. Every $z \in \mathbb{C}$ is congruent to a unique point of P_0: write $z = a + b\tau$ with $a, b \in \mathbb{R}$ (1 and τ form a real basis of $\mathbb{C}$), let n, m be the greatest integers $\le a$, $\le b$, and $w = z - n - m\tau = (a-n) + (b-m)\tau \in P_0$. If $w = a + b\tau$ and $w' = a' + b'\tau$ in P_0 are congruent, then $a - a'$, $b - b'$ are integers in $(-1, 1)$, hence 0. A **period parallelogram** is any translate $P = P_0 + h$; applying the above to $z - h$, every point is congruent to a unique point of P. Moreover

$$\mathbb{C} = \bigcup_{n,m \in \mathbb{Z}} (n + m\tau + P_0), \tag{2}$$

a disjoint union.

Proposition 1.1

Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ. Then: (i) Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram. (ii) Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram. (iii) The lattice Λ provides a disjoint covering of the complex plane, in the sense of (2). (iv) The function f is completely determined by its values in any period parallelogram.

1.1 Liouville's theorems

Theorem 1.2

An entire doubly periodic function is constant.

Proof. It is determined by its values on P_0, whose closure is compact; so it is bounded on $\mathbb{C}$, hence constant by Liouville's theorem. ∎

A non-constant doubly periodic meromorphic function is an **elliptic function**. It has finitely many zeros and poles in any period parallelogram (possibly some on ∂P_0); we count them with multiplicities.

Theorem 1.3

The total number of poles of an elliptic function in P_0 is always ≥ 2.

So f cannot have just one simple pole (a single double pole is allowed).

Proof. Suppose first f has no poles on ∂P_0. By the residue theorem $\int_{\partial P_0} f(z)\,dz = 2\pi i \sum \operatorname{res} f$, and the integral is 0 by periodicity:

$$\int_{\partial P_0} f = \int_0^1 f + \int_1^{1+\tau} f + \int_{1+\tau}^{\tau} f + \int_{\tau}^{0} f,$$

and opposite sides cancel; e.g.

$$\int_0^1 f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz = \int_0^1 f(z)\,dz + \int_1^0 f(s+\tau)\,ds = \int_0^1 f(z)\,dz - \int_0^1 f(s)\,ds = 0.$$

So $\sum \operatorname{res} f = 0$, and f (having a pole) must have at least two poles in P_0. If f has a pole on ∂P_0, choose a small h with no poles on $\partial(h + P_0)$ and argue with $P = h + P_0$. ∎

The total number of poles (with multiplicities) of an elliptic function is its **order**.

Theorem 1.4

Every elliptic function of order m has m zeros in P_0.

Proof. If f has no zeros or poles on ∂P_0, the argument principle gives $\int_{\partial P_0} \frac{f'}{f} = 2\pi i(N_z - N_p)$, and by periodicity (as above) the integral is 0, so $N_z = N_p$. Otherwise apply the argument to a translate of P_0. ∎

Consequently, for every $c \in \mathbb{C}$, $f(z) = c$ has as many solutions in P_0 as the order of f ($f - c$ is elliptic with the same poles).

1.2 The Weierstrass $\wp$ function

An elliptic function of order two. We construct an elliptic function whose only singularities are double poles at the lattice points. For one period, $\sum_{n=-\infty}^{\infty} \frac{1}{z+n}$ is unchanged under $z \mapsto z+1$ with poles at the integers but is not absolutely convergent; summing symmetrically,

$$F(z) = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \Big[\frac{1}{z+n} + \frac{1}{z-n}\Big],$$

the bracket being $O(1/n^2)$; $F(z) = \pi \cot \pi z$ (Chapter 5). Alternatively, write it as $\frac{1}{z} + \sum_{n\neq 0} \big[\frac{1}{z+n} - \frac{1}{n}\big]$, absolutely convergent as $\frac{1}{z+n} - \frac{1}{n} = O(1/n^2)$, and equal to the former since $\big[\frac{1}{z+n} - \frac{1}{n}\big] + \big[\frac{1}{z-n} -$

$\frac{1}{-n}] = \frac{1}{z+n} + \frac{1}{z-n}$. Mimicking the second way: $\sum_{\omega\in\Lambda} \frac{1}{(z+\omega)^2}$ does not converge absolutely, so with $\Lambda^* = \Lambda - \{0\}$ consider

$$\frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big], \qquad \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2-2z\omega}{(z+\omega)^2\omega^2} = O\Big(\frac{1}{|\omega|^3}\Big) \text{ as } |\omega|\to\infty.$$

Lemma 1.5

The two series

$$\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} \qquad \text{and} \qquad \sum_{n+m\tau\in\Lambda^*} \frac{1}{|n+m\tau|^r}$$

converge if $r > 2$.

(By the Note at the end of Chapter 7, absolute convergence of a double series does not depend on the order of summation; we sum first in m, then in n.)

Proof. For $n \neq 0$, by integral comparison ($1/k^r \le \int_{k-1}^{k} dx/x^r$),

$$\sum_{m\in\mathbb{Z}} \frac{1}{(|n|+|m|)^r} = \frac{1}{|n|^r} + 2\sum_{k\ge|n|+1} \frac{1}{k^r} \le \frac{1}{|n|^r} + 2\int_{|n|}^{\infty} \frac{dx}{x^r} \le \frac{1}{|n|^r} + \frac{C}{|n|^{r-1}},$$

so for $r > 2$, $\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} = \sum_{|m|\neq 0} \frac{1}{|m|^r} + \sum_{|n|\neq 0}\sum_m \frac{1}{(|n|+|m|)^r} < \infty$. For the second series it suffices that $|n|+|m| \le c|n+m\tau|$. Write $A \approx B$ if $A \lesssim B$ and $B \lesssim A$ ($x \lesssim y$ meaning $x \le ay$ for some $a > 0$). For positive A, B, $(A^2+B^2)^{1/2} \approx A + B$ (squaring gives one direction, and $A, B \le (A^2+B^2)^{1/2}$ the other). With $\tau = s + it$, $t > 0$,

$$|n+m\tau| = [(n+ms)^2 + (mt)^2]^{1/2} \approx |n+ms| + |mt| \approx |n+ms| + |m|,$$

and $|n+ms| + |m| \approx |n| + |m|$, considering separately $|n| \ge 2|m||s|$ and $|n| < 2|m||s|$. ∎

Remark

The proof shows that for $r > 2$ the series $\sum |n+m\tau|^{-r}$ converges uniformly in every half-plane $\mathrm{Im}(\tau) \ge \delta > 0$. For $r = 2$ it diverges.

The **Weierstrass $\wp$ function** is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big] = \frac{1}{z^2} + \sum_{(n,m)\neq(0,0)} \Big[\frac{1}{(z+n+m\tau)^2} - \frac{1}{(n+m\tau)^2}\Big].$$

It is meromorphic with double poles at the lattice points: for $|z| < R$ split the sum into $|\omega| \le 2R$ and $|\omega| > 2R$; the terms of the second sum are $O(1/|\omega|^3)$ uniformly, so by Lemma 1.5 it is holomorphic in $|z| < R$, and the finite first sum exhibits the double poles.

Theorem 1.6

The function $\wp$ is an elliptic function that has periods 1 and τ, and double poles at the lattice points.

Proof. It remains to prove periodicity. Differentiating termwise,

$$\wp'(z) = -2\sum_{n,m\in\mathbb{Z}} \frac{1}{(z+n+m\tau)^3},$$

which converges absolutely for $z \notin \Lambda$ ($r = 3$ in Lemma 1.5) and is clearly periodic with periods 1, τ, the terms $1/\omega^2$ having disappeared. So $\wp(z+1) = \wp(z) + a$ and $\wp(z+\tau) = \wp(z) + b$. $\wp$ is even (replace the sum over ω by the sum over $-\omega$), so $\wp(-1/2) = \wp(1/2)$ and $\wp(-\tau/2) = \wp(\tau/2)$; setting $z = -1/2$ and $z = -\tau/2$ gives $a = b = 0$. ∎

Properties of $\wp$. $\wp$ is even, so $\wp'$ is odd; with periodicity,

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\Big(\frac{1+\tau}{2}\Big) = 0,$$

e.g. $\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2+1) = -\wp'(1/2)$. Since $\wp'$ is elliptic of order 3, the **half-periods** $1/2$, $\tau/2$, $(1+\tau)/2$ are its only roots in P_0, each of multiplicity 1. Let

$$\wp(1/2) = e_1, \qquad \wp(\tau/2) = e_2, \qquad \wp\Big(\frac{1+\tau}{2}\Big) = e_3.$$

Then $\wp(z) = e_1$ has a double root at $1/2$, and since $\wp$ has order 2, no other solutions in P_0; similarly for e_2, e_3. In particular e_1, e_2, e_3 are distinct (else $\wp$ would take a value four times).

Theorem 1.7

The function $(\wp')^2$ is the cubic polynomial in $\wp$

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. $F(z) = (\wp(z)-e_1)(\wp(z)-e_2)(\wp(z)-e_3)$ has, in P_0, only double roots at $1/2$, $\tau/2$, $(1+\tau)/2$; so does $(\wp')^2$. Both have poles of order 6 at the lattice points. So $(\wp')^2/F$ is holomorphic and doubly periodic, hence constant. Near 0, $\wp(z) = \frac{1}{z^2} + \cdots$ and $\wp'(z) = \frac{-2}{z^3} + \cdots$, so the constant is 4. ∎

$\wp$ is universal:

Theorem 1.8

Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.

Lemma 1.9

Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.

Proof. A zero or pole of the even function F at 0 has even order, so for some integer m, $\wp^m F$ has no zero or pole at the lattice points; we may assume F itself has none. We construct a doubly periodic G with the same zeros and poles as F, recalling that $\wp(z)-\wp(a)$ has a single zero of order 2 if a is a half-period, and two distinct zeros a, $-a$ otherwise. If a is a zero of F, so is $-a$ (F is even); $a \sim -a$ iff a is a half-period, in which case the zero has even order. So if $a_1, -a_1, \ldots, a_m, -a_m$ (with multiplicities; for a half-period a_j, each of a_j, $-a_j$ carries half the multiplicity) are all the zeros of F in a period parallelogram, $[\wp(z) - \wp(a_1)]\cdots[\wp(z) - \wp(a_m)]$ has precisely the same roots as F. Similarly with the poles $b_1, -b_1, \ldots, b_m, -b_m$,

$$G(z) = \frac{[\wp(z) - \wp(a_1)]\cdots[\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)]\cdots[\wp(z) - \wp(b_m)]}$$

is periodic with the same zeros and poles as F. So F/G is holomorphic and doubly periodic, hence constant. ∎

Proof of Theorem 1.8. Write $f = f_{\text{even}} + f_{\text{odd}}$ with $f_{\text{even}}(z) = \frac{f(z)+f(-z)}{2}$, $f_{\text{odd}}(z) = \frac{f(z)-f(-z)}{2}$. Since $\wp'$ is odd, $f_{\text{odd}}/\wp'$ is even; apply the lemma to f_{even} and $f_{\text{odd}}/\wp'$. ∎

2 The modular character of elliptic functions and Eisenstein series

We now study the dependence on τ and write $\wp_\tau$. Since $1, \tau$ and $1, \tau+1$ generate the same lattice, $\wp_\tau = \wp_{\tau+1}$. Since $\tau = \omega_2/\omega_1$, $-1/\tau = -\omega_1/\omega_2$ (with $\operatorname{Im}(-1/\tau) > 0$) corresponds essentially to

interchanging the periods, and indeed $\wp_{-1/\tau}(z) = \tau^2\wp_\tau(\tau z)$. So we consider the group of transformations of the upper half-plane generated by $\tau \mapsto \tau+1$ and $\tau \mapsto -1/\tau$, the **modular group**; quantities attached to $\wp_\tau$ should reflect these transformations.

2.1 Eisenstein series

The **Eisenstein series of order** k is

$$E_k(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^k} = \sum_{\omega\in\Lambda^*} \frac{1}{\omega^k},$$

for integers $k \geq 3$ and $\mathrm{Im}(\tau) > 0$.

Theorem 2.1

Eisenstein series have the following properties: (i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane. (ii) $E_k(\tau) = 0$ if k is odd. (iii) $E_k(\tau)$ satisfies the following transformation relations: $E_k(\tau+1) = E_k(\tau)$ and $E_k(\tau) = \tau^{-k}E_k(-1/\tau)$.

Property (iii) is the **modular character** of the Eisenstein series.

Proof. (i) By Lemma 1.5 and the remark after it, the series converges absolutely and uniformly in every half-plane $\mathrm{Im}(\tau) \geq \delta > 0$ for $k \geq 3$. (ii) Replacing (n,m) by $(-n,-m)$ shows $E_k = -E_k$ for odd k. (iii) $n+m(\tau+1) = (n+m)+m\tau$, and we may replace $n+m$ by n. Also $(n+m(-1/\tau))^k = \tau^{-k}(n\tau - m)^k$, and rearranging, replacing $(-m,n)$ by (n,m), gives the second relation. ∎

(Because of (ii), some authors define the Eisenstein series as $\sum 1/(n+m\tau)^{2k}$, possibly with a constant factor.)

Theorem 2.2

For z near 0, we have

$$\wp(z) = \frac{1}{z^2} + 3E_4z^2 + 5E_6z^4 + \cdots = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k}.$$

Proof. Replacing ω by $-\omega$, $\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*}\left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right]$. Differentiating the geometric series, $\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty}(\ell+1)w^\ell$ for $|w|<1$, so for small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2}\sum_{\ell=0}^{\infty}(\ell+1)\Big(\frac{z}{\omega}\Big)^\ell = \frac{1}{\omega^2} + \sum_{\ell=1}^{\infty}(\ell+1)\frac{z^\ell}{\omega^{\ell+2}}.$$

Hence

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)\Big(\sum_{\omega\in\Lambda^*}\frac{1}{\omega^{\ell+2}}\Big)z^\ell = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)E_{\ell+2}z^\ell = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k},$$

since $E_{\ell+2} = 0$ for odd ℓ. ∎

Hence near 0,

$$\wp'(z) = \frac{-2}{z^3} + 6E_4z + 20E_6z^3 + \cdots, \quad (\wp'(z))^2 = \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \quad (\wp(z))^3 = \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots.$$

So $(\wp')^2 - 4\wp^3 + 60E_4\wp + 140E_6$ is holomorphic near 0 and vanishes there; being doubly periodic, it is constant (Theorem 1.2), hence 0:

Corollary 2.3

If $g_2 = 60E_4$ and $g_3 = 140E_6$, then $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

This is another version of Theorem 1.7, and expresses the symmetric functions of the e_j via Eisenstein series.

2.2 Eisenstein series and divisor functions

Consider the Fourier expansion of the periodic function $E_k(\tau)$; equivalently, the Laurent expansion in $z = e^{2\pi i\tau}$.

Lemma 2.4

If $k \geq 2$ and $\mathrm{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell \tau}.$$

Proof. One may apply Poisson summation to $f(z) = 1/(z+\tau)^k$. Alternatively, it suffices to treat $k = 2$, the other cases following by termwise differentiation. Differentiating $\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau$ (Chapter 5) gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

With $w = e^{2\pi i\tau}$ ($|w| < 1$), $\sin^2 \pi\tau = -\frac{(e^{i\pi\tau} - e^{-i\pi\tau})^2}{4} = -\frac{(1-w)^2}{4w}$, so $\frac{\pi^2}{\sin^2 \pi\tau} = -4\pi^2 \frac{w}{(1-w)^2} = (-2\pi i)^2 \sum_{r=1}^{\infty} r w^r$, using $\sum_{r \geq 1} r w^r = \frac{w}{(1-w)^2}$. ∎

The **divisor function** $\sigma_\ell(r) = \sum_{d|r} d^\ell$ is the sum of the ℓth powers of the divisors of r.

Theorem 2.5

If $k \geq 4$ is even, and $\mathrm{Im}(\tau) > 0$, then

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. $\sigma_{k-1}(r) \leq \sum_{d \leq r} d^{k-1} \leq r^k$; if $\mathrm{Im}(\tau) \geq t_0 > 0$, $|e^{2\pi i r\tau}| \leq e^{-2\pi r t_0}$, so the series converges absolutely, by comparison with $\sum r^k e^{-2\pi r t_0}$. Using the definition, k even, and the lemma with τ replaced by $m\tau$,

$$\begin{aligned} E_k(\tau) &= \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} = 2\zeta(k) + 2 \sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2\frac{(-2\pi i)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m \tau} = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i r \tau}, \end{aligned}$$

collecting the terms with $\ell m = r$. ∎

For the "forbidden" case $k = 2$, $\sum_{(n,m) \neq (0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but define

$$F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}, \qquad (n,m) \neq (0,0),$$

summed in the indicated order (first n, then m).

Corollary 2.6

The double sum defining F converges in the indicated order. We have

$$F(\tau) = 2\zeta(2) - 8\pi^2 \sum_{r=1}^{\infty} \sigma(r) e^{2\pi i r\tau},$$

where $\sigma(r) = \sum_{d|r} d$ is the sum of the divisors of r.

(The argument of Theorem 2.5 applies with $k = 2$.) $F(-1/\tau)\tau^{-2} \neq F(\tau)$; equivalently, summing first in m and then in n gives a different value. Nevertheless this forbidden Eisenstein series plays a crucial role in the four-squares theorem (Chapter 10).

Chapter 10. Applications of Theta Functions

Overview

The *theta function*

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

converges for all $z \in \mathbb{C}$ and τ in the upper half-plane. It has a dual nature: as a function of z it belongs to the world of elliptic functions (period 1, "quasi-period" τ); as a function of τ it is modular and connects with partitions and with representing integers as sums of squares. The main tools are the *triple product formula* and the *transformation law*. We then discuss partitions and prove the celebrated two-squares and four-squares theorems.

1 Product formula for the Jacobi theta function

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, **Jacobi's theta function** is

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}. \tag{1}$$

Special cases: $\theta(\tau) = \sum_n e^{\pi i n^2 \tau} = \Theta(0|\tau)$ for $\tau \in \mathbb{H}$, and $\vartheta(t) = \sum_n e^{-\pi n^2 t} = \theta(it)$ for $t > 0$. The heat kernel on the circle (Book I) is $H_t(x) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = \Theta(x|4\pi i t)$; and the functional equation of ϑ gave that of ζ (Chapter 6).

Proposition 1.1

The function Θ satisfies the following properties: (i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$. (ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$. (iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$. (iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. If $\operatorname{Im}(\tau) = t \geq t_0 > 0$ and $|z| \leq M$, then $\sum_n |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty$, uniformly; this gives (i). (ii) is clear since $e^{2\pi i n z}$ has period 1. For (iii), complete the square:

$$\Theta(z+\tau|\tau) = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} = \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1) z} e^{-2\pi i z} = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}.$$

By (ii), (iii) it suffices for (iv) to show $\Theta(1/2 + \tau/2|\tau) = 0$: $\Theta(1/2+\tau/2|\tau) = \sum_n (-1)^n e^{\pi i (n^2+n)\tau}$, and pairing $n \geq 0$ with $-n-1$, which has opposite parity and $(-n-1)^2 + (-n-1) = n^2 + n$, the terms cancel. ∎

With the standard notation $q = e^{\pi i \tau}$, the **triple product** is

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 1.2

The function $\Pi(z|\tau)$ satisfies the following properties: (i) $\Pi(z|\tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$. (ii) $\Pi(z+1|\tau) = \Pi(z|\tau)$. (iii) $\Pi(z+\tau|\tau) = \Pi(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$. (iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\mathrm{Im}(\tau) \geq t_0 > 0$, $|q| \leq e^{-\pi t_0} < 1$ and each factor is $1+O(|q|^{2n-1}e^{2\pi|z|})$; since $\sum |q|^{2n-1} < \infty$, the results on infinite products (Section 3) give (i). (ii) is clear. For (iii), since $q^2 = e^{2\pi i\tau}$,

$$\Pi(z+\tau|\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n+1}e^{2\pi iz})(1+q^{2n-3}e^{-2\pi iz}),$$

and comparing with $\Pi(z|\tau)$, $\Pi(z+\tau|\tau) = \Pi(z|\tau)\frac{1+q^{-1}e^{-2\pi iz}}{1+qe^{2\pi iz}}$; (iii) follows from $(1+x)/(1+x^{-1}) = x$ for $x \neq -1$. For (iv), a convergent product vanishes only if a factor does. $1-q^{2n} \neq 0$. $1+q^{2n-1}e^{2\pi iz} = 0$ iff $q^{2n-1}e^{2\pi iz} = -1 = e^{\pi i}$, i.e. $(2n-1)\tau+2z \equiv 1 \pmod 2$, i.e. $z \equiv 1/2+\tau/2-n\tau \pmod 1$ $(n \geq 1)$. $1+q^{2n-1}e^{-2\pi iz} = 0$ iff $(2n-1)\tau-2z \equiv 1 \pmod 2$, i.e. $z \equiv -1/2-\tau/2+n\tau \equiv 1/2+\tau/2+(n-1)\tau \pmod 1$ $(n \geq 1)$. This exhausts the zeros, and they are simple since $e^w - 1$ vanishes to order 1 at 0. ∎

Theorem 1.3 (Product formula)

For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.

Proof. Fix $\tau \in \mathbb{H}$. There is a constant $c(\tau)$ with

$$\Theta(z|\tau) = c(\tau)\Pi(z|\tau) : \tag{2}$$

$F(z) = \Theta(z|\tau)/\Pi(z|\tau)$ is entire (by (iv) of both propositions) and doubly periodic with periods 1, τ (by (ii), (iii)), hence constant. We show $c(\tau) = c(4\tau)$. Put $z = 1/2$ in (2) ($e^{\pm 2\pi iz} = -1$):

$$\sum_n (-1)^n q^{n^2} = c(\tau)\prod_{n=1}^{\infty}(1-q^{2n})(1-q^{2n-1})^2 = c(\tau)\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1}), \quad c(\tau) = \frac{\sum_n (-1)^n q^{n^2}}{\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1})}. \tag{3}$$

Put $z = 1/4$ ($e^{2\pi iz} = i$): $\Theta(1/4|\tau) = \sum_n q^{n^2} i^n$, and since $1/i = -i$, only even $n = 2m$ survive: $\Theta(1/4|\tau) = \sum_m q^{4m^2}(-1)^m$. On the other hand

$$\Pi(1/4|\tau) = \prod_{m=1}^{\infty}(1-q^{2m})(1+iq^{2m-1})(1-iq^{2m-1}) = \prod_{m=1}^{\infty}(1-q^{2m})(1+q^{4m-2}) = \prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4}),$$

the last step by splitting $2m = 4n$ and $2m = 4n-2$ in the first factor ($(1-q^{4n-2})(1+q^{4n-2}) = 1-q^{8n-4}$). Hence

$$c(\tau) = \frac{\sum_n (-1)^n q^{4n^2}}{\prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4})}, \tag{4}$$

and comparing (3) and (4) ($q \mapsto q^4$ corresponds to $\tau \mapsto 4\tau$), $c(\tau) = c(4\tau)$. So $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{\pi i 4^k \tau} \to 0$, (3) gives $c(4^k\tau) \to 1$; so $c(\tau) = 1$. ∎

Corollary 1.4

If $\mathrm{Im}(\tau) > 0$ and $q = e^{\pi i\tau}$, then

$$\theta(\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

Corollary 1.5

For each fixed $\tau \in \mathbb{H}$, the quotient

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ, and with a double pole at $z = 1/2+\tau/2$.

(Primes denote differentiation in z.)

Proof. Let $F = (\log \Theta)' = \Theta'/\Theta$. Differentiating (ii), (iii) of Proposition 1.1 gives $F(z+1) = F(z)$ and $F(z+\tau) = F(z) - 2\pi i$, so F' is doubly periodic. Θ vanishes in the fundamental parallelogram only at $1/2 + \tau/2$, simply, so F has only a simple pole there and F' only a double pole. ∎

1.1 Further transformation laws

In Chapter 9 the modular character was expressed through $T_1 : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. For Θ it is natural to use $T_2 : \tau \mapsto \tau + 2$ and S, since $\Theta(z|\tau + 2) = \Theta(z|\tau)$ but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Theorem 1.6

If $\tau \in \mathbb{H}$, then

$$\Theta(z|-1/\tau) = \sqrt{\tau/i}\, e^{\pi i z^2 \tau} \Theta(z\tau|\tau) \qquad \text{for all } z \in \mathbb{C}. \tag{5}$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to prove (5) for $z = x$ real and $\tau = it$, $t > 0$: for fixed real x both sides are holomorphic in $\tau \in \mathbb{H}$ and would agree on the positive imaginary axis; and for fixed τ both sides are entire in z and would agree on $\mathbb{R}$. For $z = x$, $\tau = it$, (5) reads

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a, this is $\sum_n e^{-\pi t(n+a)^2} = \sum_n t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}$, which is equation (3) of Chapter 4 (from Poisson summation). ∎

Corollary 1.7

If $\mathrm{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i}\,\theta(\tau)$.

(Set $z = 0$; for $\tau = it$ this is the functional equation of ϑ.) This gives precise information about θ as $\tau \to 0$; the next corollary describes θ near $\tau = 1$.

Corollary 1.8

If $\tau \in \mathbb{H}$, then

$$\theta(1 - 1/\tau) = \sqrt{\tau/i} \sum_{n=-\infty}^{\infty} e^{\pi i (n+1/2)^2 \tau} = \sqrt{\tau/i}\,(2e^{\pi i \tau/4} + \cdots).$$

The second identity means that $\theta(1 - 1/\tau) \sim 2\sqrt{\tau/i}\, e^{\pi i \tau/4}$ as $\mathrm{Im}(\tau) \to \infty$.

Proof. n and n^2 have the same parity, so $\theta(1+\tau) = \sum_n (-1)^n e^{\pi i n^2 \tau} = \Theta(1/2|\tau)$; hence $\theta(1-1/\tau) =$

$\Theta(1/2|-1/\tau)$. By Theorem 1.6 with $z=1/2$,

$$\theta(1-1/\tau)=\sqrt{\tau/i}\,e^{\pi i\tau/4}\Theta(\tau/2|\tau)=\sqrt{\tau/i}\,e^{\pi i\tau/4}\sum_n e^{\pi i n^2\tau}e^{\pi i n\tau}=\sqrt{\tau/i}\sum_n e^{\pi i(n+1/2)^2\tau}.$$

The terms $n=0,-1$ contribute $2e^{\pi i\tau/4}$, of absolute value $2e^{-\pi t/4}$ ($\tau=\sigma+it$); the others total $O(\sum_{k\ge1}e^{-\pi(k+1/2)^2t})=O(e^{-9\pi t/4})$. ∎

The **Dedekind eta function** is, for $\mathrm{Im}(\tau)>0$,

$$\eta(\tau)=e^{\frac{\pi i\tau}{12}}\prod_{n=1}^{\infty}(1-e^{2\pi i n\tau}).$$

Proposition 1.9

If $\mathrm{Im}(\tau)>0$, then $\eta(-1/\tau)=\sqrt{\tau/i}\,\eta(\tau)$.

Proof. Differentiate (5) and evaluate at $z_0=1/2+\tau/2$. By the product formula,

$$\Theta(z|\tau)=(1+qe^{-2\pi iz})\prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1}e^{2\pi iz})(1+q^{2n+1}e^{-2\pi iz}),$$

and the first factor vanishes at z_0 (where $e^{2\pi iz_0}=-q$), with derivative $-2\pi iqe^{-2\pi iz_0}=2\pi i$ there, while the remaining factors equal $(1-q^{2n})^3$. So

$$\Theta'(z_0|\tau)=2\pi iH(\tau),\qquad H(\tau)=\prod_{n=1}^{\infty}(1-e^{2\pi i n\tau})^3.$$

Replacing $-1/\tau$ by τ in (5), $\Theta(z|\tau)=\sqrt{i/\tau}\,e^{-\pi iz^2/\tau}\Theta(-z/\tau|-1/\tau)$. Differentiating and evaluating at z_0 (where $-z_0/\tau=-\frac{1}{2\tau}-\frac12$ is a zero of $\Theta(\cdot|-1/\tau)$, congruent to $\frac12+\frac{-1/\tau}{2}$), a computation gives

$$2\pi iH(\tau)=\sqrt{i/\tau}\,e^{-\frac{\pi i\tau}{4}}e^{-\frac{\pi i}{2}}e^{-\frac{\pi i}{4\tau}}\Big(-\frac{1}{\tau}\Big)2\pi iH(-1/\tau),\qquad\text{i.e.}\qquad e^{\frac{\pi i\tau}{4}}H(\tau)=\Big(\frac{i}{\tau}\Big)^{3/2}e^{-\frac{\pi i}{4\tau}}H(-1/\tau),$$

which says $\eta(\tau)^3=(i/\tau)^{3/2}\eta(-1/\tau)^3$. For $\tau=it$, $\eta(\tau)>0$, so taking cube roots gives $\eta(\tau)=\sqrt{i/\tau}\,\eta(-1/\tau)$; by analytic continuation this holds on $\mathbb{H}$. ∎

2 Generating functions

For a sequence $\{F_n\}_{n=0}^{\infty}$, its **generating function** is $F(x)=\sum_{n=0}^{\infty}F_nx^n$; properties of the sequence translate into algebraic or analytic properties of F, which can lead back to new insights. The **partition function** $p(n)$ is the number of ways n can be written as a sum of positive integers ($p(0)=1$; e.g. $p(1)=1$, $p(2)=2$, $p(3)=3$, $p(4)=5$, $p(5)=7$, $p(6)=11$, $p(7)=15$, $p(8)=22$, $p(12)=77$).

Theorem 2.1 (Euler)

If $|x|<1$, then $\displaystyle\sum_{n=0}^{\infty}p(n)x^n=\prod_{k=1}^{\infty}\frac{1}{1-x^k}$.

Formally, write $\frac{1}{1-x^k}=\sum_{m=0}^{\infty}x^{km}$ and multiply out: grouping equal parts, a partition is $n=m_1k_1+\cdots+m_rk_r$ with distinct k_j, corresponding to the term $(x^{k_1})^{m_1}\cdots(x^{k_r})^{m_r}$. The justification is as for the product formula for ζ (Section 1), using convergence of $\prod 1/(1-x^k)$, since $\frac{1}{1-x^k}=1+O(x^k)$.

Similarly $\prod 1/(1-x^{2n-1})$ generates $p_o(n)$, the number of partitions of n into odd parts, and $\prod(1+x^n)$ generates $p_u(n)$, partitions into unequal parts. Remarkably $p_o(n) = p_u(n)$, i.e.

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty}(1+x^n) :$$

indeed $\prod(1+x^n)\prod(1-x^n) = \prod(1-x^{2n})$, and by parity $\prod(1-x^{2n})\prod(1-x^{2n-1}) = \prod(1-x^n)$.

Let $p_{e,u}(n)$, $p_{o,u}(n)$ be the numbers of partitions of n into an even, resp. odd, number of unequal parts. Euler proved $p_{e,u}(n) = p_{o,u}(n)$ unless n is a **pentagonal number** $k(3k+1)/2$, $k \in \mathbb{Z}$ (namely $1, 2, 5, 7, 12, 15, 22, 26, \dots$), in which case $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$. Since multiplying out gives terms $(-1)^r x^{n_1+\cdots+n_r}$ with distinct n_j,

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{n=0}^{\infty}[p_{e,u}(n) - p_{o,u}(n)]x^n,$$

so Euler's theorem follows from:

Proposition 2.2 (Euler's pentagonal number theorem)

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{k=-\infty}^{\infty}(-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. Set $x = e^{2\pi iu}$. In the triple product take $q = e^{3\pi iu}$ and $z = 1/2 + u/2$; then

$$\prod_{n=1}^{\infty}(1-e^{2\pi i3nu})(1-e^{2\pi i(3n-1)u})(1-e^{2\pi i(3n-2)u}) = \prod_{n=1}^{\infty}(1-e^{2\pi inu}).$$

By Theorem 1.3 the product equals $\sum_n e^{3\pi in^2u}(-1)^n e^{2\pi inu/2} = \sum_n (-1)^n e^{\pi in(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}$. ∎

The growth of $p(n)$ can be analyzed via $1/\prod(1-x^n)$ as $|x| \to 1$; elementary considerations give a rough order of growth, and the transformation law of η (Proposition 1.9) leads to a precise asymptotic formula (Appendix A).

3 The theorems about sums of squares

Let $r_2(n)$ be the number of pairs $(x, y) \in \mathbb{Z}^2$ with $n = x^2 + y^2$ (e.g. $r_2(3) = 0$, $r_2(5) = 8$). *Sum of two squares:* which integers are sums of two squares, and what is $r_2(n)$? No integer $4k+3$ is a sum of two squares, and no integer $8k+7$ is a sum of three squares. Let $r_4(n)$ be the number of representations of n as a sum of four squares. *Sum of four squares:* is every positive integer a sum of four squares, and what is $r_4(n)$? These problems, going back to the third century, were first fully solved with Jacobi's theta functions.

3.1 The two-squares theorem

If $n = a^2 + b^2$ and $m = c^2 + d^2$, then with $x + iy = (a + ib)(c + id)$, $x^2 + y^2 = nm$; so divisibility properties of n matter. Let $d_1(n)$ and $d_3(n)$ be the numbers of divisors of n of the form $4k+1$ and $4k+3$.

Theorem 3.1 (Two-squares theorem)

If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.

Consequently, if $n = p_1^{a_1}\cdots p_r^{a_r}$ with distinct primes p_j, then n is a sum of two squares iff every prime p_j of the form $4k+3$ has an even exponent a_j. The starting point: for $q = e^{\pi i\tau}$, $\tau \in \mathbb{H}$,

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n, \tag{6}$$

since $\theta(\tau)^2 = \sum_{n_1}\sum_{n_2} q^{n_1^2} q^{n_2^2} = \sum_{(n_1,n_2)\in\mathbb{Z}^2} q^{n_1^2+n_2^2}$.

Proposition 3.2

The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities

$$\theta(\tau)^2 = 2\sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4\sum_{n=1}^{\infty} \frac{q^n}{1+q^{2n}}, \tag{7}$$

whenever $q = e^{\pi i\tau}$ and $\tau \in \mathbb{H}$.

Proof. Both series converge absolutely ($|q| < 1$), and they are equal since $1/(q^n + q^{-n}) = q^{|n|}/(1+q^{2|n|})$. Since $(1+q^{2n})^{-1} = (1-q^{2n})/(1-q^{4n})$, the right side of (7) is $1 + 4\sum_{n\geq 1}\left[\frac{q^n}{1-q^{4n}} - \frac{q^{3n}}{1-q^{4n}}\right]$. Now $\frac{1}{1-q^{4n}} = \sum_{m\geq 0} q^{4nm}$, so

$$\sum_{n=1}^{\infty}\frac{q^n}{1-q^{4n}} = \sum_{n=1}^{\infty}\sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k)q^k, \qquad \sum_{n=1}^{\infty}\frac{q^{3n}}{1-q^{4n}} = \sum_{k=1}^{\infty} d_3(k)q^k$$

(the series converge since $d_j(k) \leq k$). Compare with (6). ∎

Let

$$C(\tau) = 2\sum_{n=-\infty}^{\infty}\frac{1}{q^n+q^{-n}} = \sum_{n=-\infty}^{\infty}\frac{1}{\cos(\pi n\tau)}, \qquad q = e^{\pi i\tau},\ \tau\in\mathbb{H}. \tag{8}$$

We must prove $\theta^2 = C$. There is a striking parallel: θ comes from the heat equation on the line, whose kernel is the Gaussian $e^{-\pi x^2}$, its own Fourier transform, and its transformation law comes from Poisson summation; C comes from the steady-state heat equation in a strip, whose kernel is $1/\cosh \pi x$ (Section 1), again its own Fourier transform (Example 3), and its transformation law again comes from Poisson summation. Indeed, $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7); and (5) of Chapter 4 with $a = 0$ gives $\sum_n \frac{1}{\cosh \pi n t} = \frac{1}{t}\sum_n \frac{1}{\cosh(\pi n/t)}$, i.e. $C(\tau) = (i/\tau)C(-1/\tau)$ for $\tau = it$, hence for all $\tau \in \mathbb{H}$. Both θ^2 and C tend to 1 as $\mathrm{Im}(\tau) \to \infty$. At the "cusp" $\tau = 1$: by Corollary 1.8, $\theta(1-1/\tau)^2 \sim 4(\tau/i)e^{\pi i\tau/2}$; and (5) of Chapter 4 with $a = 1/2$ gives $\sum_n \frac{(-1)^n}{\cosh(\pi n/t)} = t\sum_n \frac{1}{\cosh(\pi(n+1/2)t)}$, so by analytic continuation

$$C(1-1/\tau) = \frac{\tau}{i}\sum_n \frac{1}{\cos(\pi(n+1/2)\tau)} = 4\frac{\tau}{i}e^{\pi i\tau/2} + O(|\tau|e^{-3\pi t/2}) \quad (t\to\infty),$$

the main terms being $n = 0, -1$.

Proposition 3.3

The function $C(\tau) = \sum 1/\cos(\pi n\tau)$, defined in the upper half-plane, satisfies (i) $C(\tau+2) = C(\tau)$. (ii) $C(\tau) = (i/\tau)C(-1/\tau)$. (iii) $C(\tau) \to 1$ as $\mathrm{Im}(\tau) \to \infty$. (iv) $C(1-1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\mathrm{Im}(\tau)\to\infty$. Moreover, $\theta(\tau)^2$ satisfies the same properties.

Theorem 3.4

Suppose f is a holomorphic function in the upper half-plane that satisfies: (i) $f(\tau+2)=f(\tau)$, (ii) $f(-1/\tau)=f(\tau)$, (iii) $f(\tau)$ is bounded, then f is constant.

Let $\mathcal{F}=\{\tau\in\overline{\mathbb{H}} : |\mathrm{Re}(\tau)|\le 1,\ |\tau|\ge 1\}$; the points $\tau=\pm 1$ are its **cusps** (equivalent under $\tau\mapsto\tau+2$).

Lemma 3.5

Every point in the upper half-plane can be mapped into $\mathcal{F}$ using repeatedly one or another of the following fractional linear transformations or their inverses: $T_2:\tau\mapsto\tau+2$, $S:\tau\mapsto -1/\tau$.

$\mathcal{F}$ is the **fundamental domain** for the group G generated by T_2 and S (uniqueness of representatives fails only on $\partial\mathcal{F}$). Elements $g\in G$ are represented by matrices $\begin{pmatrix} a & b\\ c & d\end{pmatrix}$ with $g(\tau)=\frac{a\tau+b}{c\tau+d}$; these have integer entries and determinant 1, so

$$\mathrm{Im}(g(\tau))=\frac{\mathrm{Im}(\tau)}{|c\tau+d|^2}. \tag{9}$$

Proof. For $\tau\in\mathbb{H}$, since c,d are integers, by (9) we may choose $g_0\in G$ with $\mathrm{Im}(g_0(\tau))$ maximal. Translations $T_2^{\pm 1}$ do not change imaginary parts, so there is $g_1\in G$ with $|\mathrm{Re}(g_1(\tau))|\le 1$ and $\mathrm{Im}(g_1(\tau))$ maximal. If $|g_1(\tau)|<1$, then $\mathrm{Im}(Sg_1(\tau))=\mathrm{Im}(-1/g_1(\tau))=\mathrm{Im}(g_1(\tau))/|g_1(\tau)|^2>\mathrm{Im}(g_1(\tau))$, contradicting maximality. So $g_1(\tau)\in\mathcal{F}$. ∎

Proof of Theorem 3.4. Suppose f is not constant. Let $g(z)=f(\tau)$ with $z=e^{\pi i\tau}$; g is well defined in the punctured unit disc (period 2) and bounded near 0, so 0 is removable and $\lim_{\mathrm{Im}\,\tau\to\infty} f(\tau)$ exists; by the maximum modulus principle $\lim_{\mathrm{Im}\,\tau\to\infty}|f(\tau)|<\sup_{\mathcal{F}}|f|$. At the cusps (by $f(\tau+2)=f(\tau)$ it suffices to treat $\tau=1$) we show $\lim_{\mathrm{Im}\,\tau\to\infty} f(1-1/\tau)$ exists and $\lim|f(1-1/\tau)|<\sup_{\mathcal{F}}|f|$, by interchanging 1 and ∞. Let U_n be the fractional linear transformation of $\begin{pmatrix} 1-n & n\\ -n & 1+n\end{pmatrix}$, which fixes 1; with $\nu(\tau)=1/(1-\tau)$ (taking 1 to ∞, with inverse $\nu^{-1}(\tau)=1-1/\tau$), $U_n=\nu^{-1}T_n\nu$, $T_n(\tau)=\tau+n$. Hence $U_nU_m=U_{n+m}$ and $U_{-1}=\begin{pmatrix} 2 & -1\\ 1 & 0\end{pmatrix}=T_2S$. So every U_n is a finite composite of T_2, S and their inverses, and f is invariant under U_n: $f(\nu^{-1}T_n\nu(\tau))=f(\tau)$. Thus $F(\tau)=f(\nu^{-1}(\tau))=f(1-1/\tau)$ has period 1, and as above $h(z)=F(\tau)$, $z=e^{2\pi i\tau}$, has a removable singularity at 0, giving the claim by the maximum principle. Hence f attains its maximum in the interior of $\mathbb{H}$, contradicting the maximum principle. ∎

Proof of the two-squares theorem. $f(\tau)=C(\tau)/\theta(\tau)^2$ is holomorphic in $\mathbb{H}$ ($\theta\neq 0$ by Corollary 1.4) and invariant under T_2 and S (Proposition 3.3). In $\mathcal{F}$ it is bounded, tending to 1 as $\mathrm{Im}(\tau)\to\infty$ or $\tau\to\pm 1$, by (iii), (iv) of Proposition 3.3, verified by both C and θ^2. So f is bounded in $\mathbb{H}$, hence constant by Theorem 3.4, and the constant is 1: $\theta^2=C$. ∎

3.2 The four-squares theorem

Let $\sigma_1^*(n)$ be the sum of the divisors of n that are not divisible by 4.

Theorem 3.6 (Four-squares theorem)

Every positive integer is the sum of four squares, and moreover $r_4(n)=8\sigma_1^*(n)$ for all $n\ge 1$.

As before, $\theta(\tau)^4=\sum_{n\ge 0} r_4(n)q^n$, $q=e^{\pi i\tau}$. Here nothing as simple as C is available; we need a subtle variant of the Eisenstein series. Recall the forbidden Eisenstein series $F(\tau)=\sum_m\sum_n \frac{1}{(n+m\tau)^2}$

(sum first in n, then m; $(n,m)\neq(0,0)$) of Corollary 2.6, and define

$$E_2^*(\tau)=F(\tau/2)-4F(2\tau),\qquad \tau\in\mathbb{H},\tag{10}$$

equivalently $E_2^*(\tau)=\sum_m\sum_n\left[\frac{1}{(n+m\tau/2)^2}-\frac{1}{(n/2+m\tau)^2}\right]$; the order of summation is critical.

Proposition 3.7

The assertion $r_4(n)=8\sigma_1^*(n)$ is equivalent to the identity $\theta(\tau)^4=-\pi^{-2}E_2^*(\tau)$, where $\tau\in\mathbb{H}$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau)=1+8\sum_{k\geq1}\sigma_1^*(k)q^k$. By Corollary 2.6, $F(\tau)=\frac{\pi^2}{3}-8\pi^2\sum_{k\geq1}\sigma_1(k)e^{2\pi ik\tau}$, $\sigma_1(k)$ the sum of divisors of k. Now $\sigma_1^*(n)=\sigma_1(n)$ if $4\nmid n$, and $\sigma_1^*(n)=\sigma_1(n)-4\sigma_1(n/4)$ if $4\mid n$ (a divisor $d=4\tilde{d}$ of $n=4\tilde{n}$ corresponds to the divisor $\tilde{d}$ of $\tilde{n}$). With (10),

$$E_2^*(\tau)=-\pi^2-8\pi^2\sum_{k=1}^{\infty}\sigma_1^*(k)e^{\pi ik\tau}.$$

■

So Theorem 3.6 reduces to $\theta^4=-\pi^{-2}E_2^*$, and the key is that E_2^* has the same modular properties as θ^4.

Proposition 3.8

The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties: (i) $E_2^*(\tau+2)=E_2^*(\tau)$. (ii) $E_2^*(\tau)=-\tau^{-2}E_2^*(-1/\tau)$. (iii) $E_2^*(\tau)\to-\pi^2$ as $\operatorname{Im}(\tau)\to\infty$. (iv) $|E_2^*(1-1/\tau)|=O(|\tau^2e^{\pi i\tau}|)$ as $\operatorname{Im}(\tau)\to\infty$. Moreover $-\pi^2\theta^4$ has the same properties.

(i) is immediate from the definition. Consider F and its *reverse* $\tilde{F}(\tau)=\sum_n\sum_m\frac{1}{(n+m\tau)^2}$ (summing first in m).

Lemma 3.9

The functions F and $\tilde{F}$ satisfy: (a) $F(-1/\tau)=\tau^2\tilde{F}(\tau)$, (b) $F(\tau)-\tilde{F}(\tau)=2\pi i/\tau$, (c) $F(-1/\tau)=\tau^2F(\tau)-2\pi i\tau$.

Proof. (a) follows from the definitions and $(n+m(-1/\tau))^2=\tau^{-2}(-m+n\tau)^2$. For (b) use $\eta(-1/\tau)=\sqrt{\tau/i}\,\eta(\tau)$ with $\eta(\tau)=q^{1/12}\prod(1-q^{2n})$, $q=e^{\pi i\tau}$. By Proposition 3.2, the logarithmic derivative in τ is

$$\frac{\eta'}{\eta}(\tau)=\frac{\pi i}{12}-2\pi i\sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}},\qquad \sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}}=\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}nq^{2nm}=\sum_{k=1}^{\infty}\sigma_1(k)q^{2k}.$$

Since $F(\tau)=\frac{\pi^2}{3}-8\pi^2\sum_k\sigma_1(k)q^{2k}$, $\frac{\eta'}{\eta}(\tau)=\frac{i}{4\pi}F(\tau)$. By the chain rule the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}\frac{\eta'}{\eta}(-1/\tau)$, which by (a) equals $\frac{i}{4\pi}\tilde{F}(\tau)$. Taking logarithmic derivatives in the functional equation for η,

$$\frac{i}{4\pi}\tilde{F}(\tau)=\frac{1}{2\tau}+\frac{i}{4\pi}F(\tau),$$

so $\tilde{F}(\tau)=F(\tau)-2\pi i/\tau$. (c) follows from (a) and (b). ■

Proof of Proposition 3.8. (ii) By (10) and Lemma 3.9(c),

$$\begin{aligned}E_2^*(-1/\tau)&=F(-1/(2\tau))-4F(-2/\tau)=[4\tau^2F(2\tau)-4\pi i\tau]-4[(\tau/2)^2F(\tau/2)-\pi i\tau]\\&=4\tau^2F(2\tau)-\tau^2F(\tau/2)=-\tau^2(F(\tau/2)-4F(2\tau))=-\tau^2E_2^*(\tau).\end{aligned}$$

(iii) In $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum \sigma_1(k) e^{2\pi i k\tau}$ the sum tends to 0 as $\operatorname{Im}(\tau) \to \infty$, so by (10) $E_2^*(\tau) \to \frac{\pi^2}{3} - 4\frac{\pi^2}{3} = -\pi^2$. (iv) We first show

$$E_2^*(1 - 1/\tau) = \tau^2 \Big[F\Big(\frac{\tau - 1}{2}\Big) - F(\tau/2) \Big]. \tag{11}$$

F has period 1, and Lemma 3.9(c) can be written $F(w) = w^{-2}[F(-1/w) + 2\pi i w]$. Let $u = (\tau - 1)/2$ and $w = u/\tau = \frac{1}{2} - \frac{1}{2\tau}$. Then $-1/w = -2 - \frac{2}{\tau - 1}$, so $F(-1/w) = F(-1/u) = u^2 F(u) - 2\pi i u$, and

$$F\Big(\frac{1}{2} - \frac{1}{2\tau}\Big) = \frac{\tau^2}{u^2}\big[u^2 F(u) - 2\pi i u + 2\pi i u/\tau\big] = \tau^2 F(u) - \frac{2\pi i \tau^2 (1 - 1/\tau)}{u} = \tau^2 F\Big(\frac{\tau - 1}{2}\Big) - 4\pi i \tau.$$

Also $4F(2 - 2/\tau) = 4F(-1/(\tau/2)) = 4[(\tau/2)^2 F(\tau/2) - 2\pi i \tau/2] = \tau^2 F(\tau/2) - 4\pi i \tau$. Subtracting gives (11), since $E_2^*(1 - 1/\tau) = F(\frac{1}{2} - \frac{1}{2\tau}) - 4F(2 - \frac{2}{\tau})$. By the expansion of F, $F(\frac{\tau - 1}{2}) - F(\tau/2) = -8\pi^2 \sum_k \sigma_1(k) e^{\pi i k \tau}((-1)^k - 1) = 16\pi^2 e^{\pi i \tau} + O(|e^{3\pi i \tau}|)$, which with (11) gives (iv). For θ^4: (i)–(iii) follow from Section 1 ($\theta(-1/\tau)^4 = (\tau/i)^2 \theta(\tau)^4 = -\tau^2 \theta(\tau)^4$, $\theta \to 1$), and (iv) from Corollary 1.8: $\theta(1 - 1/\tau)^4 \sim 16(\tau/i)^2 e^{\pi i \tau}$. ∎

Proof of the four-squares theorem. Consider $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, holomorphic in $\mathbb{H}$, invariant under T_2 and S by Proposition 3.8. Since $\theta(\tau)^4 \to 1$ and $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i \tau}(-1)$ as $\operatorname{Im}(\tau) \to \infty$, f is bounded in $\mathcal{F}$, hence in $\mathbb{H}$. By Theorem 3.4, f is constant, equal to $-\pi^2$ by (iii). So $\theta^4 = -\pi^{-2} E_2^*$, which proves the theorem; in particular $r_4(n) \geq 8 > 0$ for all $n \geq 1$. ∎

Appendix A. Asymptotics

Overview

When a function cannot be computed explicitly, one studies its asymptotic behavior; complex analysis is of crucial help. Typically we study, for large s,

$$I(s) = \int_a^b e^{-s\Phi(x)}\,dx. \tag{1}$$

Three guiding principles:

(i) *Deformation of contour.* If Φ is complex-valued, the integrand oscillates rapidly for large s and cancellations mask the behavior. When Φ is holomorphic one may change the contour so that Φ is essentially real on it, and then read off the behavior (illustrated with Bessel functions).

(ii) *Laplace's method.* If Φ is real and $s > 0$, the main contribution comes from near a minimum of Φ, giving an expansion in terms of the quadratic behavior of Φ there (Stirling's formula, the Airy function).

(iii) *Generating functions.* The asymptotics of a sequence F_n can be studied via $F_n = \int_\gamma F(e^{2\pi i z})e^{-2\pi i n z}\,dz$, γ a segment of unit length in the upper half-plane, a variant of (1); we obtain the asymptotics of the partition function $p(n)$.

1 Bessel functions

Bessel functions arise in problems with rotational symmetry (e.g. the Fourier transform of a radial function on $\mathbb{R}^d$ involves $J_{d/2-1}$). For $\nu > -1/2$,

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)} \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{2}$$

$J_{-1/2}(s) = \lim_{\nu\to -1/2} J_\nu(s) = \sqrt{2/\pi s}\cos s$ and $J_{1/2}(s) = \sqrt{2/\pi s}\sin s$; J_ν is elementary only for half-integral ν. These examples suggest:

Theorem 1.1

$$J_\nu(s) = \sqrt{\frac{2}{\pi s}}\cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2}) \text{ as } s\to\infty.$$

It suffices to study

$$I(s) = \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{3}$$

Let $f(z) = e^{isz}(1-z^2)^{\nu-1/2}$ in the plane slit along $(-\infty,-1]\cup[1,\infty)$, with the branch positive on $(-1,1)$. For $s>0$ fixed, Cauchy's theorem on the rectangle $\gamma_{\epsilon,R}$ with vertices $-1+\epsilon$, $1-\epsilon$, $1-\epsilon+iR$, $-1+\epsilon+iR$ (letting $\epsilon\to 0$, $R\to\infty$) gives

$$I(s) = -I_-(s) - I_+(s),$$

$I_\pm$ the integrals over the vertical half-lines from ± 1 upward, oriented upward for I_+ and downward for I_- appropriately. On the contour of I_+, $z = 1+iy$:

$$I_+(s) = ie^{is}\int_0^\infty e^{-sy}(1-(1+iy)^2)^{\nu-1/2}\,dy, \tag{4}$$

and similarly for I_-. In (3), e^{isx} oscillates rapidly; in (4), e^{-sy} decreases rapidly except near $y=0$, so the main contribution comes from y near 0:

Proposition 1.2

Suppose a and m are fixed, with $a > 0$ and $m > -1$. Then as $s \to \infty$

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(e^{-cs}), \tag{5}$$

for some positive c.

Proof. $m > -1$ guarantees convergence at 0. Write $\int_0^a = \int_0^\infty - \int_a^\infty$. The first equals $s^{-m-1}\Gamma(m+1)$ (change variables $x \mapsto x/s$). For the second,

$$\int_a^\infty e^{-sx} x^m \, dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m \, dx = O(e^{-cs}) \tag{6}$$

whenever $c < a$. ∎

Proof of Theorem 1.1. $(1-(1+iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2})$ for $0 \le y \le 1$, and $= O(y^{\nu-1/2} + y^{2\nu-1})$ for $y \ge 1$. Proposition 1.2 with $a = 1$, $m = \nu - 1/2$ (and $\nu + 1/2$), together with (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2}), \qquad I_-(s) = i(2i)^{\nu-1/2} e^{-is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2})$$

(with the appropriate sign conventions). Since $J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)]$ and $\Gamma(1/2) = \sqrt{\pi}$, combining the two conjugate terms gives the theorem: $(-2i)^{\nu-1/2}$ and $(2i)^{\nu-1/2}$ contribute the phases $e^{\mp i\pi(\nu-1/2)/2}$, and the sum is $2^{\nu+1/2} s^{-\nu-1/2}\Gamma(\nu+\frac{1}{2})\cos(s - \frac{\pi\nu}{2} - \frac{\pi}{4})$ up to $O(s^{-\nu-3/2})$. ∎

Under restricted circumstances Proposition 1.2 extends to $\mathrm{Re}(s) \ge 0$:

Proposition 1.3

Suppose a and m are fixed, with $a > 0$ and $-1 < m < 0$. Then as $|s| \to \infty$ with $\mathrm{Re}(s) \ge 0$,

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(1/|s|).$$

(Here s^{-m-1} is the branch that is positive for $s > 0$.)

Proof. First, for $\mathrm{Re}(s) \ge 0$, $s \ne 0$, $\int_0^\infty e^{-sx} x^m \, dx = \lim_{N\to\infty} \int_0^N e^{-sx} x^m \, dx$ exists and equals $s^{-m-1}\Gamma(m+1)$. Write $\int_0^N = \int_0^a + \int_a^N$; the first is entire in s. Since $-\frac{1}{s}\frac{d}{dx}(e^{-sx}) = e^{-sx}$, integration by parts gives

$$\int_a^N e^{-sx} x^m \, dx = -\frac{m}{s}\int_a^N e^{-sx} x^{m-1} \, dx - \Big[\frac{e^{-sx}x^m}{s}\Big]_a^N. \tag{7}$$

As $m < 0$, $\int_a^\infty x^{m-1} dx < \infty$, so letting $N \to \infty$, $\int_a^\infty e^{-sx} x^m \, dx$ is analytic in $\mathrm{Re}(s) > 0$ and continuous on $\mathrm{Re}(s) \ge 0$, $s \ne 0$; it equals $s^{-m-1}\Gamma(m+1) - \int_0^a$ for $s > 0$, hence everywhere there. Finally $\int_0^a = \int_0^\infty - \int_a^\infty$, and by (7), $\int_a^\infty e^{-sx} x^m \, dx = O(1/|s|)$. ∎

(Better error terms, or other ranges of m, require smooth cut-offs to mitigate the end-point $x = a$.)

2 Laplace's method; Stirling's formula

Consider $\int_a^b e^{-s\Phi(x)}\psi(x)\,dx$ with real Φ, and Φ, ψ indefinitely differentiable. Assume there is $x_0 \in (a,b)$ with $\Phi'(x_0) = 0$ and $\Phi''(x) > 0$ throughout $[a,b]$.

Proposition 2.1 (Laplace's method)

Under the above assumptions, with $s > 0$ and $s \to \infty$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = e^{-s\Phi(x_0)}\Big[\frac{A}{s^{1/2}} + O\Big(\frac{1}{s}\Big)\Big], \qquad A = \frac{(2\pi)^{1/2}\psi(x_0)}{(\Phi''(x_0))^{1/2}}. \tag{8}$$

Proof. Replacing Φ by $\Phi - \Phi(x_0)$, assume $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, $\frac{\Phi(x)}{(x-x_0)^2} = \frac{\Phi''(x_0)}{2}\varphi(x)$ with φ smooth, $\varphi(x) = 1 + O(x - x_0)$. Near x_0 change variables $y = (x - x_0)(\varphi(x))^{1/2}$; then $dx/dy = 1 + O(y)$ and $\psi(x) = \tilde{\psi}(y) = \psi(x_0) + O(y)$. On a small interval $[a', b'] \ni x_0$,

$$\int_{a'}^{b'} e^{-s\Phi(x)}\psi(x)\,dx = \psi(x_0)\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}\,dy + O\Big(\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}|y|\,dy\Big), \tag{9}$$

$\alpha < 0 < \beta$. With $y^2 = X$, $dy = \frac{1}{2}X^{-1/2}dX$, and (5), the first integral is $s^{-1/2}(\frac{2\pi}{\Phi''(x_0)})^{1/2} + O(e^{-\delta s})$; similarly the second is $O(1/s)$. On $[a, a']$ and $[b', b]$, $\Phi \geq c > 0$, so those integrals decay exponentially. ∎

(8) extends to complex s with $\mathrm{Re}(s) \geq 0$, using integration by parts to handle the oscillations of $e^{-s\Phi}$:

Proposition 2.2

With the same assumptions on Φ and ψ, the relation (8) continues to hold if $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$.

Proof. Proceed to (9); the first term is handled by Proposition 1.3 with $m = -1/2$. Observation: if $\Phi \geq 0$ and $|\Phi'(x)| \geq c > 0$ on $[a, b]$, then for $\mathrm{Re}(s) \geq 0$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = O\Big(\frac{1}{|s|}\Big) \qquad \text{as } |s| \to \infty; \tag{10}$$

indeed the integral is $-\frac{1}{s}\int_a^b \frac{d}{dx}(e^{-s\Phi})\frac{\psi}{\Phi'}\,dx = \frac{1}{s}\int_a^b e^{-s\Phi}\frac{d}{dx}\Big(\frac{\psi}{\Phi'}\Big)dx - \frac{1}{s}\Big[e^{-s\Phi}\frac{\psi}{\Phi'}\Big]_a^b$, and $|e^{-s\Phi}| \leq 1$. This handles $[a, a']$ and $[b', b]$, where $|\Phi'| \geq c > 0$ (as $\Phi'(x_0) = 0$ and $\Phi'' \geq c_1 > 0$). The second term of (9) has the form $\int e^{-s\frac{\Phi''(x_0)}{2}y^2}y\eta(y)\,dy$ with η differentiable; write it as $\frac{-1}{s\Phi''(x_0)}\int \frac{d}{dy}(e^{-s\frac{\Phi''(x_0)}{2}y^2})\eta(y)\,dy$ and integrate by parts to get $O(1/|s|)$. ∎

For purely imaginary $s = it$ this is the **method of stationary phase**; points with $\Phi'(x_0) = 0$ are **critical points**.

For $\delta > 0$ let $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and let $\log s$ be the principal branch.

Theorem 2.3 (Stirling's formula)

If $|s| \to \infty$ with $s \in S_\delta$, then

$$\Gamma(s) = e^{s\log s}e^{-s}\frac{\sqrt{2\pi}}{s^{1/2}}\Big(1 + O\Big(\frac{1}{|s|^{1/2}}\Big)\Big). \tag{11}$$

(With more effort the error term improves to $O(1/|s|)$, with a full asymptotic expansion in powers of $1/s$. Note $\Gamma(s) \sim \sqrt{2\pi}s^{s-1/2}e^{-s}$, the usual form.)

Proof. First let $\mathrm{Re}(s) \geq 0$, with uniform error terms off a neighborhood of 0. For $s > 0$,

$$\Gamma(s) = \int_0^\infty e^{-x}x^s\frac{dx}{x} = \int_0^\infty e^{-x+s\log x}\frac{dx}{x} = e^{s\log s}e^{-s}\int_0^\infty e^{-s\Phi(x)}\frac{dx}{x}, \qquad \Phi(x) = x - 1 - \log x,$$

after $x \mapsto sx$; by analytic continuation $\Gamma(s) = e^{s\log s}e^{-s}I(s)$ for $\mathrm{Re}(s) > 0$, $I(s) = \int_0^\infty e^{-s\Phi(x)}\frac{dx}{x}$. It suffices to show

$$I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\Big(\frac{1}{|s|}\Big) \qquad \text{for } \mathrm{Re}(s) \geq 0. \tag{12}$$

$\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$, $\Phi''(1) = 1$; so Φ is convex, positive, with minimum at 1. Apply Proposition 2.2 on $[1/2, 2]$ with $x_0 = 1$, $\psi(x) = 1/x$: this gives (12) there. On $[0, 1/2]$, since $\Phi'(x) = 1 - 1/x$,

$$\int_\epsilon^{1/2} e^{-s\Phi(x)}\frac{dx}{x} = -\frac{1}{s}\int_\epsilon^{1/2}\frac{d}{dx}(e^{-s\Phi(x)})\frac{dx}{\Phi'(x)x} = -\frac{1}{s}\Big[\frac{e^{-s\Phi(x)}}{x-1}\Big]_\epsilon^{1/2} - \frac{1}{s}\int_\epsilon^{1/2} e^{-s\Phi(x)}\frac{dx}{(x-1)^2}.$$

Since $\Phi(\epsilon) \to +\infty$ as $\epsilon \to 0$ and $|e^{-s\Phi}| \leq 1$, in the limit $\int_0^{1/2} e^{-s\Phi}\frac{dx}{x} = \frac{2}{s}e^{-s\Phi(1/2)} - \frac{1}{s}\int_0^{1/2} e^{-s\Phi}\frac{dx}{(x-1)^2} = O(1/|s|)$. $\int_2^\infty$ is treated analogously, since $\int_2^\infty (x-1)^{-2}dx < \infty$. So (11) holds for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

To pass to $\mathrm{Re}(s) \leq 0$, $s \in S_\delta$: for $\mathrm{Re}(s) \geq 0$, $s = \sigma + it$, $t \neq 0$, $\log(-s) = \log s - i\pi$ if $t > 0$ and $\log s + i\pi$ if $t < 0$. Hence for $G(s) = e^{s\log s}e^{-s}$,

$$G(-s)^{-1} = e^{s\log s}e^{-s}e^{-si\pi}\ (t > 0), \qquad G(-s)^{-1} = e^{s\log s}e^{-s}e^{si\pi}\ (t < 0). \tag{13}$$

Also

$$\Gamma(s)\Gamma(-s) = \frac{-\pi}{s\sin\pi s}, \tag{14}$$

from $\Gamma(s)\Gamma(1-s) = \pi/\sin\pi s$ and $\Gamma(1-s) = -s\Gamma(-s)$ (Theorem 1.4, Lemma 1.2). Combining (13), (14) and $(1 + O(|s|^{-1/2}))^{-1} = 1 + O(|s|^{-1/2})$ extends (11) to S_δ (in S_δ, $\sin\pi s$ is comparable to $e^{\pm i\pi s}/(\mp 2i)$ for large $|t|$, and near the negative real axis s is excluded). ∎

3 The Airy function

The Airy function appeared in optics (intensity of light near a caustic):

$$\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(x^3/3+sx)}\,dx, \qquad s \in \mathbb{R}. \tag{15}$$

The integral converges and is continuous in s: $\frac{1}{i(x^2+s)}\frac{d}{dx}e^{i(x^3/3+sx)} = e^{i(x^3/3+sx)}$, so for $a \geq 2|s|^{1/2}$,

$$\int_a^R e^{i(x^3/3+sx)}\,dx = \int_a^R \frac{1}{i(x^2+s)}\frac{d}{dx}e^{i(x^3/3+sx)}\,dx, \tag{16}$$

and integrating by parts and letting $R \to \infty$ shows uniform convergence for $|s| \leq a^2/4$; similarly on $(-\infty, -a]$. Replacing $\mathbb{R}$ by $L_\delta = \{x + i\delta : x \in \mathbb{R}\}$, $\delta > 0$, improves matters: by Cauchy's theorem on rectangles γ_R with $f(z) = e^{i(z^3/3+sz)}$, which is $O(e^{-\delta x^2})$ on L_δ and $O(e^{-yR^2})$ on the vertical sides,

$$\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{L_\delta} e^{i(z^3/3+sz)}\,dz,$$

and since $f(z) = O(e^{-\delta x^2})$ on L_δ for every complex s, Ai extends to an entire function. It satisfies

$$\mathrm{Ai}''(s) = s\,\mathrm{Ai}(s)\,: \tag{17}$$

$\mathrm{Ai}''(s) - s\,\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{L_\delta}(-z^2 - s)e^{i(z^3/3+sz)}dz = \frac{i}{2\pi}\int_{L_\delta}\frac{d}{dz}(f(z))\,dz = 0$, since f vanishes at infinity along L_δ. Comparing (17) with $y'' = Ay$ (A large) suggests $\mathrm{Ai}(s)$ decreases rapidly as $s \to \infty$ (it stays bounded by (16), excluding $e^{\sqrt{A}s}$) and oscillates as $s \to -\infty$.

Theorem 3.1

Suppose $u > 0$. Then as $u \to \infty$, (i) $\mathrm{Ai}(-u) = \pi^{-1/2}u^{-1/4}\cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4}))$. (ii) $\mathrm{Ai}(u) = \dfrac{1}{2\pi^{1/2}}u^{-1/4}e^{-\frac{2}{3}u^{3/2}}(1 + O(1/u^{3/4}))$.

Proof. (i) Change variables $x \mapsto u^{1/2}x$ with $s = -u$: $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$, where

$$I_-(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{it(x^3/3-x)}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx \tag{19}$$

with $s = -it$, $\Phi(x) = x^3/3 - x$. $\Phi'(x) = x^2 - 1$: two critical points $x_0 = \pm 1$; $\Phi''(x) = 2x$, $\Phi(\pm 1) = \mp 2/3$. Split into $[-2, 0]$, $[0, 2]$ (one critical point each) and $(-\infty, -2]$, $[2, \infty)$. On $[0, 2]$, Proposition 2.2 with $s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi(1) = -2/3$, $\Phi''(1) = 2$ gives $\frac{1}{2\sqrt{\pi}}e^{-i\frac{2}{3}t}\left[\frac{1}{(-it)^{1/2}} + O(\frac{1}{|t|})\right]$; the interval $[-2, 0]$ gives $\frac{1}{2\sqrt{\pi}}e^{i\frac{2}{3}t}\left[\frac{1}{(it)^{1/2}} + O(\frac{1}{|t|})\right]$ (there Φ has a maximum; apply the proposition to $-\Phi$ with $s = it$). The complementary integrals are $O(1/|t|)$ by integration by parts ($\Phi' \neq 0$ there), e.g. $\int_{-\infty}^{-2} e^{it\Phi}dx = \lim_N \frac{1}{it}\int_{-N}^{-2}\frac{d}{dx}(e^{it\Phi})\frac{dx}{\Phi'(x)}$. Adding and inserting into $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$ gives (i).

(ii) With $s = u$, $\mathrm{Ai}(u) = u^{1/2}I_+(u^{3/2})$,

$$I_+(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx, \qquad F(x) = -i(x^3/3 + x). \tag{20}$$

There is no real critical point; repeated integration by parts shows rapid decay, but not its exact order. Follow the Bessel-function principle: deform the line to a contour γ on which (a) $\mathrm{Im}(F) = 0$, and (b) $\mathrm{Re}(F)$ has a non-degenerate minimum at some z_0 (second derivative along γ positive); then $F'(z_0) = 0$. Through such z_0 pass two orthogonal curves γ_1, γ_2 on which F is real, $\mathrm{Re}(F)|_{\gamma_1}$ having a minimum and $\mathrm{Re}(F)|_{\gamma_2}$ a maximum at z_0; take $\gamma = \gamma_1$—the **method of steepest descent** ($-\mathrm{Re}\,F$ has a saddle at z_0). Here $\mathrm{Re}(F) = x^2y - y^3/3 + y$, $\mathrm{Im}(F) = -x^3/3 + xy^2 - x$, $F'(z) = -i(z^2 + 1)$, critical points $z_0 = \pm i$. At $z_0 = i$ the curves with $\mathrm{Im}(F) = 0$ are $\gamma_1 = \{y^2 = x^2/3 + 1\}$ and $\gamma_2 = \{x = 0\}$; on γ_2, $\mathrm{Re}(F)$ has a maximum at i, so take $\gamma = \gamma_1$, the branch $y = (x^2/3 + 1)^{1/2}$ of a hyperbola, asymptotic to the rays $re^{i\pi/6}$, $re^{i5\pi/6}$. Then

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx = \frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz, \tag{21}$$

by Cauchy's theorem on the region between $\mathbb{R}$ and γ cut off by the circle of radius R (there $e^{-sF(z)} = O(e^{-cyx^2})$, so the arcs contribute $O(\int_0^{\pi/6} e^{-cR^2\sin\theta}R\,d\theta) = O(1/R)$). On γ, $\Phi(x) = \mathrm{Re}(F) = y(x^2 - y^2/3 + 1) = (\frac{8}{9}x^2 + \frac{2}{3})(x^2/3 + 1)^{1/2}$, and $dz = dx + i\frac{x}{3}(x^2/3 + 1)^{-1/2}dx$; since Φ is even and $x(x^2/3 + 1)^{-1/2}$ odd,

$$\frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx. \tag{22}$$

As $(1 + u)^{1/2} = 1 + u/2 + O(u^2)$, $\Phi(x) = \frac{2}{3} + x^2 + O(x^4)$, so $\Phi(0) = 2/3$, $\Phi''(0) = 2$. Proposition 2.1 on $[-c, c]$ with $\psi = 1/2\pi$ gives $e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$; the rest is $O(e^{-2s/3}e^{-\delta s})$, since there $\Phi(x) \geq \frac{2}{3} + c_1x^2$. So $I_+(s) = e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$, which gives (ii). ∎

4 The partition function

Theorem 4.1 (Hardy–Ramanujan)

If p denotes the partition function, then (i) $p(n) \sim \dfrac{1}{4\sqrt{3}\,n} e^{Kn^{1/2}}$ as $n \to \infty$, where $K = \pi\sqrt{2/3}$. (ii) A much more precise assertion is that

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn}\Big(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\Big) + O\big(e^{\frac{K}{2}n^{1/2}}\big).$$

(Since $(n-\frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, $e^{K(n-1/24)^{1/2}} \sim e^{Kn^{1/2}}$, and (ii) implies (i).)

Heuristic principle: if $\sum F_n w^n$ has the unit disc as disc of convergence, the asymptotic behavior of F_n is determined by the location and nature of the "singularities" of F on the unit circle, each contributing according to its "order". (E.g. if F is meromorphic in a larger disc with only a pole of order r at $w = 1$ on the circle, then $F_n = P(n) + O(e^{-\epsilon n})$ for a polynomial P of degree $r-1$, $\sum P(n)w^n$ being essentially the principal part.)

By Euler's formula (Theorem 2.1), with $w = e^{2\pi i z}$, $z = x + iy$, $y > 0$,

$$\sum_{n=0}^{\infty} p(n) e^{2\pi i n z} = f(z) = \prod_{n=1}^{\infty} \frac{1}{1 - e^{2\pi i n z}}, \qquad p(n) = \int_{\gamma} f(z) e^{-2\pi i n z}\, dz, \tag{23}$$

γ the segment from $-1/2 + i\delta$ to $1/2 + i\delta$, $\delta > 0$ to be chosen in terms of n. f is largest near $z = 0$: $|f(x+iy)| \le f(iy)$ (positive coefficients), increasing as y decreases. Fortunately f is a variant of the Dedekind eta function $\eta(z) = e^{\pi i z/12}\prod(1 - e^{2\pi i n z})$: $f(z) = e^{\pi i z/12}\eta(z)^{-1}$ (the $1/12$ explains the $1/24$ in the formula). By $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$ (Proposition 1.9),

$$f(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} f(-1/z), \tag{24}$$

and as $\mathrm{Im}(-1/z) \to \infty$, $f(-1/z) \to 1$ rapidly, since

$$f(z) = 1 + O(e^{-2\pi y}), \qquad z = x + iy,\ y \ge 1. \tag{25}$$

So $f_1(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}}$ approximates f near 0, and by (24) $p(n) = p_1(n) + E(n)$ with

$$p_1(n) = \int_{\gamma} \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz, \qquad E(n) = \int_{\gamma} \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}(f(-1/z) - 1)\, dz.$$

The error term. For $z \in \gamma$, since $\mathrm{Re}(i/z) = \delta/(\delta^2 + x^2)$,

$$\Big|\sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\Big| \le c e^{2\pi n\delta} e^{\frac{\pi}{12}\frac{\delta}{\delta^2+x^2}}. \tag{26}$$

By (25) with z replaced by $-1/z$,

$$|f(-1/z) - 1| \le c e^{-2\pi\frac{\delta}{\delta^2+x^2}} \qquad \text{if } \frac{\delta}{\delta^2+x^2} \ge 1; \tag{27}$$

and since $|f(z)| \le f(iy) \le C e^{\frac{\pi}{12y}}$ for $y \le 1$ (by (24)),

$$|f(-1/z) - 1| = O\big(e^{\frac{\pi}{12}\frac{\delta^2+x^2}{\delta}}\big) = O\big(e^{\frac{\pi}{48\delta}}\big) \qquad \text{if } \frac{\delta}{\delta^2+x^2} \le 1, \tag{28}$$

as $|x| \le 1/2$. Using (26), (27) when $\frac{\delta}{\delta^2+x^2} \ge 1$ gives $O(e^{2\pi n\delta})$ (as $2\pi > \pi/12$); using (26), (28) otherwise gives $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Choose δ to minimize, $2\pi n\delta = \frac{\pi}{48\delta}$, i.e. $\delta = \frac{1}{4\sqrt{6}} n^{-1/2}$; then

$$E(n) = O\big(e^{\frac{\pi}{\sqrt{6}}n^{1/2}}\big) = O\big(e^{\frac{K}{2}n^{1/2}}\big).$$

The main term. Add to γ the end-segments $[-1/2, -1/2+i\delta]$ and $[1/2+i\delta, 1/2]$, forming Γ; on them $\sqrt{z/i}\,e^{\frac{\pi i}{12z}} = O(1)$, so they contribute $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4}n^{1/2}})$, absorbed in $E(n)$. So

$$p_1(n) = \int_\Gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz. \tag{29}$$

Change variables $z \mapsto \mu z$ so that the exponentials combine to $e^{Ai(\frac{1}{z}-z)}$: this requires $A = 2\pi\mu(n-\frac{1}{24})$ and $A = \frac{\pi}{12\mu}$, i.e.

$$A = \frac{\pi}{\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{1/2}, \qquad \mu = \frac{1}{2\sqrt{6}}\Big(n-\frac{1}{24}\Big)^{-1/2}.$$

Then

$$p_1(n) = \mu^{3/2}\int_{\Gamma'} e^{-sF(z)}\sqrt{z/i}\, dz, \qquad F(z) = i(z-1/z), \quad s = A, \tag{30}$$

$\Gamma' = \mu^{-1}\Gamma$ the union of $[-a_n, -a_n+i\delta']$, $[-a_n+i\delta', a_n+i\delta']$, $[a_n+i\delta', a_n]$, with $a_n = \frac{1}{2}\mu^{-1} \approx \sqrt{6}\, n^{1/2}$, $\delta' = \mu^{-1}\delta \to 1$. Steepest descent: F has the critical point $z = i$ in $\mathbb{H}$; through it, F is real on the imaginary axis (maximum, rejected) and on the unit circle (minimum at i). By Cauchy's theorem replace Γ' by Γ'': the segments $[-a_n, -1]$, $[1, a_n]$ and the upper unit semicircle from -1 to 1. On the segments F is purely imaginary, so they contribute $O(a_n^{3/2}\mu^{3/2}) = O(1)$. On the semicircle, $z = e^{i\theta}$ (θ from π to 0), $dz = ie^{i\theta}d\theta$, $i(z-1/z) = -2\sin\theta$ and $\sqrt{z/i} = e^{i(\theta/2-\pi/4)}$; with $\theta = \varphi + \pi/2$ the integrand becomes $-e^{2s\cos\varphi}e^{3i\varphi/2}$, and the contribution is

$$\mu^{3/2}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\big(\cos(3\varphi/2) + i\sin(3\varphi/2)\big)\, d\varphi.$$

Laplace's method with $\Phi(\varphi) = -\cos\varphi$, $\varphi_0 = 0$, $\Phi(0) = -1$, $\Phi''(0) = 1$, $\psi(\varphi) = \cos(3\varphi/2) + i\sin(3\varphi/2)$, $\psi(0) = 1$, gives $\mu^{3/2}e^{2s}\frac{\sqrt{2\pi}}{(2s)^{1/2}}(1+O(s^{-1/2}))$. Since $2s = K(n-\frac{1}{24})^{1/2}$ and $\mu = \frac{1}{2\sqrt{6}}(n-\frac{1}{24})^{-1/2}$,

$$p(n) = \frac{1}{4\sqrt{3}\,n}e^{Kn^{1/2}}(1+O(n^{-1/4})),$$

which is (i).

The precise formula (ii). Write $p_1(n) = \frac{d}{dn}q(n) + e(n)$, where

$$q(n) = \frac{1}{2\pi}\int_\Gamma (z/i)^{-1/2} e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz$$

(differentiating $e^{-2\pi i n z}$ in n produces $-2\pi i z$, and $\frac{-2\pi i z}{2\pi}(z/i)^{-1/2} = (z/i)^{1/2}$), and $e(n)$ comes from the dependence of $\Gamma = \Gamma_n$ on n; by Cauchy's theorem it is $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4}n^{1/2}})$. As before,

$$q(n) = \frac{\mu^{1/2}}{2\pi}\int_{\Gamma''} e^{-sF(z)}(z/i)^{-1/2}\, dz, \tag{31}$$

the segments contributing $O(a_n^{1/2}\mu^{1/2}) = O(1)$ to $\frac{d}{dn}q(n)$. On the semicircle the integral equals

$$\frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\big(\cos(\varphi/2) + i\sin(\varphi/2)\big)\, d\varphi = \frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi}\cos(\varphi/2)\, d\varphi,$$

the odd part vanishing. With $\cos\varphi = 1 - 2\sin^2(\varphi/2)$ and $x = \sin(\varphi/2)$, $\cos(\varphi/2)d\varphi = 2dx$, this is

$$\frac{\mu^{1/2}}{\pi}e^{2s}\int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\, dx, \qquad \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\, dx = \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}),$$

and the same holds after differentiating in s. Collecting the error terms,

$$p(n) = \frac{d}{dn}\Big(\frac{\mu^{1/2}e^{2s}}{2\sqrt{\pi}\,s^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

Since $\mu^{1/2}/s^{1/2} = (2\pi)^{-1/2}(n-\frac{1}{24})^{-1/2}$ and $2s = K(n-\frac{1}{24})^{1/2}$, this is (ii).

Appendix B. Simple Connectivity and Jordan Curve Theorem

Overview

Simple connectivity underlies many basic results of complex analysis. Recall (Chapter 3) that a region Ω is *simply connected* if any two curves in Ω with the same end-points are homotopic; from this we deduced that

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every f holomorphic in Ω. In this appendix we prove:

(I) Ω is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all f holomorphic in Ω.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) Ω is simply connected if and only if no curve in Ω winds around a point of Ω^c.

In the second part we answer: given a simple closed curve γ, what is the region "enclosed by γ"? We prove the Jordan curve theorem for piecewise-smooth curves: γ^c has exactly two components, the bounded one is simply connected and both have boundary γ. This yields a general form of Cauchy's theorem extending the version for toy contours of Chapter 2. As in Chapter 1, "curve" means "piecewise-smooth curve" unless stated otherwise.

1 Equivalent formulations of simple connectivity

Theorem 1.1

A region Ω is holomorphically simply connected if and only if Ω is simply connected.

Proof. If Ω is simply connected, it is holomorphically simply connected by Corollary 5.3. Conversely, let Ω be holomorphically simply connected. If $\Omega = \mathbb{C}$ it is simply connected. Otherwise the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim: the only use of simple connectivity there is to construct a holomorphic logarithm (hence square root) of a non-vanishing holomorphic function g, namely $\log g(z) = c + \int_{z_0}^{z} g'/g$, which is well defined precisely because (1) holds. Hence Ω is conformally equivalent to $\mathbb{D}$. Since $\mathbb{D}$ is simply connected and a conformal map carries homotopies to homotopies, Ω is simply connected. ∎

Theorem 1.2

If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if Ω^c is connected.

Boundedness is needed: an infinite strip is simply connected but its complement has two components. If complements are taken in the Riemann sphere, the conclusion holds for all regions.

Proof that Ω^c connected $\Rightarrow$ Ω simply connected. By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and f holomorphic in Ω. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

K_δ is compact (Ω is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small δ since η is compact. We show that K_δ^c is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with O_1, O_2 disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

Step F_1 is closed. Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ (Ω^c is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large n as O_2 is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise F_2 is closed.

Step $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. Pick $w \in O_1$. Since $w \notin K_\delta$, there is $z \in \Omega^c$ with $|w - z| < \delta$. Every point of the segment $[w, z]$ is within distance $< \delta$ of $z \in \Omega^c$, hence lies in K_δ^c. So the connected segment lies in $O_1 \cup O_2$ and meets both ($w \in O_1$, $z \in O_2$), a contradiction. Likewise $F_2 \neq \emptyset$.

Thus Ω^c is disconnected, contrary to hypothesis; so K_δ^c is connected. By Runge's theorem (Theorem 5.7) there are polynomials $P_n \to f$ uniformly on K_δ, hence on η. Since $\int_\eta P_n = 0$ (polynomials have primitives), $\int_\eta f = \lim \int_\eta P_n = 0$. ∎

The converse is proved after Theorem 1.4, using winding numbers.

Winding numbers

If γ is closed and $z \notin \gamma$, each loop of γ around z changes $\frac{1}{2\pi} \arg(\zeta - z)$ by ± 1. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

Definition (Winding number)

The **winding number** (or **index**) of a closed curve γ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for e^{-ikt} the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

Lemma 1.3

Let γ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If z, w lie in the same connected component of γ^c, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If z lies in the unbounded component of γ^c, then $W_\gamma(z) = 0$.

Proof. (i) Let $\gamma : [0,1] \to \mathbb{C}$ parametrize the curve and $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$. G is continuous and, except at finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H(t) = (\gamma(t) - z)e^{-G(t)}$ is continuous with $H' = \gamma' e^{-G} - (\gamma - z)G' e^{-G} = 0$ except at finitely many points, so H is constant: $H(1) = H(0) = \gamma(0) - z$. As $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$, so $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) W_γ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on z, uniformly on γ), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since W_γ is constant on the unbounded component by (ii), it is 0 there. ∎

Theorem 1.4

A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices to show Ω^c is connected. Suppose not: $\Omega^c = F_1 \cup F_2$ with F_1, F_2 disjoint, closed, non-empty. Only one of them can be unbounded (Ω is bounded, so Ω^c contains the connected set $\{|z| > R\}$), so we may assume F_1 is bounded, hence compact. We construct a closed curve $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using the following lemma. ∎

Lemma 1.5

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) w is in the interior of Q_1;

(ii) the interiors of Q_j and Q_k are disjoint for $j \neq k$;

(iii) F_1 is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from F_2;

(v) the boundary of $\bigcup_j Q_j$ lies in Ω and is a finite union of disjoint simple closed polygonal curves.

Proof of Theorem 1.4, assuming Lemma 1.5. Orient each ∂Q_j positively. Since w is interior to Q_1 and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

A side shared by two squares of $\mathcal{Q}$ is traversed twice in opposite directions and cancels. The remaining sides, oriented with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_{j=1}^{M} W_{\gamma_j}(w) = 1$. Hence $W_{\gamma_{j_0}}(w) \neq 0$ for some j_0, while $\gamma_{j_0} \subset \Omega$ and $w \notin \Omega$: a contradiction. ∎

Proof of Lemma 1.5. F_1 is compact and F_2 closed, so $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ such that w is the center of a square R_1, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be the squares of $\mathcal{G}_0$ meeting F_1.

Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$. (i), (ii) are clear. Every point of $\bigcup R_j$ is within $\sqrt{2}\delta < d$ of F_1, giving (iv). If $z \in F_1$, every square of $\mathcal{G}_0$ containing z meets F_1 and so belongs to $\mathcal{R}$; these squares cover a neighborhood of z, giving (iii). A *boundary side* (a side of exactly one square of $\mathcal{R}$) is a side of a square not in $\mathcal{R}$, so it misses F_1; it also misses F_2 by (iv). Hence $\partial \bigcup R_j \subset \Omega$.

Step Bad vertices. The boundary of $\bigcup R_j$ is the union of the boundary sides; their end-points are boundary vertices. At a grid vertex P the four surrounding squares give 0, 2 or 4 boundary sides at P, and 4 occurs exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not. Call such a P *bad*.

Step Refinement. Let $\mathcal{G}$ be the grid obtained by cutting each square of $\mathcal{G}_0$ into 9 equal subsquares, and take all $9m$ subsquares of squares of $\mathcal{R}$, with Q_1 the middle subsquare of R_1 (so w is its center). At each bad vertex P add the two subsquares of $\mathcal{G}$ having P as a corner and lying in the two missing squares of $\mathcal{G}_0$. Call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. Now all four small squares at P are in $\mathcal{Q}$, so P is interior. An added subsquare s at P has three other corners. At the two lying on sides of $\mathcal{G}_0$, three of the four surrounding small squares are in $\mathcal{Q}$. At the one inside a big square, only s is. So none of them is bad. Added squares at different bad vertices sit at different corners of a big square, so they do not touch. Hence $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist: the added squares lie in squares of $\mathcal{G}_0$ adjacent to $\mathcal{R}$, within $3\delta < d$ of F_1. The new boundary sides lie in squares of $\mathcal{G}_0$ not in $\mathcal{R}$, so they miss F_1, and also F_2. So $\partial \bigcup Q_j \subset \Omega$.

Step (v). Every boundary vertex of $\bigcup Q_j$ now meets exactly two boundary sides. With the orientation leaving $\bigcup Q_j$ on the left, one of them enters the vertex and the other leaves it (check the

cases of one, two adjacent, or three surrounding squares). Starting from a boundary side $[a_1, a_2]$ we get a unique continuation $[a_2, a_3], [a_3, a_4], \ldots$. There are finitely many sides, so some vertex repeats; let a_m be the first repetition, $a_m = a_k$ with $k < m$. If $k > 1$, then a_k meets the three distinct sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, which is impossible. So $k = 1$ and $a_1 \cdots a_m$ is a simple closed polygon. Removing its sides and repeating exhausts the boundary sides, giving finitely many disjoint simple closed polygons (disjoint because each vertex lies on only one of them). ■

Proof of Theorem 1.2, converse. Let Ω be bounded and simply connected. If Ω^c were disconnected, the proof of Theorem 1.4 produces a closed curve $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$. This contradicts the first part of Theorem 1.4, since Ω is simply connected. ■

2 The Jordan curve theorem

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves γ_η below).

Theorem 2.1 (Jordan arc theorem)

Let γ be a simple (non-closed) piecewise-smooth curve in the plane. Then γ^c is an open connected set whose boundary is precisely γ.

Theorem 2.2 (Jordan curve theorem)

Let γ be a simple closed piecewise-smooth curve in the plane. Then γ^c consists of two disjoint connected open sets. Exactly one of them is bounded and simply connected; it is the **interior** Ω_γ of γ. The other is unbounded; it is the **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of γ,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

Remark

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

Theorem 2.3

If f is holomorphic in the interior Ω_γ of a simple closed curve γ, then $\int_\eta f(\zeta)\, d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

Proof (supplied). Ω_γ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3. ■

Idea of the proof of Theorem 2.1. γ^c is open, so it suffices to show it is pathwise connected. Join $z, w \notin \gamma$ by a curve. If it meets γ, first go from z to a point z' and from w to a point w' near γ without meeting γ. Then join z' to w' by traveling "parallel" to γ, going around an end-point if necessary. If γ parametrizes a smooth piece, $\gamma'(t) \neq 0$ is tangent to it, $i\gamma'(t)$ is normal, and $\gamma(t) + i\eta\gamma'(t)$ is a "parallel" curve.

In the next three lemmas and two propositions, γ_0 is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'(t)| = 1$ (every smooth curve has one). An *interior point* of γ_0 is a point $\gamma(t)$ with $0 < t < L$ (not to be confused with the interior of Theorem 2.2).

Lemma 2.4

For $\eta \in \mathbb{R}$ let γ_η be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

Proof.

Step Local version. For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of γ' there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \ge |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |t - s + i\eta| - \frac{|t - s|}{2} \ge \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \ne \gamma(s)$ when $|t - s| < \delta$ and $\eta \ne 0$.

Step Global version. Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \le \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \ge \delta/2\}$ (intersected with $[0, L]$). The I_k cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \ne 0). \tag{3}$$

Since γ_0 is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |\gamma(t) - \gamma(s)| - |\eta| \ge d_k - |\eta| \ge d_k/2 \quad \text{if } |\eta| \le d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ■

Lemma 2.5

Suppose $z \notin \gamma_0$ and $z_0 \in \gamma_0$ is a point closest to z with $z_0 = \gamma(t_0)$, $0 < t_0 < L$ (this holds if z is closer to an interior point of γ_0 than to either end-point). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$ for some real $\eta \ne 0$, i.e. $z \in \gamma_\eta$.

Proof. Near t_0, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \le |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\,[(z - \gamma(t_0))\overline{\gamma'(t_0)}] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \ne 0$ because $z \notin \gamma_0$. ■

If $z \in \gamma_\eta$ and $w \in \gamma_\epsilon$ ($\eta, \epsilon \ne 0$) are obtained as in Lemma 2.5, we say z, w are on the **same side** of γ_0 if η, ϵ have the same sign, and on **opposite sides** otherwise. We do not define "the two sides of γ_0"; the notion refers to the chosen representations, and we do not claim the two cases are exclusive. Points on the same side can be joined almost directly by a curve parallel to γ_0. Points on opposite sides require going around an end-point.

Proposition 2.6

Let A, B be the end-points of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A, B\}$ (or a subset thereof). If $z, w \notin \gamma_0$ lie on the same side of γ_0, and each is closer to an interior point of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous

curve in the complement of $K \cup \gamma_0$.

Proof. By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to z, w,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with η_0, ϵ_0 of the same sign, say positive, and say $t_0 \le s_0$.

Step Normal segments. The half-open segment from z to z_0 lies in $(K \cup \gamma_0)^c$. If p is on it with $p \ne z_0$ and $p \in K \cup \gamma_0$, then $|z - p| < |z - z_0| = d(z, K \cup \gamma_0)$, which is absurd. So for all small $\eta > 0$, z can be joined to $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and likewise w to $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

Step Parallel curve. Choose $\eta < \eta_1$ (Lemma 2.4) and η smaller than the distance from K to $\{\gamma(t) : t_0 \le t \le s_0\}$. This distance is positive since this arc avoids A, B. Then $\{\gamma_\eta(t) : t_0 \le t \le s_0\}$ joins z_η to w_η. It misses γ_0 by Lemma 2.4, and it misses K since $|\gamma_\eta(t) - \gamma(t)| = \eta$. ∎

Lemma 2.7

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \le \theta \le \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from γ_0.

Proof. As in Lemma 2.4,

$$\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)]\,du + (L - t + \eta e^{i\theta})\gamma'(L).$$

Choose δ with $|\gamma'(u) - \gamma'(L)| < 1/2$ for $|u - L| < \delta$. Since $\cos\theta \ge 0$, $|L - t + \eta e^{i\theta}|^2 \ge (L-t)^2 + \eta^2 \ge \frac{1}{2}(L - t + \eta)^2$. So for $L - \delta < t \le L$,

$$|\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t)| \ge \frac{L - t + \eta}{\sqrt{2}} - \frac{L - t}{2} \ge \frac{\eta}{\sqrt{2}} > 0.$$

Thus $\gamma(t) \notin N$ for $L - \delta < t \le L$. Finally, take η_2 less than the distance from $\gamma(L)$ to $\{\gamma(t) : 0 \le t \le L - \delta\}$, which is positive since γ_0 is simple. ∎

Proposition 2.8

Let A be an end-point of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A\}$. If $z, w \notin \gamma_0$ are closer to interior points of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $\gamma_0 \cup K$.

Proof. If z, w are on the same side, apply Proposition 2.6. So let them be on opposite sides, and (reversing the parametrization if needed) $A = \gamma(0)$, so the other end-point $B = \gamma(L) \notin K$. Write

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) - i\epsilon_0\gamma'(s_0), \qquad \eta_0, \epsilon_0 > 0,$$

and let $\tau = \min(t_0, s_0) > 0$. Choose $\eta > 0$ smaller than η_1, η_2 (Lemmas 2.4, 2.7), and the distance from K to $\{\gamma(t) : \tau \le t \le L\}$ (positive since this arc avoids A). As in Proposition 2.6, join z to $\gamma_\eta(t_0)$ by a normal segment. Then follow γ_η from t_0 to L to reach $\gamma(L) + i\eta\gamma'(L)$. Then follow the half-circle $\gamma(L) + \eta e^{i\theta}\gamma'(L)$, θ from $\pi/2$ down to $-\pi/2$, inside N, to reach $\gamma(L) - i\eta\gamma'(L)$. Then follow $\gamma_{-\eta}$ from L back to s_0, and the normal segment to w. The parallel pieces miss γ_0 by Lemma 2.4; the half-circle misses γ_0 by Lemma 2.7. All pieces lie within η of $\{\gamma(t) : \tau \le t \le L\}$, so they miss K. ∎

Proof of Theorem 2.1.

Step Boundary. $O = \gamma^c$ is open and $\partial O \subset \gamma$. Every point of a smooth piece of γ is a limit of points of O: by Lemma 2.4 (applied to a small sub-arc, and noting the rest of γ is at positive distance), $\gamma(t) + i\eta\gamma'(t) \notin \gamma$ for small $\eta \neq 0$. Since ∂O is closed and the smooth points are dense in γ, $\partial O = \gamma$.

Step Connectedness: induction on the number of smooth pieces. First let γ be smooth, with end-points A, B, and let $Z, W \notin \gamma$. Join Z to W by a polygonal path Λ avoiding A and B. If Λ misses γ we are done. Otherwise let p and q be the first and last points of Λ on γ; they are interior points of γ. Points z of Λ just before p satisfy $d(z, \gamma) \leq |z - p| \to 0$, while $|z - A|, |z - B|$ stay bounded below. So we may take such a $z \notin \gamma$ closer to an interior point of γ than to A, B, joined to Z along Λ without meeting γ. Similarly take w just after q. Proposition 2.8 with $K = \emptyset$ joins z to w in γ^c.

Now suppose the result holds for curves of $n-1$ smooth pieces, and let $\gamma = K \cup \gamma_0$ with K the union of $n-1$ consecutive pieces and γ_0 smooth. Then K is compact and $K \cap \gamma_0$ is a single end-point A of γ_0. Given $Z, W \notin \gamma$, the induction hypothesis gives a curve from Z to W missing K. Since K^c is open and connected, we may take it polygonal and avoiding the other end-point of γ_0. If it meets γ_0, it does so at interior points. As above, take z (resp. w) just before the first (resp. after the last) meeting point; it is closer to an interior point of γ_0 than to K or to the end-points, since its distance to K is bounded below. Proposition 2.8 joins z to w in $(\gamma_0 \cup K)^c = \gamma^c$. ∎

Proof of Theorem 2.2.

Step At most two components. Fix W outside a large disc containing γ, and let $\mathcal{U}$ be the set of points joined to W by a continuous curve in γ^c. $\mathcal{U}$ is open and connected, and contains the exterior of the disc. Let $\Omega = \gamma^c - \mathcal{U}$; we show Ω is connected. Remove a smooth piece γ_0 (with end-points A, B) from γ, and let K be the remaining arc, so $K \cap \gamma_0 = \{A, B\}$. For $Z \in \Omega$, Theorem 2.1 applied to K gives a curve Λ_Z from Z to W missing K. Since $Z \notin \mathcal{U}$, Λ_Z meets γ_0, necessarily at interior points. As in the proof of Theorem 2.1, choose $z, w \in \Lambda_Z$ closer to interior points of γ_0 than to K or to A, B, such that the pieces of Λ_Z from Z to z and from w to W lie in γ^c. Then z, w are on opposite sides of γ_0. Otherwise Proposition 2.6 would join z to w in γ^c, hence Z to W, contradicting $Z \notin \mathcal{U}$. If $Z_1 \in \Omega$ is another point, with corresponding z_1, w_1 on opposite sides, then z and z_1 are on the same side. Otherwise z_1 is on the same side as w, and Proposition 2.6 joins Z_1 to W in γ^c, a contradiction. So Proposition 2.6 joins z to z_1 in γ^c, and Z, Z_1 lie in the same component. Thus γ^c has at most two components, $\mathcal{U}$ and (if non-empty) Ω.

Step Exactly two components. By Lemma 1.3 it suffices to find points with different winding numbers. Parametrize γ and fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$. For $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small ϵ these avoid γ, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \leq \epsilon\}$ lie in γ^c (Lemma 2.4 for the piece, plus the positive distance from $\gamma(t_0)$ to the rest of γ). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t - t_0) + o(|t - t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v\gamma'(t)}{a(t)^2 + \epsilon^2 v^2} = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2}\,(1 + \rho(t, \epsilon)),$$

where, given $\sigma > 0$, there is $\delta > 0$ with $|\rho(t, \epsilon)| < \sigma$ for $|t - t_0| \leq \delta$, uniformly in ϵ. (Indeed $a^2 + \epsilon^2 v^2 = v^2[(t - t_0)^2 + \epsilon^2] + o((t - t_0)^2)$ and $o((t - t_0)^2)/[(t - t_0)^2 + \epsilon^2] \to 0$.) Split the integral defining $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ into $|t - t_0| \geq \delta$ and $|t - t_0| < \delta$. On the first part $\gamma(t)$ stays at positive distance from $\gamma(t_0)$, while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part tends to 0 as $\epsilon \to 0$. On the second,

$$\frac{1}{2\pi i}\int_{|t-t_0|<\delta} \frac{2i\epsilon\, dt}{(t - t_0)^2 + \epsilon^2} = \frac{1}{\pi}\int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2 + 1} = \frac{2}{\pi}\arctan(\delta/\epsilon) \to 1,$$

and the ρ-term is at most $\frac{\sigma}{\pi}\int_{\mathbb{R}}\frac{ds}{s^2+1}=\sigma$ in absolute value. Letting $\epsilon\to 0$ gives $|\Delta-1|\le\sigma$ for every $\sigma>0$, so $\Delta=1$.

Hence z_ϵ and w_ϵ lie in different components, so γ^c has exactly the two components $\mathcal{U}$ and Ω. $\mathcal{U}$ is unbounded with $W_\gamma=0$ there (Lemma 1.3(iii)), and Ω is bounded. Since $\Delta=1$, $W_\gamma=\pm1$ on Ω, and after possibly reversing orientation $W_\gamma=1$ there. Every smooth point of γ is a limit of the z_ϵ and of the w_ϵ, hence of points of both components. Smooth points are dense in γ, and the boundary of each component lies in γ. So $\gamma=\partial\Omega=\partial\mathcal{U}$. Write $\Omega_\gamma=\Omega$ and $\mathcal{U}_\gamma=\mathcal{U}$.

Step Ω_γ is simply connected. By Theorem 1.2 it suffices that $\Omega_\gamma^c=\gamma\cup\mathcal{U}_\gamma$ is connected. Suppose $\Omega_\gamma^c=F_1\cup F_2$ with F_1,F_2 closed, disjoint, non-empty, and let $O_i=\mathcal{U}_\gamma\cap F_i$. The sets O_1,O_2 are disjoint and $\mathcal{U}_\gamma=O_1\cup O_2$.

O_1 is open. Let $z\in O_1$; small balls B at z lie in $\mathcal{U}_\gamma\subset F_1\cup F_2$. If every such ball met F_2, then $z\in F_2$ (F_2 closed), contradicting $F_1\cap F_2=\emptyset$. So some ball lies in F_1, hence in O_1. Likewise O_2 is open.

$O_1\neq\emptyset$. Otherwise $F_1\subset\gamma$ and $\mathcal{U}_\gamma\subset F_2$. Take $z\in F_1\subset\gamma$. Since $\gamma=\partial\mathcal{U}_\gamma$, every ball at z meets $\mathcal{U}_\gamma\subset F_2$, so $z\in F_2$, a contradiction. Likewise $O_2\neq\emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ∎

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9

If f is holomorphic in an open set containing a simple closed piecewise-smooth curve γ and its interior, then $\int_\gamma f=0$.

Let $\mathcal{O}\supset\gamma\cup\Omega_\gamma$ be open with f holomorphic in $\mathcal{O}$. The idea is to build a closed curve $\Gamma\subset\Omega_\gamma$ so close to γ that $\int_\gamma f=\int_\Gamma f$. The right side vanishes by Theorem 2.3, since f is holomorphic in the simply connected set Ω_γ. Near smooth parts of γ, Γ is essentially a parallel curve γ_η (Lemma 2.4); near the junctions of smooth parts, Γ follows arcs of circles.

Lemma 2.10

Let $\gamma:[0,1]\to\mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon>0$ the circle C_ϵ of center $\gamma(0)$ and radius ϵ meets γ in exactly one point.

Proof. Assume $\gamma(0)=0$. Since $\gamma(1)\neq0$, the intermediate value theorem applied to $|\gamma(t)|$ gives at least one intersection point for $\epsilon<|\gamma(1)|$. Suppose the claim fails: there are $\epsilon_j\to0$ such that $|\gamma(t)|=\epsilon_j$ has two distinct solutions. Since $|\gamma(t)|\ge c(\tau)>0$ for $t\ge\tau$ (simplicity), both solutions tend to 0. By the mean value theorem applied to $h(t)=|\gamma(t)|^2$, there are $t_j\to0$, $t_j>0$, with $h'(t_j)=2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}]=0$. But $\gamma(t)=\gamma'(0)t+t\varphi(t)$ and $\gamma'(t)=\gamma'(0)+\psi(t)$ with $\varphi,\psi\to0$ as $t\to0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}]=|\gamma'(0)|^2t+o(t),$$

which is $\neq0$ for small $t>0$ since $\gamma'(0)\neq0$. This is a contradiction. ∎

Proof of Theorem 2.9. Choose ϵ so small that the open set U of points at distance $<\epsilon$ from γ is contained in $\mathcal{O}$. Let $P_1,\dots,P_n$ be the consecutive junction points of the smooth parts of γ. Choose $\delta<\epsilon/10$ so small that each circle C_j centered at P_j of radius δ meets γ in exactly two points. This is possible by Lemma 2.10, applied to the two smooth parts meeting at P_j; the rest of γ is at positive distance from P_j. The two points determine two arcs of C_j, exactly one of which, $\mathcal{C}_j$, has its interior in Ω_γ. Indeed, for a smooth part with end-point P_j and small η, the curves γ_η

and $\gamma_{-\eta}$ lie on opposite sides of γ, hence in different components of γ^c (proof of Theorem 2.2), and both meet C_j. The disc D_j of center P_j and radius 2δ lies in U, hence in $\mathcal{O}$.

Let B_j be the end-point of $\mathcal{C}_j$ on the smooth part γ_j from P_j to P_{j+1}, and A_{j+1} the end-point of $\mathcal{C}_{j+1}$ on γ_j. Take a chain of discs $\mathcal{D} = \{D_0, \dots, D_K\}$ in U covering γ, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs D_j among them. By Lemma 2.4 there is a continuous curve Γ_j in Ω_γ and in the union of the discs, running parallel to γ_j, from a point near B_j on $\mathcal{C}_j$ to a point near A_{j+1} on $\mathcal{C}_{j+1}$. Since γ is only C^1, Γ_j need not be smooth; replacing it by a nearby polygonal line we may assume it is. Joining Γ_j to Γ_{j+1} along a piece of $\mathcal{C}_{j+1}$, and so on, gives a closed piecewise-smooth curve $\Gamma \subset \Omega_\gamma$.

f has a primitive on each disc of $\mathcal{D}$ (Theorem 2.1). Arguing as in the proof of Theorem 5.1, and comparing γ and Γ disc by disc, gives $\int_\gamma f = \int_\Gamma f$: on consecutive discs the primitives differ by constants, and the endpoint contributions telescope. Since $\Gamma \subset \Omega_\gamma$ and Ω_γ is simply connected, $\int_\Gamma f = 0$ by Theorem 2.3. Hence $\int_\gamma f = 0$. ■